# 2 Amplitude Modulation

## Objective

After completing this chapter user will be able to understand:

- What is modulation and why it is needed?
- Amplitude modulation and its types
- Transmission and reception of amplitude modulated signal
- Hilbert transform
- Vestigial sideband
- Frequency division multiplexing

## Introduction

- Communication, in general, is an exchange of information between sender and receiver.
- There exists a channel through which the information is conveyed.
- Similarly, communication system is an electronic way to send, receive and process the information.
- There are numerous ways to fulfill this objective. We will be studying some of the most efficient ways of effective communication.

## Basic Communication System Block Diagram

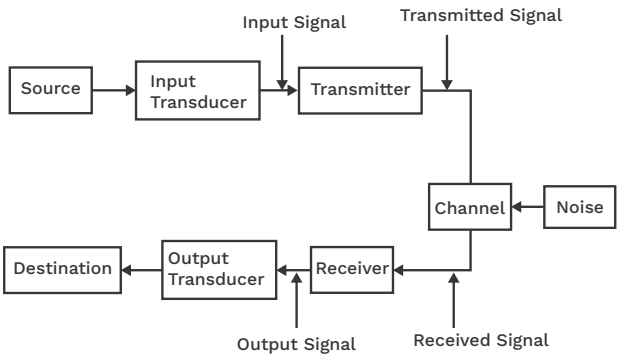



**Fig. 2.1 Block Diagram of Communication System**

**Transmitter section:**

- It consists of source, input transducer and transmitter.
- The source gives the information that can be in the form of speech, audio, video, picture, text etc.
- The transducer converts these signals into electrical form which is referred to as message or baseband signal.
- The transmitter block converts this baseband signal to a modified one for efficient transmission.
- It also has various sub-systems present in it. E.g. modulator and/or an encoder.

**Channel:**

A channel is a medium through which the information is sent to the receiver. It can be done in multiple ways, keeping in mind the distance, feasibility, efficiency etc.

Commonly there are two types of channels:

1. Wired or physical channel: There is an actual physical wire running through the transmitter and receiver section.

   Examples: Coaxial cables, twisted pair (copper) or optical fibers.

2. Wireless channels: There is no actual physical contact present between transmitter and receiver sections.

   Examples: Radio frequency link, satellite communication etc. This mainly uses the electromagnetic spectrum for the transmission of information.

**Receiver section:**

- The receiver section distinctly has three functions to perform.
- Firstly, it demodulates the signal, then removes the signal distortion due to noise and lastly, converts it back to its original physical from with the help of an output transducer.
- The signal is then sent to its destination (e.g., Loudspeaker).

The block diagram for wireless communication system is given below:

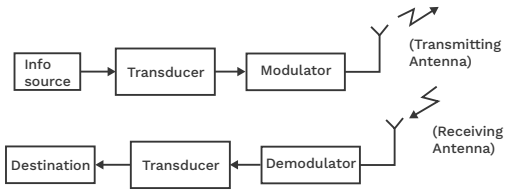



**Fig. 2.2 Block Diagram of Wireless Communication System**

## Electromagnetic Spectrum

- The wireless channel uses the electromagnetic (EM) spectrum to communicate data.
- Different frequencies are used for different applications.
- The electromagnetic spectrum is listed below

**Types of propagation:**

There are mainly three types of propagation:

**Groundwave propagation:**

Signals follow contour of the Earth; usually a large sized antenna is required.

**Skywave propagation:**

Signals are reflected by the ionosphere at an altitude back to Earth.

**Free space propagation:**

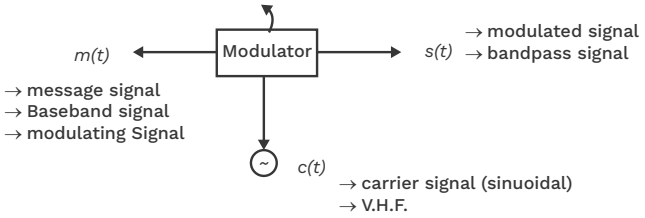
Signals are not influenced by the Earth or the atmosphere.

**Modes of communication:**

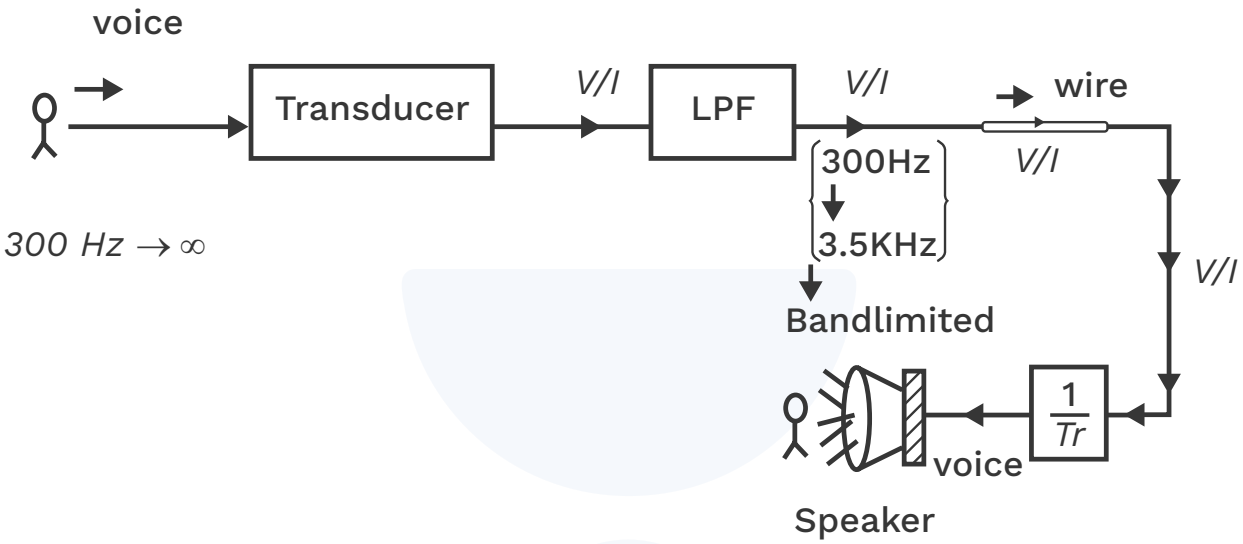
- **Simplex:** One way communication.

| Frequency | Frequency Band | Application |
|---|---|---|
| 30–300 Hz | Extremely Low Frequency (ELF) | Power line transmission |
| 0.3–3 kHz | Voice Frequency (VF) | Audio/speech communication |
| 3–30 kHz | Very Low Frequency (VLF) | Submarine communication |
| 30–300 kHz | Low Frequency (LF) | Aeronautical communication |
| 0.3–3 MHz | Medium Frequency (MF) | AM radio broadcasting |
| 3–30 MHz | High Frequency (HF) | Telephony |
| 30–300 MHz | Very High Frequency (VHF) | TV, FM broadcast |
| 0.3–3 GHz | Ultra High Frequency (UHF) | TV, cellular phones |
| 3–30 GHz | Super High Frequency (SHF) | RADAR, microwave |
| 30–300 GHz | Extremely High Frequency (EHF) | Satellite communication |
| 300 GHz–430 THz | Infrared | TV remote, LED , LASER |
| 430-750 THz | Visible light | Optical communication |
| 750–3000 THz | Ultra violet | Industrial usage |
| >3000 THz | X-rays, gamma rays and cosmic rays | Medical applications |

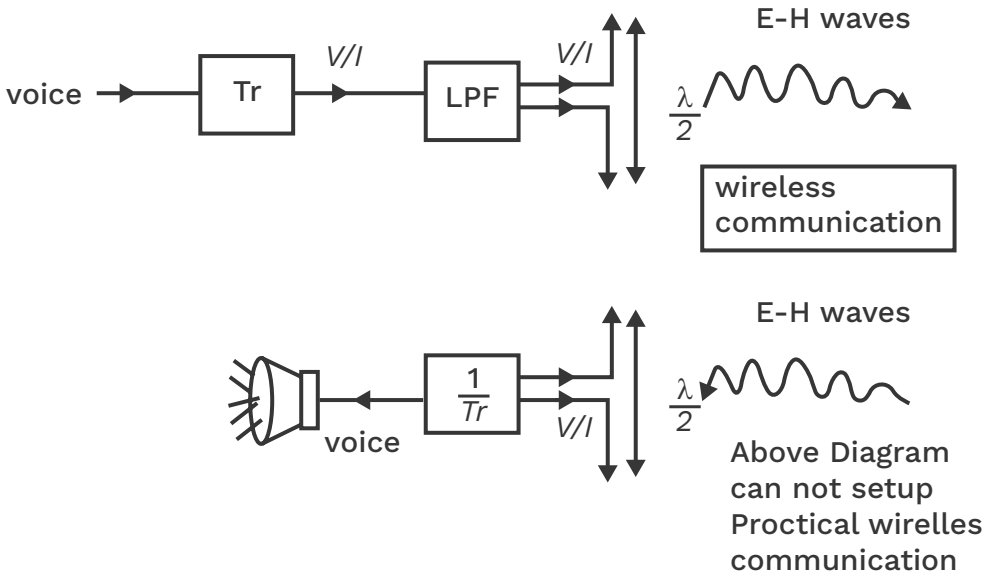
- **Half-duplex:** Two-way communication; one section at a time (sender or receiver).
- **Full-duplex:** Two-way communication; sender and receiver can send and receive data simultaneously.
- **Modulator:** Device which converts baseband signal into bandpass signal with the help of high frequency carrier.

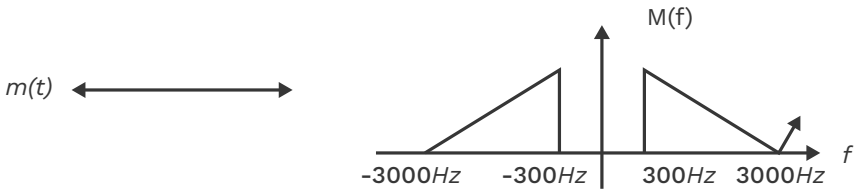


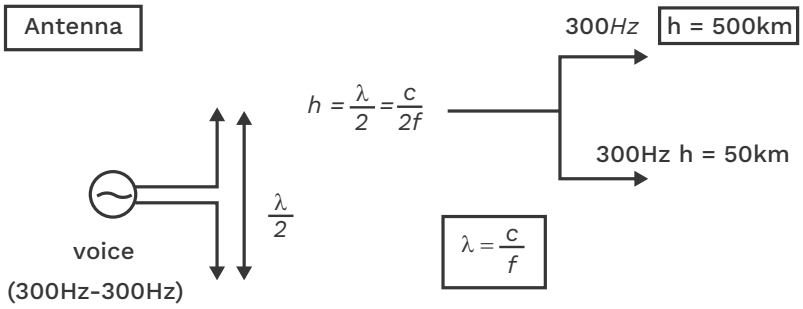

**Wired Electronic Communication**



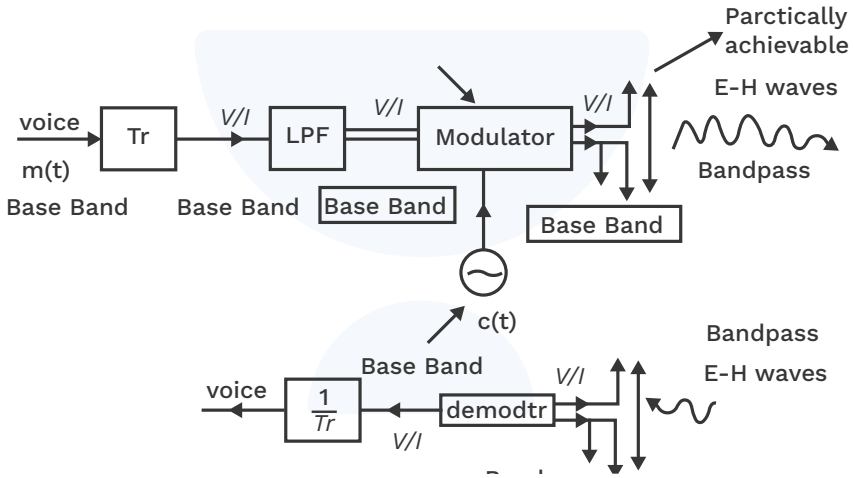

**Wireless communication from wired communication:**

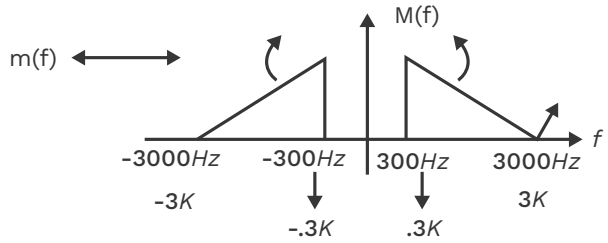





**Modified wireless communication system:**



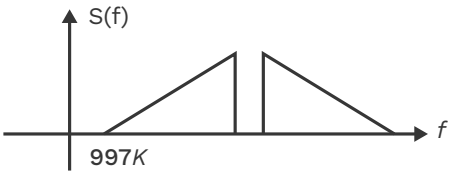




$S(f)=m(f)\cos(2\pi\times10^{6}t) \leftrightarrow$

$\downarrow$

1MHz

1000KHz




$h=\frac{\lambda}{2}=\frac{c}{2}=\frac{3\times10^{8}}{2\times997\times10^{3}}\approx150m$

**Note:** Electronic wireless communication is not possible without modulator:

**Wireless Communication System**

**Carrier signal:** In analog communication systems:

Carrier signal is high frequency sinusoidal signal which modifies itself in order to convert a baseband signal into bandpass signal.

$C(t) = A_c \cos(\omega_c t + \phi)$ or $A_c \sin(\omega_c t + \phi)$

$A_c$: $\rightarrow$ Amplitude of carrier

$\omega_c t + \phi$ PHASE of carrier

$\omega_c \rightarrow$ frequncy of carrier.

**Modulation process:** 'It is the process in which some characteristics of carrier signal is varied linearly w.r.t. baseband message signal in order to convert baseband message signal into bandpass signal.

**Effect:**

Baseband $\longrightarrow$ Bandpass

Low frequency $\longrightarrow$ High frequency

Wideband $\longrightarrow$ Narrowband

**Generalised Expression of Modulated Signal**

$\Rightarrow c(t) = A_c \cos(\omega_c t + \phi) \Rightarrow A_c \cos \omega_c t$

$\rightarrow$ Carrier Signal

$\rightarrow$ Carrier before modulation

Modulation signal:

$\rightarrow \boxed{S(t) = A(t)\cos(\omega_c t + \phi(t))}$

$\rightarrow$ Bandpass Signal.

$\rightarrow A(t) \rightarrow$ Instantaneous amplitude.

$\rightarrow \theta(t) = \omega_c t + \phi(t)$

$\rightarrow$ Instantaneous phase or angle

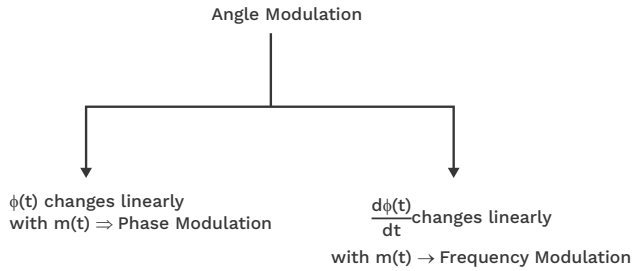
$\rightarrow \dfrac{d\theta(t)}{dt} = \omega(t) \rightarrow$ "Instantaneous frequency"

$\omega_c + \dfrac{d\phi(t)}{dt}$ Indirect

**Case 1:** A(t)varies linearly with m(t) and $\theta(t), \omega(t)$ does not change with m(t)

Amplitude modulation

**Case 2:** $\theta(t)$ Changes with $m(t)$ & $A(t)$ does not charge with $m(t)$




**Case 3:** A(t) and $\theta(t)$ changes with m(t)

$\boxed{\begin{array}{c}\text{Angle Amplitude Modulation} \\ \text{or} \\ \text{Ampplitude- Angle Modulation}\end{array}} \rightarrow$ A.A.M.

Hence, due to modulation process, baseband signals can be converted to bandpass signals which means signals are translated from lower frequencies to higher frequencies. Modulation methods can be classified into two types according to the number of frequency components present in the message signals.

- **Single-tone modulation:** Message signal is having single frequency component.
- **Multi-tone modulation:** Message signal is having multiple frequency components.

For signal tone modulation:

$$m(t) = A_m \cos 2\pi f_m t \quad \ldots(1)$$

Frequency spectrum:

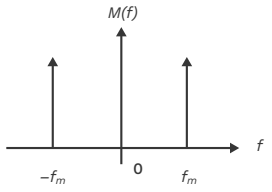



**Fig. 2.3 Frequency Spectrum of Single Tone Message Signal**

For dual tone modulation:

$$m(t) = A_{m_1} \cos 2\pi f_{m_1} t + A_{m_2} \cos 2\pi f_{m_2} t \quad \ldots(2)$$

Frequency spectrum:

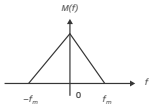


**Fig. 2.4 Frequency Spectrum of Multi–tone Message Signal**

## Amplitude Modulation

Amplitude modulation is a method by which the amplitude of the high frequency carrier signal is varied according to variation in amplitude of the message signal.

→ DSB-FC (Double side band-full carrier)

Or

Conventional AM.

→ DSB-SC (Double-sideband suppressed-carrier)

→ SSB-SC (Single-sideband suppressed-carrier)

→ VSB-SC (Vestigial-sideband suppressed-carrier)

**Double sideband full carrier (DSB-FC):**

Amplitude modulated signal

$c(t) = A_c \cos \omega_c t$ → Carrier before modulation

$$\boxed{s(t) = A(t)\cos\omega_c t}$$

$$A(t) = (A_c + m(t))$$

Expression of DSB-FC:

$$S_{AM}(t) = \overline{(A_c + m(t))}\cos\omega_c t$$

↓

carrier after modulation

$$\boxed{S_{AM}(t) = A_c \cos\omega_c t + m(t)\cos\omega_c t}$$

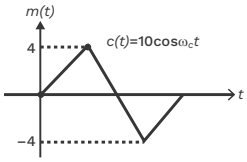
**Modulation Index**

1. It is defined as $\mu = \dfrac{|m(t)|_{max}}{Ac}$

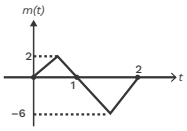
**Case 1:**

$m(t)$ is having zero offset (dc value = 0)

$c(t) = \underset{\downarrow}{10}\cos\omega_c t$ : $\mu = \dfrac{|m(t)|}{A_c}\max = \dfrac{4}{10} = 0.4$




**Case 2:** Proakis



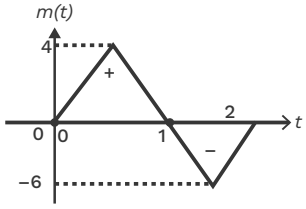

$$\boxed{\mu = \frac{|m(t)|_{max}}{A_c}}$$

$$\mu = \frac{6}{10} = 0.6$$

$$\mu = A_c$$

m(t) does not have zero offset → dc value ≠ 0

**Case 3:** $c(t) = 10\cos\omega_c t$




$$\mu = \frac{|m(t)|}{A_c}\max$$

$$\boxed{\mu = \frac{4}{10} = 0.4}$$

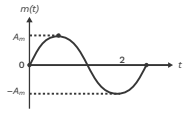
**Time domain analysis of DSB-FC:**

→ $c(t) = A_c \cos 2\pi f_c t$

$\rightarrow \underset{\text{DSB–FC}}{S(t)} = (A_c + m(t))\cos 2\pi f_c t$

$A(t) = A_c + m(t)$

**Case 1: Under modulation** $\Rightarrow \mu < 1$

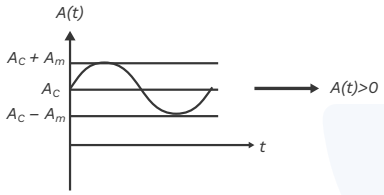


$\rightarrow \mu = \frac{A_m}{A_c}$

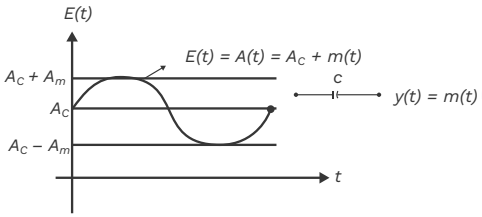
$\rightarrow \mu < 1 \Rightarrow \frac{A_m}{A_c} < 1$

$\rightarrow A_m < A_c \Rightarrow A_c - A_m > 0$

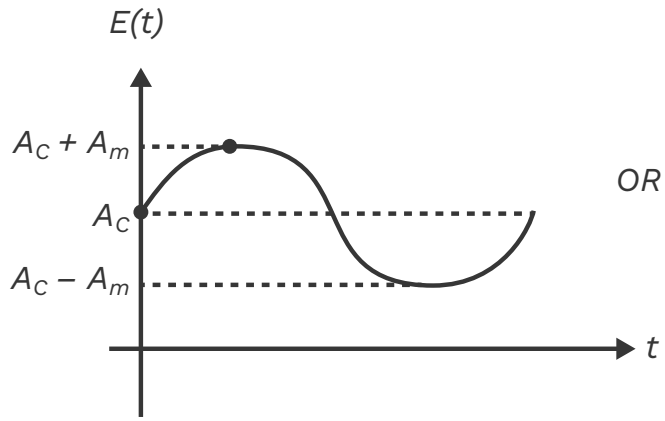
$$\boxed{s(t) = A(t)\cos 2\pi f_c t}$$

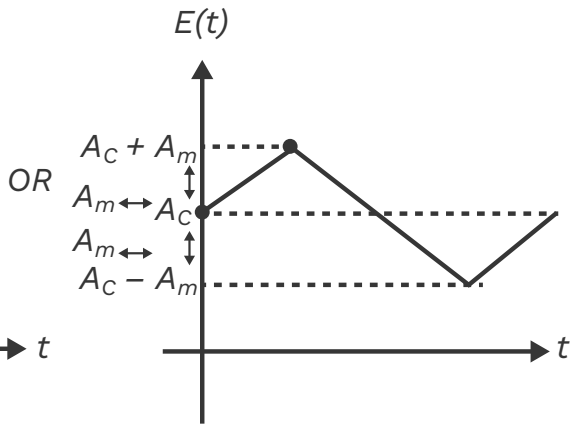



$A(t) = A_c + m(t)$




**General format of envelope of DSB-FC for** $\mu < 1$




$\left\{E(t)\right\}_{max} = A_c + A_m = A_c(1+\mu)$

$\left\{E(t)\right\}_{min} = (A_c - A_m) = A_c(1-\mu)$

$$\boxed{\mu = \frac{\left\{E(t)\right\}_{max} - \left\{E(t)\right\}_{min}}{\left\{E(t)\right\}_{max} + \left\{E(t)\right\}_{min}}}$$

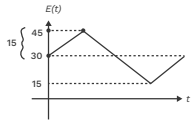
$\rightarrow$ Peak +ve value of s(+)

$\Rightarrow \left\{s(t)\right\}_{max} = \left\{E(t)\right\}_{max} = A_c(1+\mu)$

$\rightarrow$ Min +ve value of

$s(+) \Rightarrow \left\{s(t)\right\}_{min} = \left\{E(t)\right\}_{min} = A_c(1-\mu)$

**Example:** Envelope of DSB-FC is given:




1. $\mu =$
2. $A_m =$
3. $A_c =$

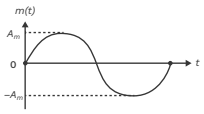
**Solution:**

(1) $\mu = \frac{45 - 15}{45 + 15} = 0.5$

(2) $A_m = 15 \quad \rightarrow \mu = \frac{15}{30} = 0.5$

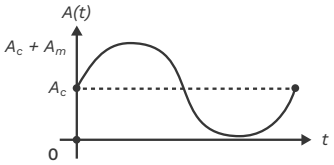
(3) $A_c = 30$

**Case 2: Critical modulation ($\mu = 1$)**

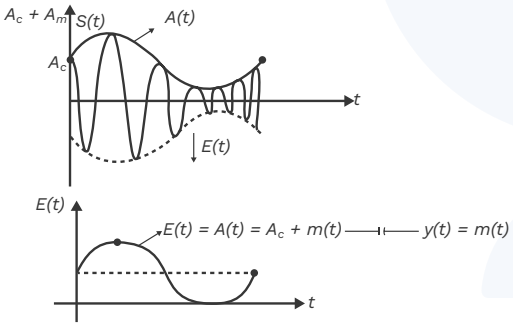



$$\mu = \frac{A_m}{A_c}$$

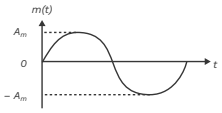
$$\boxed{\mu = 1 \Rightarrow A_c - A_m = 0}$$




$$\begin{cases} S(t) = A(t)\cos\omega_c t \\ A(t) = A_z + m(t) \end{cases}$$

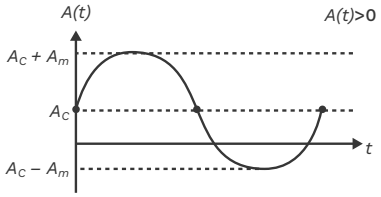



**Case 3: Over modulation ($\mu > 1$)**

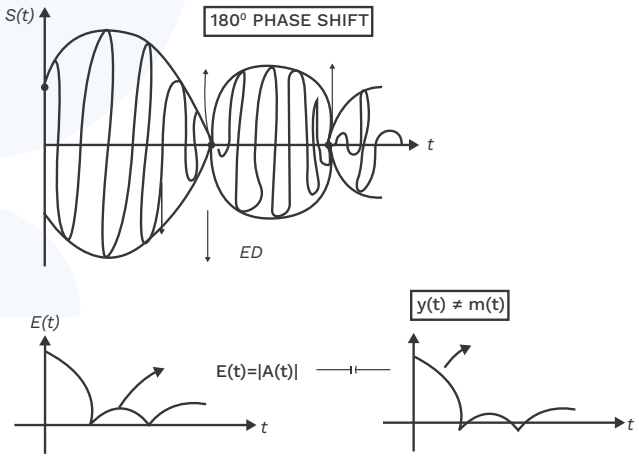



$$\mu = \frac{A_m}{A_c} > 1$$

$$\boxed{A_c - A_m < 0}$$




$$\boxed{\begin{aligned} S(t) &= A(t)\cos\omega_c t \\ A(t) &= A_c + m(t) \end{aligned}}$$




| $\mu$ | A(t) | E(t) | m(t): Recovery with ED |
|---|---|---|---|
| $\mu < 1$ | $A(t) > 0$ | $E(t) = A(t) \longrightarrow$ | Possible |
| $\mu = 1$ | $A(t) \geq 0$ | $E(t) = A(t) \longrightarrow$ | Possible |
| $\mu > 1$ | $A(t) \not< 0$ | $E(t) \lvert A(t) \rvert \longrightarrow$ | Not possible |

**Example:**



$$S(t) = \left(2 + 4\sin 4t\right)\cos \omega_c t$$

**Solution:**

**Method 1:**

$$y(t) = |A(t)| = \left|2 + \underbrace{4\sin 4t}_{\substack{-4 \le \ \le 4 \\ -2 \le \ \le 6}}\right| \longrightarrow A(t) \lessgtr 0$$

$$y(t) = |2 + 4\sin 4t|$$

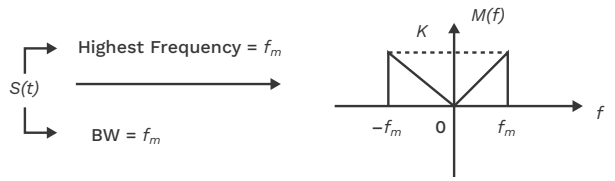
**Method 2:**

$$S(t) = \left(2 + 4\sin 4t\right)\cos \omega_c t$$

$$S(t) = \left(A_c + m(t)\right)\cos \omega_c t$$

$$\mu = \frac{|m(t)|_{max}}{A_c} = \frac{4}{2} = 2$$

$$\mu > 1 E(t) = |A(t)| = |2 + 4\sin 4t|$$




$S(t) \quad \longleftrightarrow$

$DSB - FC$

$\rightarrow BW$

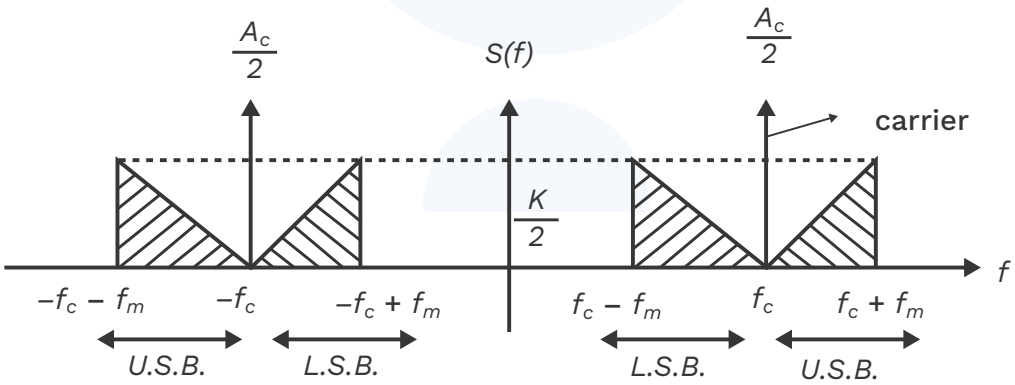
$$\left(f_c + f_m\right) - \left(f_c - f_m\right) = 2f_m$$

**Frequency Related Parameters**

(1) $m(t)$ → $BW = -f_m$, $f_{max} = f_m$

(2) $c(t) \longrightarrow f_{max} = f_c$

(3) $S(t)$ → $f_{max} = f_c + f_m$, $f_{min} = f_c - f_m$, $BW = 2f_m$ = {2×max frequency Of m(t)}




**Frequency domain analysis of DSB-FC:**

$$S(t) = \left[A_c + m(t)\right]\cos \omega_c t$$

$$S(t) = A_c \cos \omega_c t + m(t)\cos \omega_c t \qquad f_c >>> f_m$$

$\updownarrow F.T.$

$$S(t) = \frac{A_c}{2}\left[S\left(f - f_c\right) + \delta\left(f + f_c\right)\right] + \underbrace{\left\{\frac{M\left(f - f_c\right) + M\left(f + f_c\right)}{2}\right\}}$$

\# $P_{AM} = P_C + P_{SB}$

$$P_{SB} = \frac{P_m}{2} \qquad P_{USB} = P_{LSB} = \frac{P_m}{4}$$

→ Modulation efficiency:

$$\eta = \frac{P_{SB}}{P_{AM}} = \frac{P_m / 2}{P_C + \frac{P_m}{2}}$$

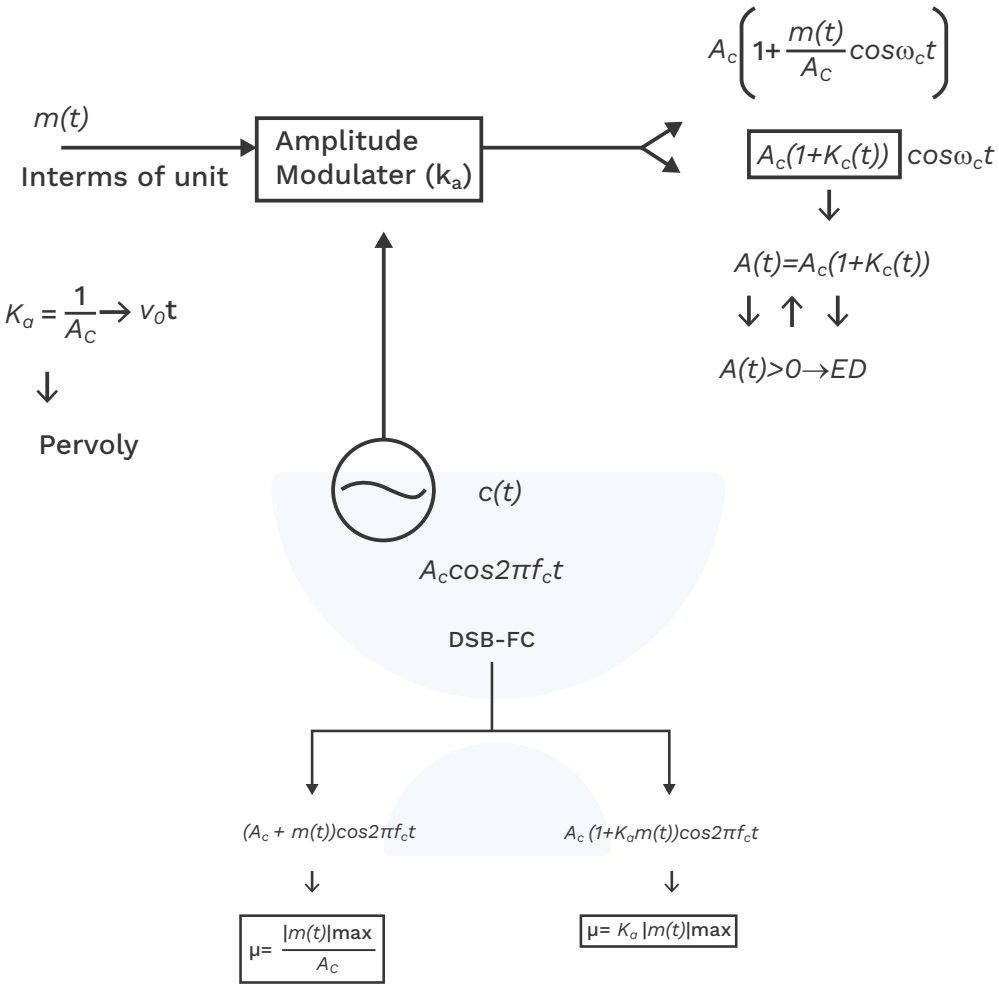
Share of SB power in total power

$\%\eta \Rightarrow$

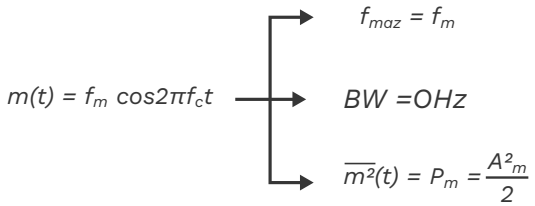
**Amplitude Sensitivity of Amplitude Modulator** $(K_a)$

$K_a$(Per volt) $\qquad A(t) > 0 \longrightarrow ED$

$$\left[A_c + m(t)\right]\cos\omega_c t$$




**Case 1:** When m(t) is single sinusoidal signal




\# $\quad c(t) = A_c \cos 2\pi f_c t$

$$\mu = \frac{A_m}{A_c}$$

**Time domain analysis:**

$$S_{AM}(t) = \left(A_c + m(t)\right)\cos\omega_c t$$

$$S_{AM}(t) = \left(A_c + A_m \cos 2\pi f_m t\right)\cos 2\pi f_c t$$

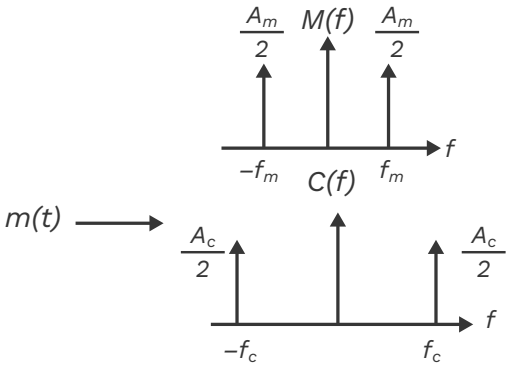
$$S_{Am}(t) = A_c\left(1+\frac{A_m}{A_c}\cos 2\pi f_m t\right)\cos 2\pi f_c t$$

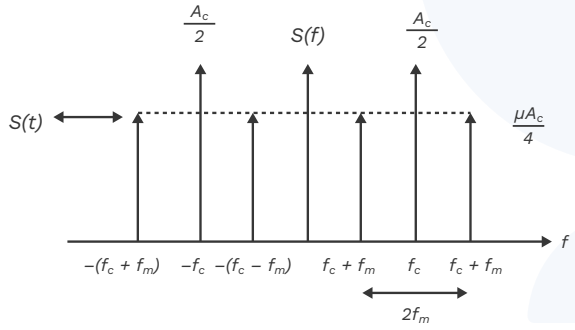
$$S_{Am}(t) = A_c\left(1+\mu\cos 2\pi f_m t\right)\cos 2\pi f_c t$$

$$S_{AM}(t) = A_c \cos 2\pi f_c t + \frac{\mu A_c}{2}\cos\left(2\pi(f_c + f_m)t\right) + \frac{\mu A_c}{2}\cos\left[2\pi(f_c - f_m)t\right] \longrightarrow DSB - FC$$

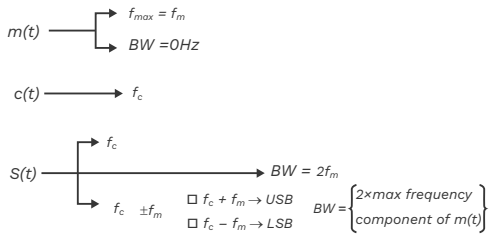
carrier → USB → LSB

## Frequency domain analysis:




## Frequency related parameters:




| Frequency | Strength |
|---|---|
| $f_c$ ⟶ | $A_c / 2$ |
| $f_c + f_m$ ⟶ | $A_c\mu / 4$ |
| $f_c - f_m$ ⟶ | $\frac{A_c\mu}{4}$ |

## Power related parameters:

$$m(t) = A_m \cos 2\pi f_m t \longrightarrow P_m = \frac{A_m^2}{2}$$

$$C(t) = A_c \cos 2\pi f_c t \longrightarrow P_c = \frac{A_c^2}{2}$$

$$S(t) \longrightarrow P_{AM} = P_c + \frac{P_m}{2}$$

$$P_{AM} = \frac{A_c^2}{2} + \frac{A_m^2}{4} = \frac{A_c^2}{2}\left\{1 + \frac{\mu^2}{2}\right\} = P_c + \overset{P_{SB}}{\frac{P_c\,\mu^2}{2}}$$

$$\eta = \frac{P_{SB}}{P_{Am}} = \frac{P_m / 2}{P_c + \frac{P_m}{2}} = \frac{P_c\mu^2 / 2}{P_C + \frac{P_C\mu^2}{2}}$$

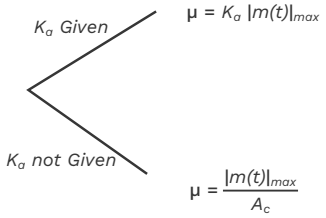
$$\%\eta = \frac{\mu^2}{2 + \mu^2} \times 100\%$$

## Use of Formula

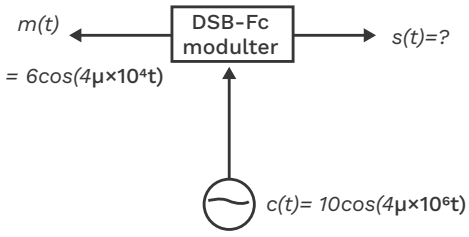
$m(t) \longrightarrow$ Single tone sinusoidal

$$P_{Am} = P_C \frac{\left(1 + \mu^2\right)}{2}$$

$$\eta = \frac{\mu^2}{2 + \mu^2}$$




## Example:




## Solution:

$$\mu = \frac{|m(t)|_{max}}{A_c} = 0.6 \rightarrow \text{undermodulation}$$

1. $c(t) = A_c \cos 2\pi f_c t$
2. $m(t) = A_m \cos 2\pi f_m t$
3. $S_{AM}(t) = [A_c + m(t)]\cos 2\pi f_c t$
   $S_{AM}(t) = A_c(1 + \mu\cos 2\pi f_m t)\cos 2\pi f_c t$

4. Frequency $m(t) \longrightarrow f_{max} = 20KHz$

$S(t) \longrightarrow f_c \rightarrow 2000KHz$

$f_c \pm f_m \longrightarrow 2000 \pm 20k$

$\rightarrow 1980\,KHz$

$\rightarrow 2020KHz$

$(BW)_{AM} = 2 \times 20KHz = 40KHz$

5. Power $P_{AM} = P_c\left(1+\frac{\mu^2}{2}\right)$

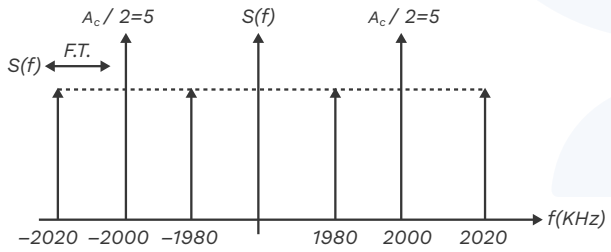
$$P_{AM} = \frac{A_c^2}{2}\left(1+\frac{\mu^2}{2}\right) = \frac{(100)}{2}\left(1+\frac{(0.6)^2}{2}\right)$$

$P_{AM} = 59\,watt \qquad Pc = \frac{A_c^2}{2} = 50\,watt$

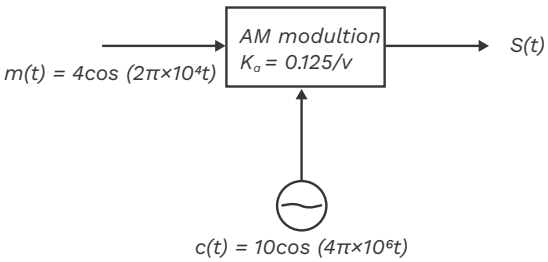
$P_{SB} = P_{AM} - P_C = 9\,watt$

$P_{USB} = P_{LSB} = 4.5watt$

$$\eta = \frac{\mu^2}{2+\mu^2} \times 100\% = \frac{9}{59} \times 100\% = 15.25\%$$




**Example:**




**Solution:**

$\mu = Ka\left|m(t)\right|_{max} = 0.125 \times 4 = 0.5 \rightarrow$ Under modulated

1. $c(t) = A_C \cos(2\pi f_c t)$ $\nearrow A_c = 10V$ $\searrow f_c = 2000K$

2. $m(t) = A_m \cos(2\pi fmt)$ $\nearrow A_m = 4$ $\searrow f_m = 10K$ $\qquad \mu = 0.5$

3. $\boxed{S_{AM}(t) = A_C(1+\mu\cos 2\pi f_m t)\cos(2\pi f_c t)}$

4. Frequency $m(t) \rightarrow f_{max} = 10KHz = f_m$

$s(t) \rightarrow f_c = 200K$

$f_c \pm f_m = 2000 \pm 10$ $\nearrow 2010K$ $\searrow 1990K$

$\boxed{BW = 2 \times f_{max} = 20KHz}$

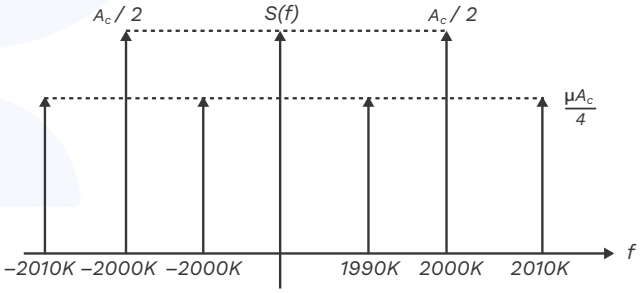
5. Power $Pc = \frac{A_c^2}{2} = 50\,watt$

$P_{AM} = Pc\left(1+\frac{\mu^2}{2}\right) = 50\left(1+\frac{\mu^2}{2}\right) = 56.25W$

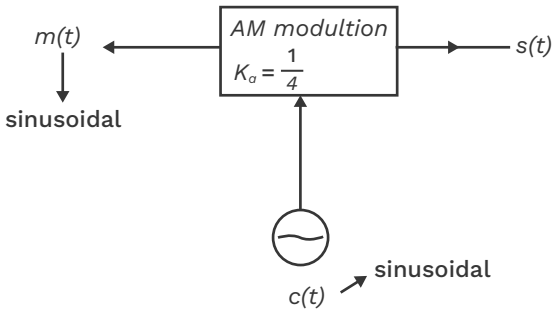
$P_{SB} = 6.25W$

$P_{USB} = P_{LSB} = 3.125W$

$\boxed{\eta = \frac{\mu^2}{2+\mu^2} = \frac{P_{SB}}{P_{AM}}} \longrightarrow \boxed{\%\eta = 11.11\%}$




**Example:**




$s(t) = 20\cos(2000\,\pi t) +$

$5\cos(1000\,\pi t) + 5\cos(2200\,\pi t)$

$f_c - f_m = 900Hz \qquad f_c + f_m = 1100Hz$

$\mu = ?$
$c(t) = ?$
$m(t) = ?$

**Solution:**

$$S(t) = A_c \cos 2\pi f_c + \frac{Ac\mu}{2}\cos(2\pi(f_c + f_m)t +$$

$$\frac{A_c\mu}{2}\cos(2\pi(f_c - f_m)t)$$

$A_c$ = 20v $\qquad f_m = 100Hz$

$f_c$ = 1000Hz

$$\frac{\mu Ac}{2} = 5 \Rightarrow \boxed{\mu = \frac{10}{A_c} = 0.5}$$

$$\mu = K_a |m(t)|_{max} \Rightarrow \frac{1}{2} = \frac{1}{4}|m(t)|_{max}$$

$$m(t) = A_m \cos(2\pi f_m t)$$

$|m(t)|_{max} = A_m$ $\qquad$ (i) $\mu = 0.5$

$\mu = Ka\,|m(t)|_{max}$ $\qquad$ (ii) $c(t) = A_c \cos 2\pi f_c t$

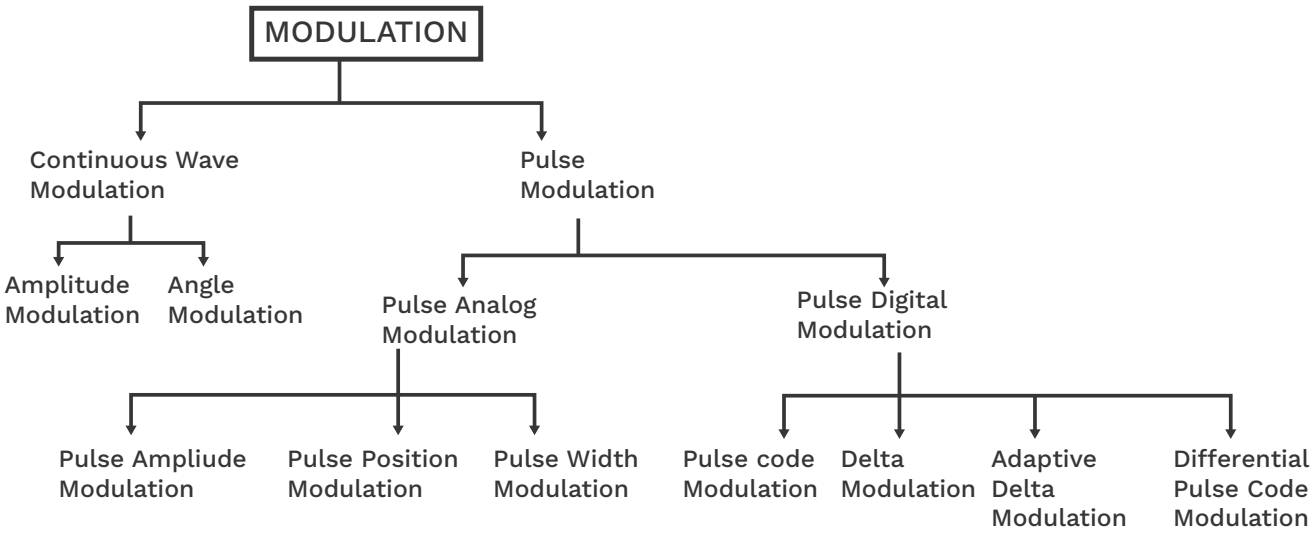
$A_m = 2$ $\qquad$ (iii) $m(t) = A_m \cos 2\pi f_m t$

## Necessity of Modulation

1. **To reduce antenna height:**
   a) As the transmission of signal has to be done for long distance and for effective transmission, the practical height of the antenna should be at least $\lambda / 4$.
   b) Therefore, in the process of modulation, the frequency of the modulation, the frequency of the modulated signal is increased leading to less antenna height requirements.

   Wavelength $(\lambda) = \frac{\text{speed of light (c)}}{\text{frequency}}$

   Height of antenna $= \frac{\lambda}{4} = \frac{c}{4f}$
2. **Multiplexing:**
   a) The process of transmitting a number of signals through a common channel is called multiplexing.
   b) For multiplexing, especially in wired communication, modulation is a requirement.
3. To reduce noise and interference.
4. To increase range or distance of communication.
5. For improving signal to noise ratio (S/N ratio)



**Example:**

Calculate the antenna height for the baseband signal of 100 kHz.

**Solution:** As, height of antenna $= \frac{\lambda}{4} = \frac{c}{4f}$

$H = 3 \times 10^8 / \left(4 \times 100 \times 10^3\right) = 0.75 \times 10^3 = 750\,\text{m}.$

## SUMMARY

$m(t) = A_m \cos 2\pi f_m t$

1. $S_{AM}(t) = A_C\left(1 + \mu \cos 2\pi f_m t\right)\cos 2\pi f_c t$

2. $S_{AM}(t) = A_C \cos 2\pi f_c t + \frac{\mu A_c}{2}\cos\left(2\pi\left(f_c + f_m\right)t\right) + \frac{\mu A_c}{2}\cos\left[2\pi\left(f_c - f_m\right)t\right]$

$P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right) \qquad \eta = \frac{\mu^2}{2+\mu^2}$

## Important Points

1. $P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right) \qquad \eta = \frac{\mu^2}{2+\mu^2}$

| $\mu = 0$ | $\mu = 1$ | % charge |
|---|---|---|
| $P_{AM} = P_c$ | $P_{AM} = 1.5P_c$ | 50% |
| $\eta = 0$ | $\eta = \frac{1}{3} = 33.33\%$ | 0% *to* 33.33% |

2. as $\mu \uparrow \rightarrow \eta \uparrow$

$P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right)$

If μ = Constant:

$P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right) \Rightarrow$ as $P_c \uparrow \Rightarrow P_{AM} \uparrow$

If $P_c$ = constant:

$P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right) \Rightarrow$ as $\mu \uparrow \Rightarrow P_{AM} \uparrow$

3. $\left(P_{AM}\right)_{max} = \left(P_c\right)_{max}\left(1 + \frac{\mu^2}{2}\right)$

Or

$\left(P_{AM}\right)_{max} = P_c\left(1 + \frac{\mu_{max}^2}{2}\right)$

Or

$\left(P_{AM}\right)_{max} = \left(P_c\right)_{max}\left(1 + \frac{\mu_{max}^2}{2}\right)$

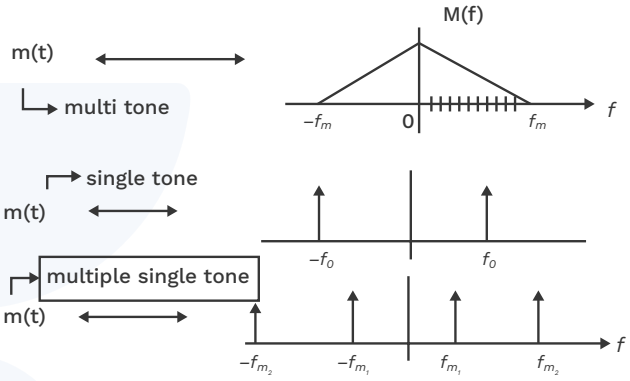
4. $P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right)$

$P_{AM} \rightarrow$ constant if $P_c \uparrow$ *and* $\mu \downarrow$

Or

$P_c \downarrow$ *and* $\mu \uparrow$

**Case 2:** Let m(t) is multiple single tone signal




$m(t) = A_{m_1}\cos 2\pi f_{m_1} + A_{m_2}\cos 2\pi f_{m_2} t$

$f_{m_1}, f_{m_2}$

Let $f_{m_1} < f_{m_2}$

$A_{m_1} > A_{m_2}$

$S_{AM}(t) = \left[A_c + m(t)\right]\cos 2\pi f_c t$

**Time domain analysis:**

$S(t) = \left[A_c + m(t)\right]\cos 2\pi f_c t$

$S(t) = \left[A_c + A_{m_1}\cos 2\pi f_{m_1} t + A_{m_2}\cos 2\pi f_{m_2} t\right].\cos 2\pi f_c t$

$y(t) = \left(m(t) + c(t)\right)\left[\frac{1}{2} + \frac{2}{\pi}\cos\omega_c t - \frac{2}{3\pi}\cos 3\omega_c t + \frac{2\cos 5\omega_c t}{5\pi} + \ldots\right].$

$$\mu_1 = \frac{A_{m_1}}{A_c} \qquad \mu_2 = \frac{A_{m_2}}{A_c} \Rightarrow \mu_1 > \mu_2$$

$$S(t) = A_c\left(1+\mu_1\cos 2\pi f_{m_1}t + \mu_2\cos 2\pi f_{m_2}t\right)\cos 2\pi f_c t.$$

$$S(t) = A_c\cos 2\pi f_c t + \left\{\begin{matrix}\frac{A_c\mu_1}{2}\cos\left[2\pi\left(f_c+f_{m_1}\right)t\right]+\\ \frac{A_c\mu_1}{2}\cos\left[2\pi\left(f_c-f_{m_1}\right)t\right]\end{matrix}\right\}$$

$$+\left\{\begin{matrix}\frac{A_c\mu_2}{2}\cos\left[2\pi\left(f_c+f_{m_2}\right)t\right]+\\ \frac{A_c\mu_2}{2}\cos\left[2\pi\left(f_c-f_{m_2}\right)t\right]\end{matrix}\right\}$$

$$c(t) \to \overline{c^2(t)} = P_c = \frac{A_c^2}{2}$$

$$P_{AM} = P_c + \frac{A_m}{2} = \frac{A_c^2}{2} + \frac{A_{m_1}^2}{4} + \frac{A_{m_2}^2}{4}$$

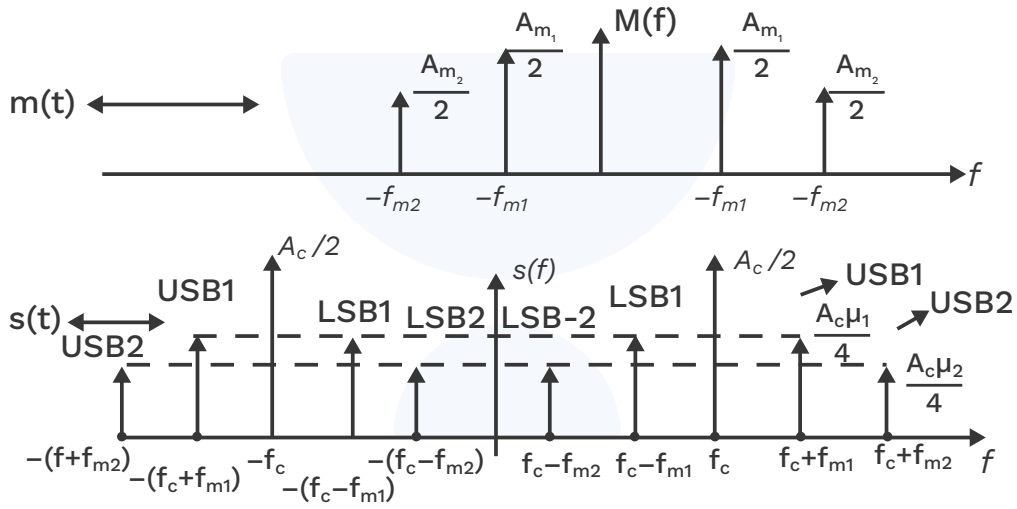
$$= \frac{A_c^2}{2}\left\{1+\frac{\mu_1^2}{2}+\frac{\mu_2^2}{2}\right\} = P_c\left(1+\frac{\mu_T^2}{2}\right)$$

$$\mu_T^2 = \mu_1^2 + \mu_2^2$$

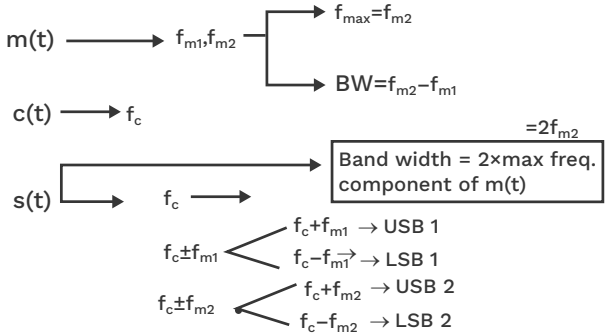
$$P_{SB} = \frac{P_c\mu_T^2}{2}$$

$$P_{USB_1} + P_{LSB_1} = \frac{P_c\,\mu_1^1}{2} \Rightarrow P_{USB_1} = P_{LSB_1} = \frac{P_c\,\mu_1^2}{4}$$

**Frequency domain analysis:**




**Frequency related calculation:**




**Power related parameter:**

$$m(t) \to \overline{m^2(t)} = P_m = \frac{A_{m_1}^2}{2} + \frac{A_{m_2}^2}{2}$$

$$P_{USB_2} + P_{LSB_2} = \frac{P_c\,\mu_2^2}{2} \Rightarrow P_{USB_2} = P_{LSB_2} = \frac{P_c\,\mu_2^2}{4}$$

**Modulation efficiency:**

$$\eta = \frac{P_{SB}}{P_{AM}} = \frac{\mu_T^2}{2+\mu_T^2}$$

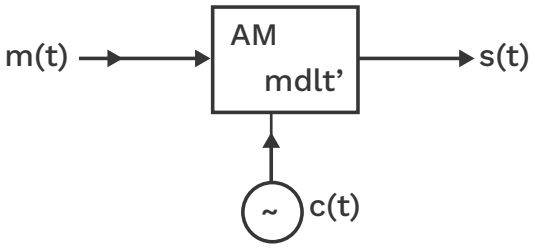
**Note:** Total modulation index:

$$\mu_T = \sqrt{\mu_1^2+\mu_2^2+\mu_3^2+\ldots}$$

**Total mobility:**

$$\frac{1}{\mu_T} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{1}{\mu_2} + \ldots$$

**Example:**




$$S(t)=\begin{bmatrix}10+8\cos\left(2\pi\times10^{3}t\right)+\\6\cos\left[2\pi\times10^{4}t\right]\end{bmatrix}\cos\left[2\pi\times10^{6}t\right]$$

**Solution:**

1. m(t): Multiple single tone.
2. $S(t)=A_c\left(1+\mu_1\cos 2\pi f_{m_1}t+\mu_2\cos 2\pi f_{m_2}t\right)\cos 2\pi f_c t$

$$S(t)=10\left(\begin{matrix}1+0.8\cos\left(2\pi\times10^{3}t\right)+\\0.6\cos\left(2\pi\times10^{4}t\right)\end{matrix}\right)\cos 2\pi\times10^{6}t$$

$A_c=10V$

$\mu_1=0.8 \qquad \mu_2=0.6$

$f_{m_1}=1kHz \qquad f_{m_2}=10\,kHz$

$f_c=1000\,kHz$

**Frequency:**

$m(t)\rightarrow 10\,kHz,\ 1kHz$

$c(t)\rightarrow 1000\,kHz$

$S(t)\rightarrow 1000\,k$

$$1000\,k\pm 1k\begin{cases}999\,k\\1001k\end{cases}$$

$$1000\,k\pm 10\,k\begin{cases}1010\,k\\990\,k\end{cases}$$

$BW\ of\ S(t)=20\,kHz$

**Power related:**

$$P_{AM}=P_c\left(1+\frac{\mu_T^2}{2}\right) \qquad \mu_T^2=\mu_1^2+\mu_2^2$$

$\mu_T=1$

$$P_{AM}=\frac{A_c^2}{2}\left(1+\frac{1}{2}\right)=75watt$$

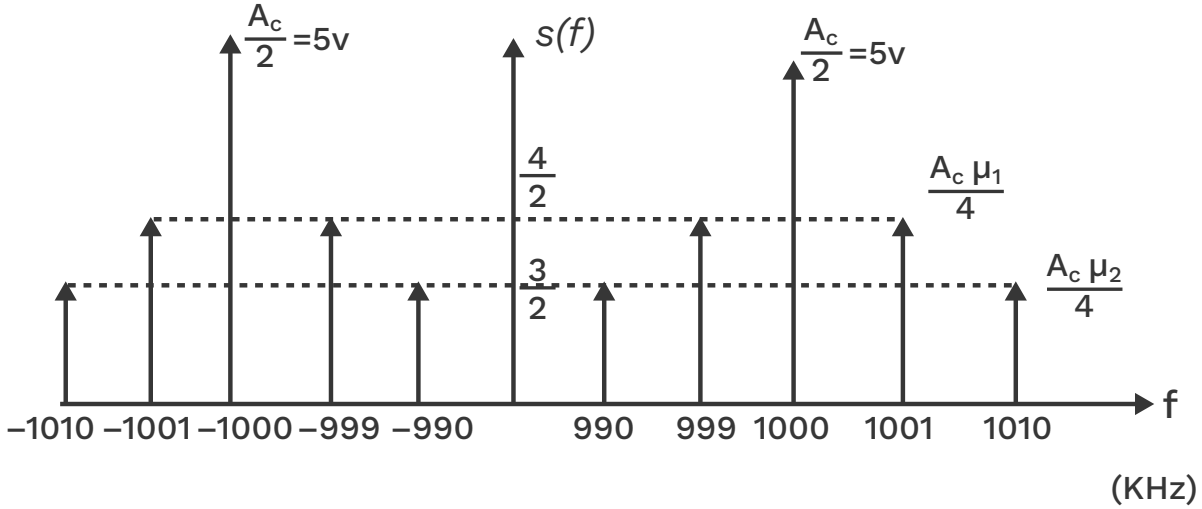
$P_c=50watt \quad P_{SB}=25watt=P_{SB_1}+P_{SB_2}$

$$P_{SB_1}=\frac{P_c\mu_1^2}{2}=16\begin{cases}P_{USB_1}=8watt\\P_{LSB_1}=8watt\end{cases}$$

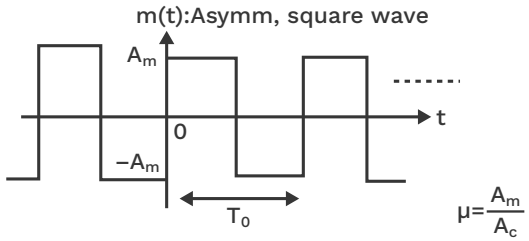
$$P_{SB_2}=\frac{P_c\mu_2^2}{2}=9\begin{cases}P_{USB_1}=4.5watt\\P_{LSB_2}=4.5watt\end{cases}$$

$$\%\eta=\frac{\mu_T^2}{2+\mu_T^2}=\frac{1}{2+1}\times 100\%$$

$\%\eta=33.33\%$



**Case 3:**




$$c(t) = A_c \cos 2\pi f_c t$$

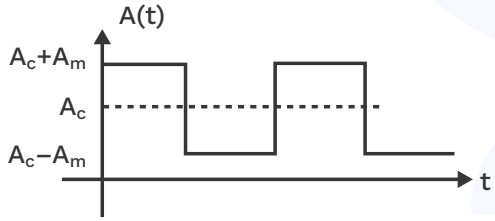
$$s(t) = \left(A_c + m(t)\right)\cos 2\pi f_c t$$

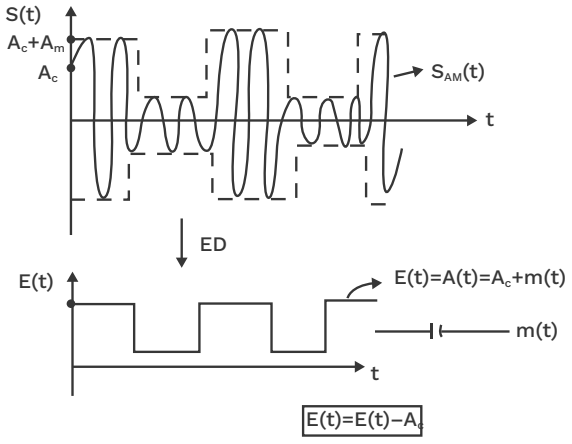
**Time domain analysis:**

$$(\mu < 1) \qquad \rightarrow \frac{A_m}{A_c} < 1$$

$$s(t) = \left[A_c + m(t)\right]\cos\left(2\pi f_c t\right) \quad \rightarrow A_c - A_m > 0$$

$$A(t) = A_c + m(t)$$




$$m(t) = E(t) - A_c$$

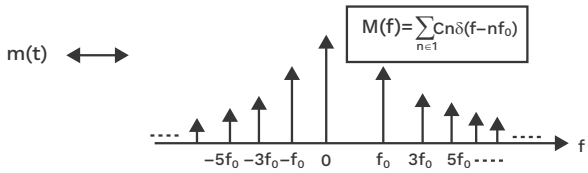
**Frequency related calculation:**

1. m(t): non-sinusoidal periodic
   Calculate $|C_n|$
   Determine absent harmonic by $|C_n| = 0$
   m(t): Asymmetric-square wave: $f_0, 3f_0, 5f_0$ ...

**Fourier transform of periodic signals:**

$$m(t) \leftrightarrow M(\omega) = 2\pi \sum C_n \delta(\omega - n\omega_0)$$

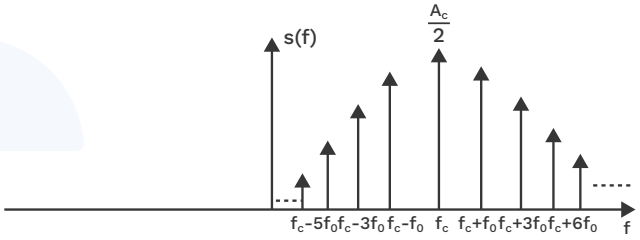
$$T_0 = 2\pi / \omega_0 \leftrightarrow M(f) = \sum C_n \delta(f - nf_0)$$




$$f_{max} \rightarrow \infty$$

Let m(t) is bandlimited to 5th harmonic ...

$$f_{max} = 5f_0$$




$$m(t) \rightarrow f_0, 3f_0, 5f_0, 7f_0 \text{ ----}$$

$$c(t) \rightarrow f_c$$

$$s(t) \rightarrow f_c \qquad \Rightarrow (BW)_{AM} = 2(5f_0) = 10f_0$$

$$|f_c \pm f_0|$$

$$|f_c \pm 3f_0|$$

$$|f_c \pm 5f_0|$$

**Power related calculation:**

$$m(t) \rightarrow \overline{m^2(t)} = P_m = {A_m}^2$$

$$c(t) \rightarrow P_c = \frac{A_c^2}{2}$$

$$s(t) \rightarrow P_{AM} = P_c + \frac{P_m}{2} = \frac{A_c^2}{2} + \frac{A_m^2}{2}$$

$$P_{AM} = \frac{A_c^2}{2}\left(1 + \frac{A_m^2}{A_c^2}\right)$$

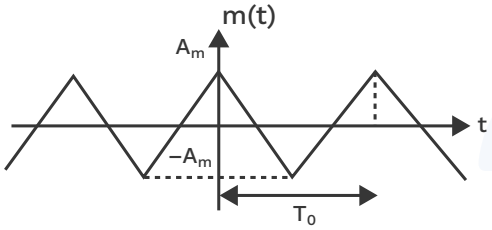
$$P_{AM} = P_c\left(1 + \mu^2\right)$$

$$P_{SB} = P_c\mu^2$$

$$\eta = \frac{P_{SB}}{P_{AM}} = \frac{\mu^2}{1+\mu^2}$$

| | $\mu = 0$ | $\mu = 1$ |
|---|---|---|
| $P_{AM}$ | $P_c$ | $2P_c \rightarrow P_{AM} \uparrow$ *by* 100% |
| $\eta$ | 0 | $\frac{1}{2}$ *or* 50% $\rightarrow \uparrow$ *From* 0% *to* 50% |

**Case 4: m(t): Triangular pulse:**

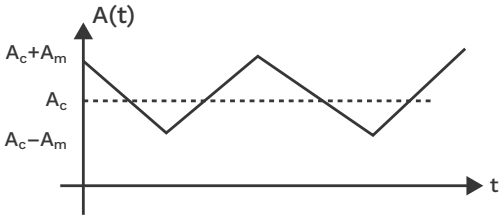


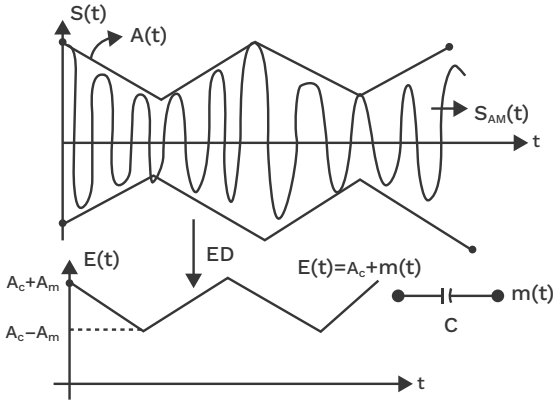

$$c(t) = A_c \cos 2\pi f_c t$$

**Time domain analysis for μ < 1:**

$$\mu < 1 \rightarrow A_c - A_m > 0$$

$$s(t) = \underbrace{\left(A_c + m(t)\right)}_{A(t)} \cos \omega_c t$$




**Frequency related parameters:**

$$m(t) \rightarrow f_0,\ 3f_0,\ 5f_0 \text{ – – – – –}$$

$$C(t) = f_c$$

$$s(t) \rightarrow f_c \qquad \Rightarrow (BW) = 2 \times f_{max}$$

$$\left|f_c \pm f_0\right|$$

$$\left|f_c \pm 3f_0\right|$$

$$\left|f_c \pm 5f_0\right|$$

**Power related parameters:**

$$m(t) \rightarrow \overline{m^2(t)} = P_m = \frac{A_m^2}{3}$$

$$C(t) = \frac{A_c^2}{2}$$

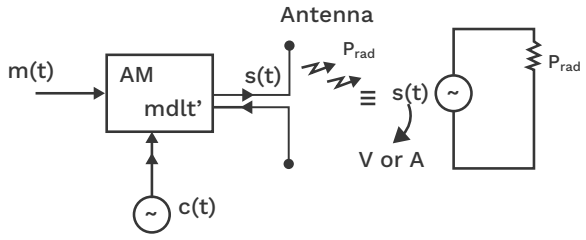
$$s(t) = P_c + \frac{P_m}{2} = P_c\left(1 + \frac{\mu^2}{3}\right) \qquad \mu = A_m \,/\, A_c$$

$$P_{SB} = \frac{P_c\mu^2}{3}$$

$$\eta = \frac{\mu^2}{3+\mu^2}$$

| | $\mu = 0$ | $\mu = 1$ |
|---|---|---|
| $P_{AM}$ | $P_c$ | $\frac{4}{3}P_c$ |
| $\eta$ | 0% | 25% |

**Important discussion 1:**




Power $\delta(t) = \overline{S^2(t)}$

$$P_{AM} = P_{rad} = \frac{\overline{S^2(t)}}{R} \text{ or } \overline{s^2(t)} \times R$$

**Important points to avoid common mistake:**

| m(t) [volt] | $P_{AM}$ | $P_{rad}$ |
|---|---|---|
| **Sinusoidal** | $P_c\left(1+\frac{\mu^2}{2}\right)$ | $\frac{P_c}{R}\left(1+\frac{\mu^2}{2}\right)$ |
| **Square wave** | $P_c\left(1+\mu^2\right)$ | $\frac{P_c}{R}\left(1+\mu^2\right)$ |
| **Triangular wave** | $P_c\left(1+\frac{\mu^2}{3}\right)$ | $\frac{P_c}{R}\left(1+\frac{\mu^2}{3}\right)$ |

**Note:** Do not directly use any formula blindly, first check the message signal then choose accordingly.

**Important discussion 2:**

$$P_{AM} = p_c\left(1+\frac{\mu^2}{2}\right)$$

$$\overline{S_{AM}^2(t)} = \frac{A_c^2}{2}\left(1+\frac{\mu^2}{2}\right)$$

$$S_{AM}(t) = V_{AM}(t) \quad V$$

$$\overline{V_{AM}^2(t)} = \frac{A_c^2}{2}\left(1+\frac{\mu^2}{2}\right)$$

$$\sqrt{V_{AM}^2(t)} = \frac{A_c}{\sqrt{2}}\sqrt{1+\frac{\mu^2}{2}}$$

$\downarrow$ $(V_c)_{rms}$

$\downarrow$ $(V_{AM})_{rms}$

$$\overline{S_{AM}^2(t)} = \frac{A_c^2}{2}\left(1+\frac{\mu^2}{2}\right)$$

$$S_{AM}(t) = I_{AM}(t)\,A$$

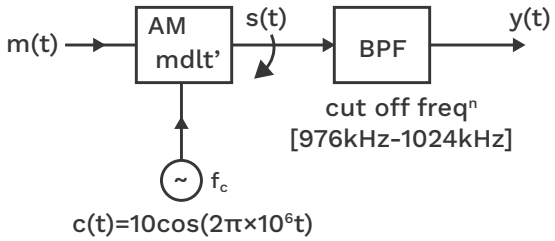
$$\overline{I_{AM}^2(t)} = \frac{A_c^2}{2}\left(1+\frac{\mu^2}{2}\right)$$

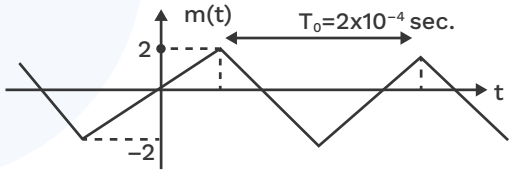
$$\sqrt{I_{AM}^2(t)} = \frac{A_c}{\sqrt{2}}\sqrt{1+\frac{\mu^2}{2}}$$

$\downarrow$ $(I_{AM}(t))_{rms} = I_c\sqrt{1+\frac{\mu^2}{2}}$

| If $S_{AM}(t) \to Volt$ | If $S_{AM}(t) \to A_{mp}$; |
|---|---|
| $V_{AM} = V_c\sqrt{1+\frac{\mu^2}{2}}$ | $I_{AM} = I_C\sqrt{1+\frac{\mu^2}{2}}$ |
| $V_{AM} = V_c\sqrt{1+\mu^2}$ | $I_{AM} = I_C\sqrt{1+\mu^2}$ |
| $V_{AM} = V_c\sqrt{1+\frac{\mu^2}{3}}$ | $I_{AM} = I_C\sqrt{1+\frac{\mu^2}{3}}$ |
| $V_{AM} = rms\,value\ of\ S_{AM}(t)$ | $I_{AM} = rms\ value\ of\ S_{AM}(t)$ |
| $V_C \to rms\ value\ of\ c(t)$ | $I_C \to rms\ value\ of\ c(t)$ |

**Example:**




1. Power of $S(t)$
2. Frequency components in $S(t)$
3. Frequency components in y(t)
4. Calculate the bandwidth of m(t) if it contains maximum freqⁿ component up to 3rd harmonic.

**Solution:**

$$m(t) \to f_0, 3f_0$$

5k, 15k

B.W of m(t) = 15-5 = 10k

$$P_{s(t)} = P_c\left(1+\frac{\mu^2}{3}\right) \qquad \mu = \frac{|m(t)|_{max}}{A_c} = \frac{2}{20} = 0.2$$

$$P_{s(t)} = 50.67\,watt \qquad P_c = \frac{A_c^2}{2} = 50$$

$$m(t) \to 5k, 15k, 25k, 35k - - - - -$$

$$c(t) \to f_c = 1000\,k$$

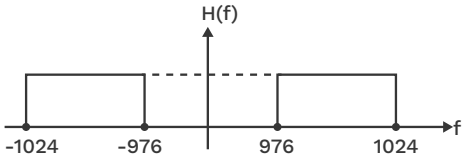
$s(t) = 1000k$

$1000k \pm 5k = 995k, 1005k$

$1000k \pm 15k = 985k, 1015k$

$1000k \pm 25k = 975k, 1025k$

$1000k \pm 35k = 965k, 1035k$

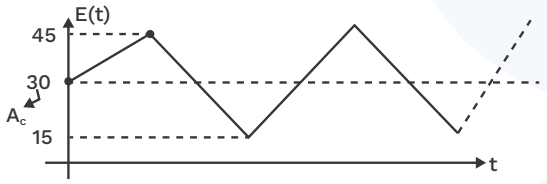



Freqⁿ in y(t) $\rightarrow$ 1000 k

995 k, 1005 k

985 k, 1015 k

**Example:** The envelope of AM signal is shown below:




1. Calculate m(t)
2. Calculate $\eta$

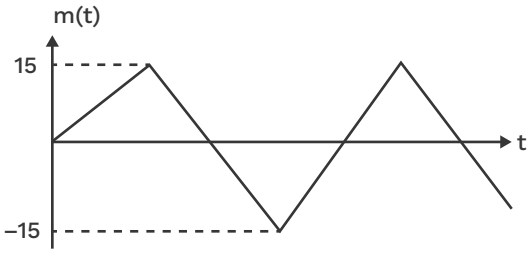
   E(t) =?

**Solution:**

$\mu = 0.5 \rightarrow \mu < 1: E(t) = A(t) = A_c + m(t)$

$A_c = 30V$

$m(t) = E(t) - 30V$




1. $\eta = \dfrac{\mu^2}{3+\mu^2} = 7.69\%$

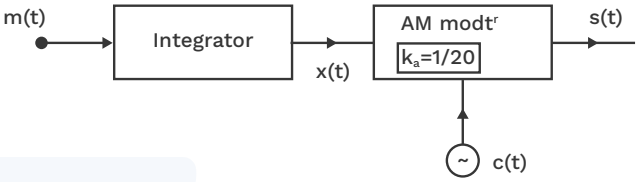
   $\mu = \dfrac{1}{2}$

2. $\eta = \dfrac{P_{SB}}{P_{AM}} \rightarrow 7.69\%$

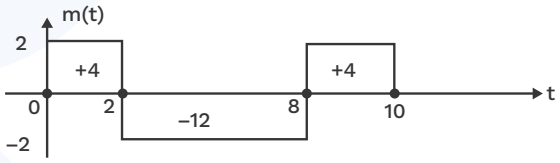
   $P_{SB} = \dfrac{P_m}{2} = \dfrac{A_m^{\ 2}}{6} = \dfrac{225}{6}$

   $P_{AM} = P_c + \dfrac{P_m}{2} = \dfrac{(30)^2}{2} + \dfrac{225}{12}$

**Example:**




Calculate modulation index of s(t).



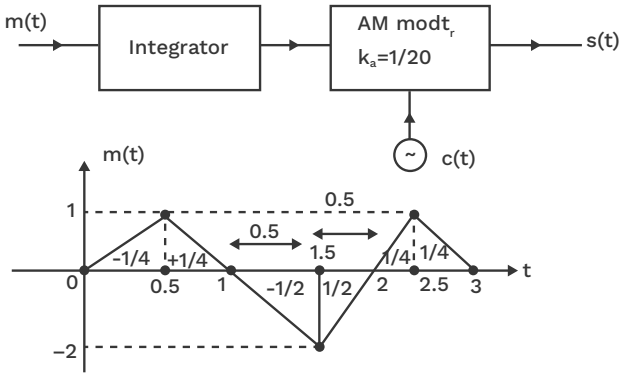

**Solution:**

$\mu = K_a \left| x(t) \right|_{max} = \dfrac{8}{20} = 0.4$

$\left| Area \right|_{max} = 8$

| Up to | Area | $\lvert Area \rvert$ |
|---|---|---|
| 0 | 0 | 0 |
| 2 | 4 | 4 |
| 8 | -8 | 8 |
| 10 | -4 | 4 |

**Example:** $\mu$ of $s(t) = ?$

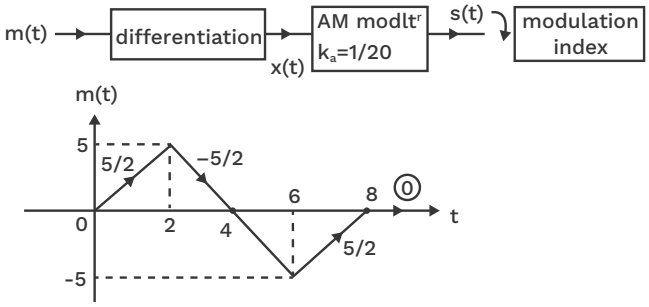



**Solution:**

| Up to | Area | $\lvert Area \rvert$ |
|---|---|---|
| 0 | 0 | 0 |
| 0.5 | $\frac{1}{4}$ | $\frac{1}{4}$ |
| 1 | $\frac{1}{2}$ | $\frac{1}{2}$ |
| 1.5 | 0 | 0 |
| 2 | $\frac{-1}{2}$ | $\frac{1}{2}$ |
| 2.5 | $\frac{-1}{4}$ | $\frac{1}{4}$ |
| 3 | 0 | 0 |

$$|Area|_{max} = \frac{1}{2} \quad \mu = k_a |m(t)|_{max} = \frac{1}{4a}$$

**Example:**




**Solution:**

$$\mu = k_a |x(t)|_{max} \qquad x(t) = \frac{dm(t)}{dt}$$

$$\mu = k_a \left|\frac{dm(t)}{dt}\right|_{max} \qquad \left|\frac{dm(t)}{dt}\right|_{max} = \frac{5}{2}$$

$$\mu = \frac{1}{20} \times \frac{5}{2} = \frac{1}{8}$$

## Test Your Understanding

**1.** Consider an amplitude modulated signal

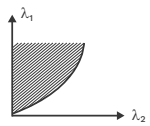
$$x_{AM}(t) = 10(1 + 0.5\cos 2000\pi t + 0.5\cos 4000\pi t)\cos 20{,}000\pi t.$$

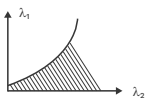
The power contained in carrier is $P_1$ and that inside bands is $P_2$. The ratio $(P_1 / P_2)$ is

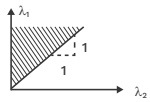
(A) 0.25
(B) 4
(C) 0.5
(D) 2

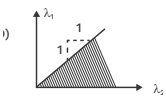
**Solution: (B)**

**2.** Consider amplitude modulated (AM) signal $X_{AM}(t) = \lambda_1 \cos \omega_c t + \lambda_2 \cos \omega_c t \cos \omega_x t$. The region on $\lambda_2 - \lambda_1$ plane in which the signal can be demodulated using envelope detector, is

(A) 

(B) 

(C) 

(D) 

**Solution: (C)**

**3.** Assume that an AM transmitter is modulated with a video testing signal given by, $m(t) = -0.2 + 0.6 \sin \omega t$
The amplitude modulation sensitivity is $k_a = 1$. If the carrier amplitude be , what will be the percentages of positive and negative modulation?

| Positive modulation | Negative modulation |
|---|---|
| (A) 29% | 40% |
| (B) 40% | 80% |
| (C) 80% | 40% |
| (D) 40% | 29% |

**Solution: (B)**

**4.** In a broadcast transmitter, the RF output is represented as

$$s(t) = 50[1 + 0.89\cos 5000t + 0.30\sin 9000t]\cos\left(6x10^6 t\right)V$$

What are the sidebands of the signals in radians?

(A) $5x10^3$ and $9x10^3$

(B) $5.991x\ 10^6$, $5.995\ x\ 10^6$, $6.005\ x\ 10^6$ and $6.009\ x\ 10^6$

(C) $4\ x\ 10^3$, $1.4\ x\ 10^4$

(D) $1x\ 10^6$, $1.1x\ 10^7$, $3\ x\ 10^6$ and $1.5\ x10^7$

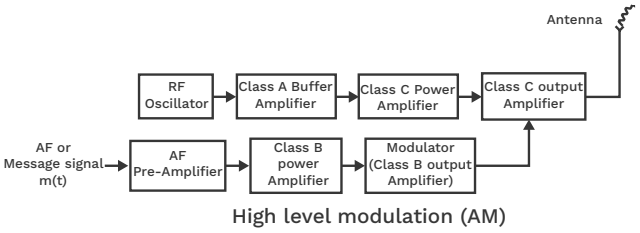
**Solution: (B)**

**Generation of AM**

There are two methods of modulation depending upon carrier power:

1. Low level modulation
2. High level modulation

The block diagrams are as follows:



High level modulation (AM)

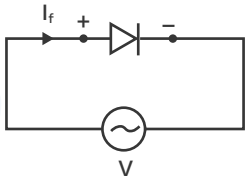
**DSB-FC (AM) Modulator**

The generation or modulation of AM can be done by two methods:

1. Square law modulator
2. Switching modulator

**'Square' Law Device (SLD):**

Let A diode is connected in F.B.



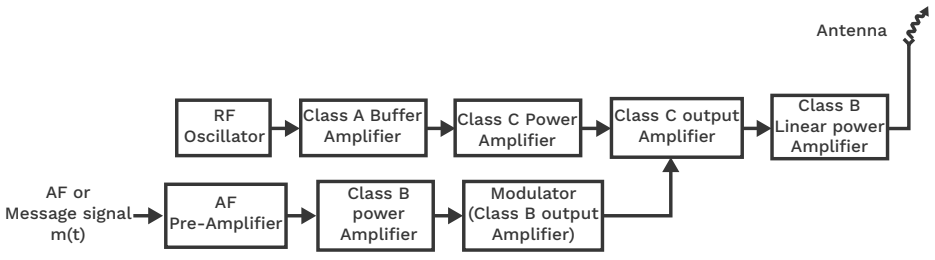

D: FB

$I_f$ = Forward current

$$I_f = I_p\left[e^{\frac{v}{\eta v_T}} - 1\right]$$

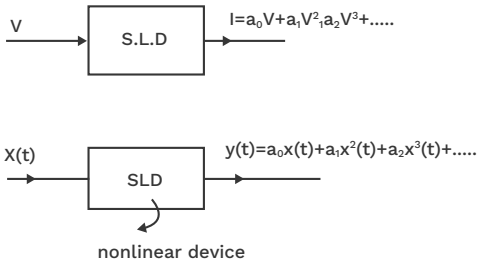
$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} .....$$

$$I_f = I_0\left[1 + \frac{v}{\eta v_T} + \frac{\left(\frac{v}{\eta v_T}\right)^2}{2!} + \frac{\left(\frac{v}{\eta v_T}\right)^3}{3!} + ... - 1\right]$$



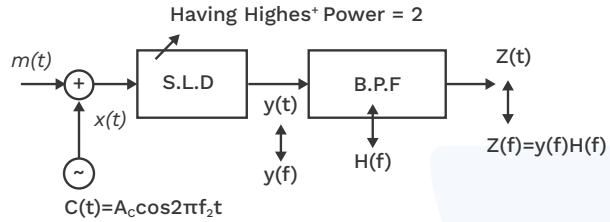
Low level modulation (AM)

$$\underset{\text{Output}}{I_f} = \left(\frac{I_0}{\eta v_T}\right)\underset{\text{Input}}{v} + \left(\frac{I_0}{2\eta^2 v_T{}^2}\right)\underset{\text{Input}}{v^2} + (\quad)v^3 + \ldots$$




**Square law modulator:**

**Block diagram:**




$$\rightarrow x(t) = m(t) + c(t) = m(t) + A_c \cos 2\pi f_c t$$

$$\rightarrow y(t) = a_0 x(t) + a_1 x^2(t)$$

$$y(t) = a_0\left[m(t) + A_c \cos 2\pi f_c t\right] + a_1\left[m(t) + A_c \cos 2\pi f_c t\right]^2$$

$$y(t) = a_0 m(t) + a_0 A_c \cos 2\pi f_c t + a_1\left[m^2(t) + \frac{A_c^2}{2} + \frac{A_c^2}{2}\cos 4\pi f_c t + 2m(t)A_c \cos 2\pi f_c t\right]$$

$$y(t) = a_0 m(t) \rightarrow (1)$$

$$+$$

$$a_0 A_c \cos 2\pi f_c t \rightarrow (2)$$

$$+$$

$$a_1 m^2(t) \rightarrow (3)$$

$$+$$

$$a_1 \frac{A_c^2}{2} \rightarrow (4)$$

$$+$$

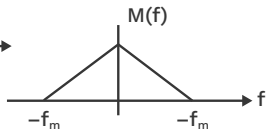
$$a_1 \frac{A_c^2}{2}\cos 4\pi f_c t \rightarrow (5)$$

$$+$$

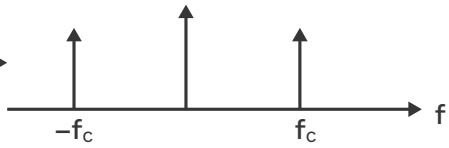
$$2a_1 m(t) A_c \cos 2\pi f_c t \rightarrow (6)$$

**Note:** Terms (2) and (6) are desirable.

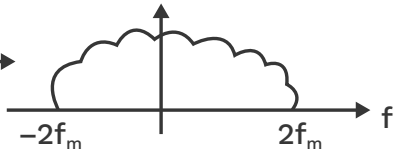
1. $a_0 m(t)$ ⟷



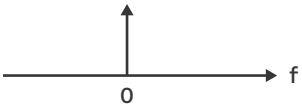

2. $a_0 A_c \cos 2\pi f_c t$ ⟷




3. $a_1 m^2(t) = a_1[m(t) \times m(t)]$ ⟷



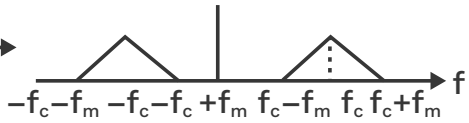

4. $\frac{a_1 A c^2}{2}$ ⟷

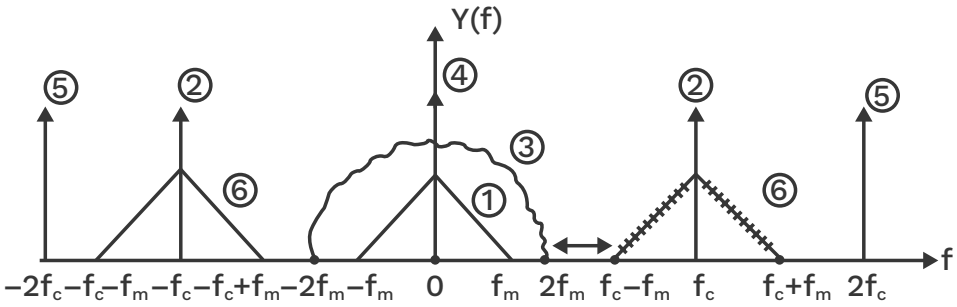


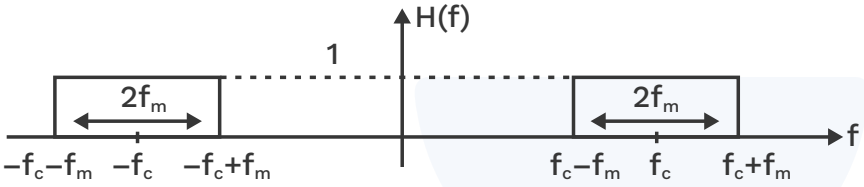
5. $\frac{a_1 Ac^2}{2}(t)\cos 4\pi f_c t \longleftrightarrow$

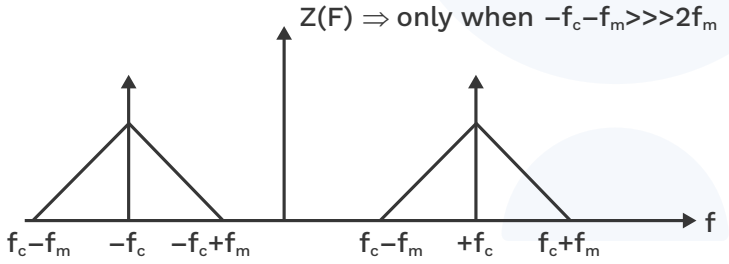



6. $2a_1\, m(t)\cos 2\pi f_c t \longleftrightarrow$




$z(f) \rightarrow$ Amplitude modulated signal.

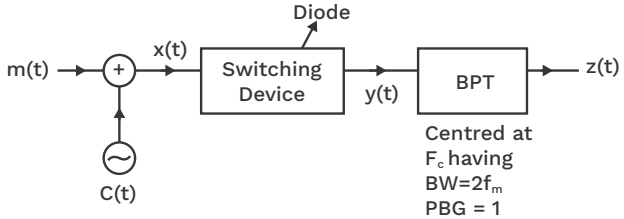
$$z(f) \longleftrightarrow z(t) = a_0 A_c \cos 2\pi f_c + 2a_1 m(t)\cos 2\pi f_c + A_c$$

$$z(t) = a_0 A_c \cos 2\pi f_c + \left[1 + \frac{2a_1}{a_0} m(t)\right]$$

$\downarrow$ DSB–FC

## Conclusions

1.



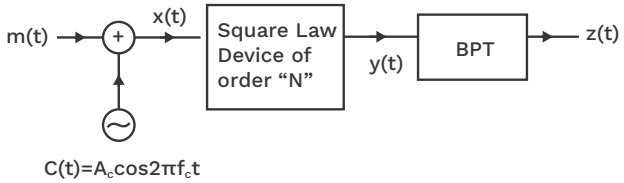
$z(t) = A_c'\left(1 + k_a m(t)\right)\cos 2\pi f_c t$ only when

$$\boxed{\begin{aligned} A_c' &= a_0 A_c \\ k_a &= \frac{2a_1}{a_0} \end{aligned}}$$

as $\mu \uparrow \rightarrow \eta \uparrow$

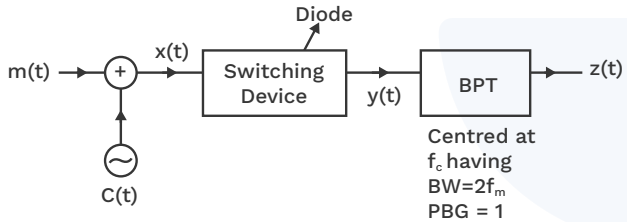
$$\boxed{\mu = k_a \left|m(t)\right|_{max}}$$

2.

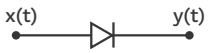



If z(t) is standard AM signal $\Rightarrow \boxed{f_c >>> (N+1) f_m}$

**Switching Modulator**




$x(t) = m(t) + c(t)$




Let $c(t) >>> m(t) \rightarrow$ Switching of diode is controlled by $c(t)$

$c(t) > 0 : D \Rightarrow FB \rightarrow s.c. \Rightarrow y(t) = x(t)$

$c(t) < 0 : D \Rightarrow FB \rightarrow o.c. \Rightarrow y(t) = 0$

$$y(t) = \begin{cases} x(t) & c(t) > 0 \\ 0 & c(t) < 0 \end{cases}$$

$$\boxed{y(t) = x(t)\, x \begin{cases} 1: & c(t) > 0 \\ 0: & c(t) < 0 \end{cases}} \Rightarrow y(t) = x(t) \times g_p(t)$$

$$g_p(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos n\omega_0 t + b_n \sin \omega_0 t\right)$$

$\omega_0 = \omega_c$

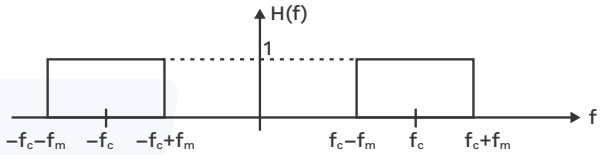
$$g_p(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty}\left[\frac{(-1)^{n-1}}{(2n-1)}\cos(2n-1)\omega_c t\right]$$

$y(t) = x(t) \cdot g_p(t)$

$$y(t) = \left(m(t) + c(t)\right)\left[\frac{1}{2} + \frac{2}{\pi}\cos\omega_c t - \frac{2}{3\pi}\cos 3\omega_c t + \frac{2\cos 5\omega_c t}{5\pi} + \ldots\right]$$

$$y(t) = \left\{\frac{m(t)}{2} + \frac{2}{\pi}m(t)\cos\omega_c t - \frac{2}{3\pi}m(t)\cos 3\omega_c t + \cdots\right\} + \left\{\frac{c(t)}{2} + \frac{2}{\pi}c(t)\cos\omega_c t - \frac{2}{3\pi}c(t)\cos 3\omega_c t + \ldots\right\}$$

BPF:

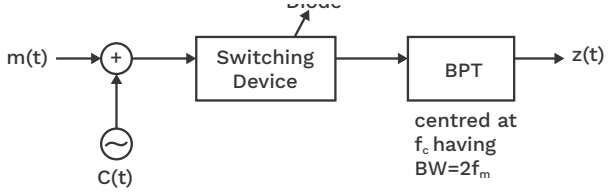



Output of BPF:

$$\Rightarrow z(t) = \frac{A_c}{2}\cos 2\pi f_c t + \frac{2}{\pi}m(t)\cos 2\pi f_c t$$

$$Z(t) = \frac{A_c}{2}\left[1 + \frac{4}{\pi A_c}m(t)\right]\cos 2\pi f t \rightarrow DSB - FC$$

## Conclusions




$Z(t) = Ac'\left(1 + k_a m(t)\cos 2\pi f_c t\right)$

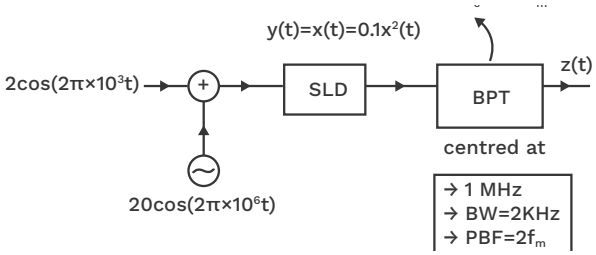
$$Ac' = \frac{Ac}{2}$$

$$k_a = \frac{4}{\pi Ac}$$

$\mu = K_a \left|m(t)\right|_{max}$

| Method | S(t) | Ac' | $K_a$ | $\mu$ |
|---|---|---|---|---|
| **Square law device** | $A_c'\left(t+k_a m(t)\right)\cos 2\pi f_c t$ | $a_o A_c$ | $\frac{2a_1}{a_o}$ | $k_a \lvert m(t)\rvert_{max}$ |
| **Switching modulator** | $A_c'\left(t+k_a m(t)\right)\cos 2\pi f_c t$ | $\frac{A_c}{2}$ | $\frac{4}{\pi Ac}$ | $k_a \lvert m(t)\rvert_{max}$ |

**Q:**




**Solution:**

$$m(t) = A_m \cos 2\pi f_m t \rightarrow \begin{matrix} A_m = 20v \\ f_m = 1KHz \end{matrix}$$

$$\boxed{\begin{matrix} a_0 = 1 \\ a_1 = 0.1 \end{matrix}}$$

$$c(t) = A_c \cos 2\pi f_c t \rightarrow \begin{matrix} A_c = 20v \\ f_c = 1000KHz \end{matrix}$$

$$z(t) = A_c'\left(1+k_a m(t)\cos 2\pi f_c t\right) \rightarrow \begin{matrix} f_c \\ f_c \pm f_m \end{matrix}$$

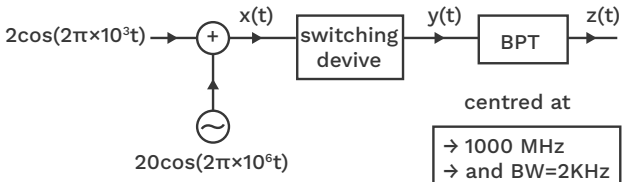
$$A_c' = (a_0 A_c)\,x\ 2 = 20x1x2 = 40 \quad BW = 2f_m$$

$$k_a = \frac{2a_1}{a_o} = \frac{2x0.1}{1} = 0.2$$

$$\mu = k_a \lvert m(t)\rvert_{max} = 0.4$$

$$P_{AM} = Pc'\left(1+\frac{\mu^2}{2}\right) \qquad P_c' = \frac{A_c'^2}{2}$$

**Example:**




Z(t) = ?

**Solution:**

$$z(t) = A_c'\left(1+k_a m(t)\right)\cos 2\pi f_c t \rightarrow P_{AM} = P_c'\left(1+\frac{\mu^2}{2}\right)$$

$$A_c' = \frac{A_c}{2} = \frac{20}{2} = 10V$$

$$k_a = = \frac{4}{\pi A_c} = \frac{4}{\pi 20} = \frac{1}{5\pi}$$

$$\mu = k_a \lvert m(t)\rvert_{max} = 0.12$$

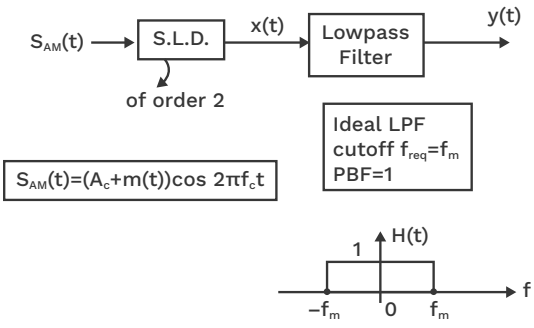
**Demodulation**

The detection techniques for demodulation of AM are:

1. Square law demodulation $\{\text{for } \mu \le 1\}$
2. Envelope or diode detector $\{\text{for } \mu \le 1\}$
3. Synchronous detector $\{\text{for any value of } \mu\}$

The third one that is synchronous detector being complex and square law detector having some drawbacks, so envelope detector is generally used for demodulation as it is cheaper and simpler.

**1. Square law demodulator:**




$$x(t) = a_0 s(t) + a_1 s^2(t)$$

$$x(t) = a_0\left[\left(A_c + m(t)\right)\cos 2\pi f_c t\right] +$$

$$a_1\left[\left(A_c + m(t)\right)\cos 2\pi f_c t\right]^2$$

$$x(t) = a_0\left[A_c\cos 2\pi f_c t + m(t)\cos 2\pi f_c t\right]$$

$$+$$

$$a_1\left[A_c\cos 2\pi f_c t + m(t)\cos 2\pi f_c t\right]^2$$

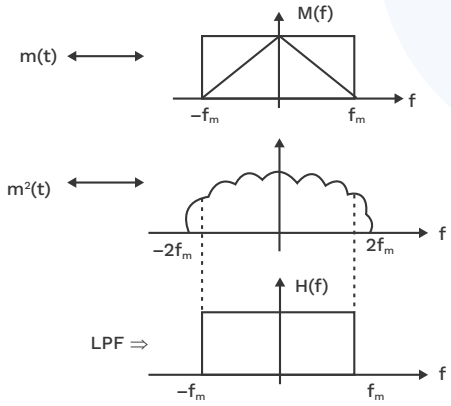
$$x(t) = a_0 A_c\cos 2\pi f_c t + a_0 m(t)\cos 2\pi f_c t$$

$$+$$

$$a_1\left[\frac{A_c^2}{2} + \frac{A_c^2}{2}\cos 4\pi f_c t + \frac{m^2(t)}{2} + \frac{m^2(t)}{2}\cos\left(4\pi f_c t\right) + A_c m(t) + A_c m(t)\cos 4\pi f_c t\right]$$

**LPF output:**

$f_0, 3f_0, 5f_0$




$T_0 = 2\pi / \omega_0 \leftrightarrow M(f) = \sum C_n \delta(f - n$



$$y_1(t) = \underbrace{a_1 A_c m(t)}_{\text{signal}} + \underbrace{a_1\frac{m^2(t)}{2}}_{\text{Interface}}$$

$$\left(\frac{S}{I}\right) = \frac{a_1 A_c m(t)}{a_1\frac{m^2(t)}{2}} = \frac{2}{\left(\frac{m(t)}{A_c}\right)}$$

$$\left(\frac{S}{I}\right)_{min} = \frac{2}{\left(\frac{|m(t)|_{max}}{A_c}\right)} = \frac{2}{\mu}$$

$$P_{SB} \uparrow \rightarrow \eta \uparrow \rightarrow \mu \uparrow \rightarrow \left(\frac{S}{I}\right)_{min} \downarrow\downarrow \rightarrow$$

'Method not used practically'

$$y(t) = a_1 A_c m(t) + \frac{a_1 m^2(t)}{2}$$

Let: $m(t)$ is sinusoidal: $m(t) = A_m\cos 2\pi f_m t$

$$y(t) = a_1 A_c A_m\cos 2\pi f_m t + \frac{a_1}{2}\left[A_m^2\cos^2 2\pi f_m t\right]$$

$$y(t) = a_1 A_c A_m\cos 2\pi f_m t + \frac{a_1 A_m^2}{4} + \frac{a_1 A_m^2}{4}\cos 4\pi f_m t$$

$$y(t) = B_0 + B_1\cos\omega_0 t + B_2\cos 2\omega t$$

**2nd Harmonic distortion** $\Rightarrow D_2 = \left|\frac{B_2}{B_1}\right|$

$$D_2 = \frac{\mu}{4}$$

$$(D_2)_{max}\% = 25\%$$

## Conclusion

1. $\left(\frac{S}{I}\right)_{min} = \frac{2}{\mu}$
2. Not used practically
3. $D_2 = \frac{\mu}{4} \Rightarrow (D_2)_{max}\% = 25\%$

   If the output required m(t) gets interfered with harmonics of higher order, i.e., $m^2(t)$, $m^3(t)$, $m^4(t)$, $m^5(t)$ ...

   Then m(t) can be lost, this is called harmonic distortion. Hence, value of $\mu$ should be small. As $\frac{S}{N} \alpha \frac{1}{\mu}$ and if $\mu$ values are small then power efficiency will compromise.

### Points To Remember

If $\frac{S}{I} \gg 1$, m(t) can be reconstructed and if $\frac{S}{I} < 1$, m(t) can't be reconstructed.

**Envelope detector:**

Envelope of signal: Trace of +ve peaks

| $P_c$ | Envelope |
|---|---|
| $A\cos\omega_c t$ | A |
| $\sqrt{A^2+B^2}\ \cos\omega_c t$ | $\sqrt{A^2+B^2}$ |

1 $x(t) = A\cos\omega_0 t + B\sin\omega_0 t$

$x(t) = \sqrt{A^2+B^2}\cos(\omega_0 t - \phi) \rightarrow E(t)$

$= \sqrt{A^2+B^2}$

2. $x(t) = A\cos(\omega_0 t + \theta) + B\sin\omega_0 t$

$x(t) = A\cos\omega_0 t\cos\theta - A\sin\omega_0 t\sin\theta -$

$B\sin\omega_0 t$

$x(t) = \cos\omega_0 t\,(A\cos\theta) + \sin(\omega_0 t)$

$(B - A\sin\theta) \Rightarrow E(t) = \sqrt{X^2+Y^2}$

$E(t) = \sqrt{A^2\cos^2\theta + (B - A\sin\theta)^2}$

$= \sqrt{A^2+B^2-2AB\sin\theta}$

3. $x(t) = A(t)\cos\omega_c t$

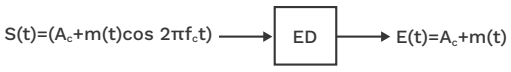
$C(t) = f_c$

4. $x(t) = (A_c + m(t))\cos\omega_c t$

$$E(t) = |A_c + m(t)| = \begin{cases} A_c + m(t) & : \ \mu \le 1 \\ |A(t) + m(t)| & : \ \mu > 1 \end{cases}$$

**Importance of carrier signal in AM expression:**

**Case 1:** $\overset{FC}{\uparrow}\{A_c\cos 2\pi f_c t\} + \overset{DBS}{\uparrow}\{m(t)\cos 2\pi f_c t\}$





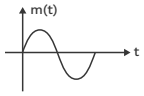

$s(t) = A_c +$

$\rightarrow [A_c + m(t)] \ge 0$

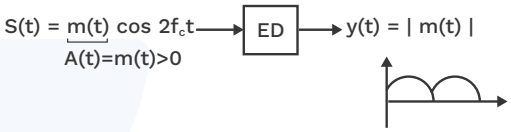
$\rightarrow \mu \le 1$

**Case 2:**

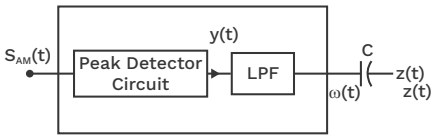
$$S(t) = \overset{0}{\uparrow} A_c\cos 2\pi f_c t + A_c m(t)\cos 2\pi f_c t$$



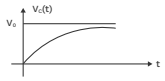

$$s(t) = \overset{0}{\uparrow}(A_c + m(t))\cos 2\pi f_c t$$




**Envelope detector:**

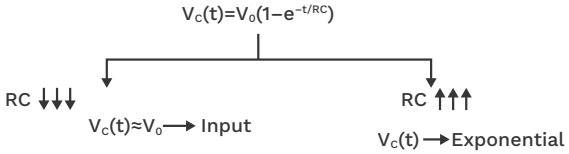



1. Capacitor: It does not allow sudden changes in voltage.

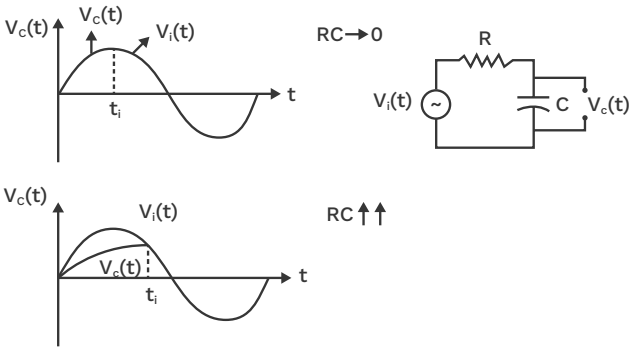



2. Charging Eqⁿ of capacitor:

$R_C = T_c \rightarrow$ charging time constant



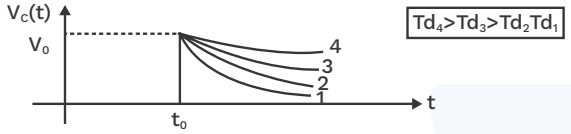
3.




4. Discharging Eqⁿ of capacitor:

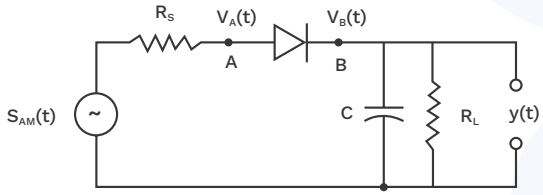
$t = 0: \quad V_c(t) = V_0 e^{-t/RC} \quad T_d = RC$

$t = 0: \quad V_c(t) = V_0 e^{-(t-t_0)/RC}$




As $RC \uparrow\uparrow$ discharging above becomes flatter.

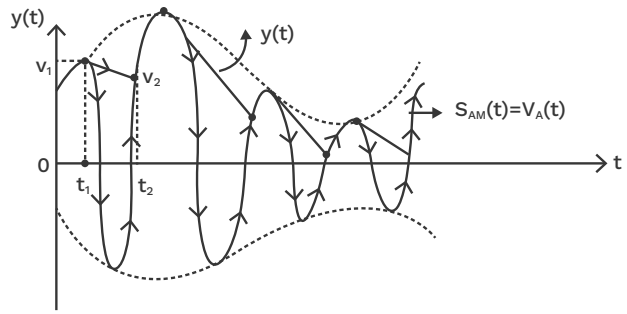
**Peak detector circuit:**




1 $V_A(t) \approx S_{AM}(t) \quad R_s \to Small$

2. $V_B(t) = y(t) =$ voltage across capacitor

$P_{AM}$




1 $t = 0^-$ $\quad \boxed{\begin{matrix} V_A(t) = 0 \\ V_B(t) = 0 \end{matrix}}$

$t = 0^+ \quad \begin{matrix} V_A(t) > 0 \\ V_B(t) = 0 \end{matrix} \to V_A > V_B \Rightarrow \boxed{D : FB}$

D: FB → Capacitor starts charging:

$T_c = (R_s \| R_L) C$

$P_c(1+\mu^2)$ $\qquad$ $R_s <<< R_L$

$T_c = R_s C$

2. $t = t_1^- \to$ $\quad \boxed{\begin{matrix} V_A = V_1 \\ V_B = V_1 \end{matrix}}$

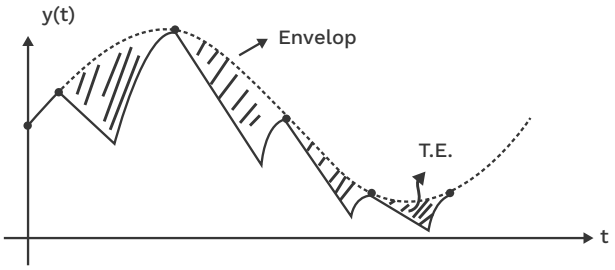
$t = t_1^+ \to \begin{matrix} V_A < V_1 \\ V_B = V_1 \end{matrix} \to V_A < V_B \Rightarrow \boxed{D : RB}$

$D : RB \to$ Capacitor starts discharging through $R_L$

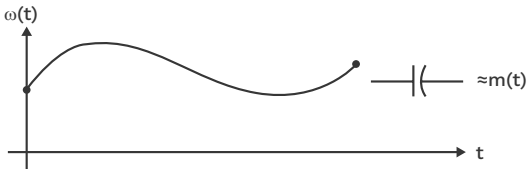
3. $t = t_2^- \to$ $\quad \boxed{\begin{matrix} V_A(t) = V_2 \\ V_B(t) = V_2 \end{matrix}}$

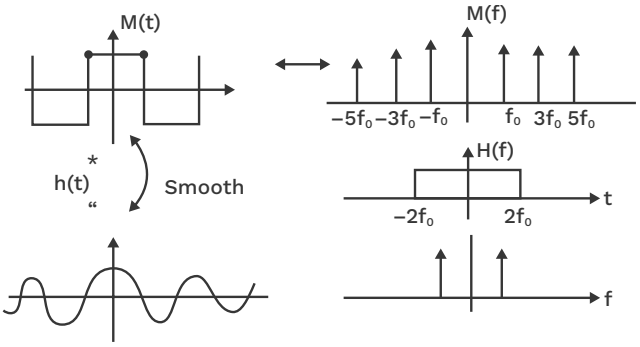
$t = t_2^+ \to \boxed{\begin{matrix} V_A(t) < V_2 \\ V_B(t) = V_2 \end{matrix}} \to V_A > V_B \to \boxed{D : FB}$

D: FB → Capacitor charges

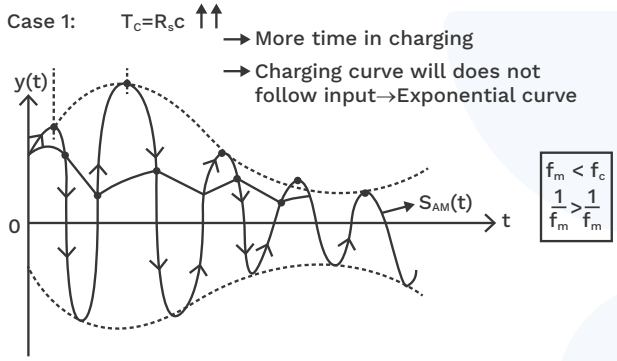



1. Output of peak detector circuit tracks the envelope of the modulated signal with some tracking error.
2. Low pass filter smoothens the curve of $y(t)$ between 2 peaks in order to minimise tracking error:




**Effect of time constant on the output waveform y(t):**




1. In this case, peak of the supply was not detected.
2. For detection of peaks of the supply

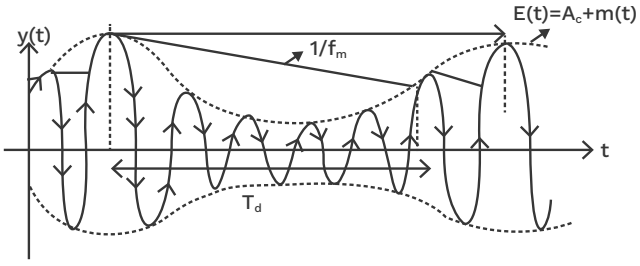
$$P_{s(t)} = P_c\left(1+\frac{\mu^2}{3}\right)$$ Peak will be detected

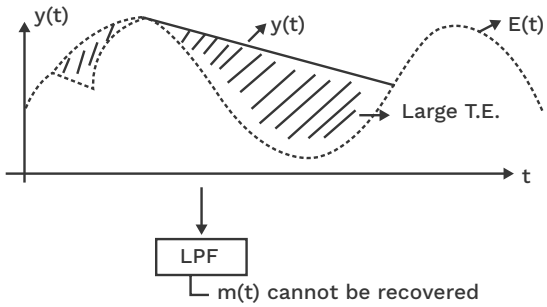
**Case 2:** $P_{s(t)} = 50$.

→ Let $T_d$ is very high

→ Discharging time is large.

→ Discharging curve: Flat




**Diagonal clipping or negative clipping:**

1. Very high $T_d = R_L C$
2. $T_d = R_L C \approx \frac{1}{f_m}$

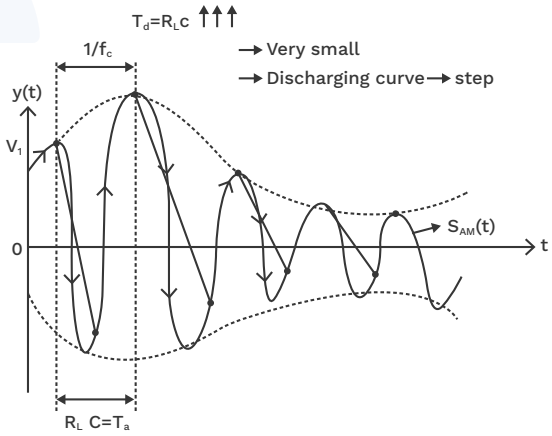
To remove diagonal clipping: →

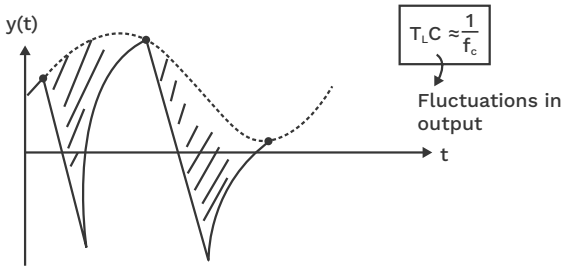
$$R_L C <<<< \frac{1}{f_m} \rightarrow Condition$$

$$R_L C \leq \frac{\sqrt{1-\mu^2}}{2\pi f_m \mu} \rightarrow Formula$$

$$\mu \leq \frac{1}{\sqrt{1+\left(2\pi f_m R_L C\right)^2}}$$

**Case 3:**



## Points To Remember

Envelope detector:

1. Used only when message signal is stored in envelope $(\mu \le 1)$
2. $T_c = R_s C <<<< \frac{1}{f_c}$
3. $T_c = R_s C >>> \frac{1}{f_c} \rightarrow$ Peaks are not detected
4. Diagonal clipping $\rightarrow R_L C \approx \frac{1}{f_m}$

   To remove diagonal clipping
   $R_L C <<< 1/f_m$

   $R_L C \le \frac{\sqrt{1-\mu^2}}{\omega_m \mu}$

5. $T_d = R_L C \approx \frac{1}{f_c} \rightarrow$ Fluctuation in output

   To remove fluctuations: $R_L C >>> \frac{1}{f_c}$

6. Proper choice of discharging time constant $R_L C$

   $No\ fluctuation \leftarrow \frac{1}{f_c} <<< R_L C <<< \frac{1}{f_m} \rightarrow$

   $no$ diagonal clipping

7. m(t): Multitone

   $f_m \rightarrow f_{max}\ freq^n\ component\ of\ m(t)$

   $f_c > f_m$

   $\frac{1}{f_c} < \frac{1}{f_m}$

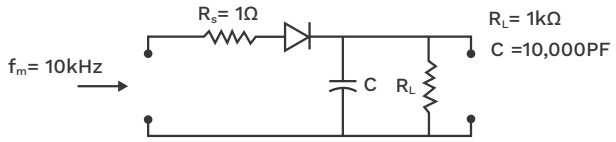
**Desired**

$R_s C <<< \frac{1}{f_c}$

$R_L C \approx \frac{1}{f_m} D.C. \Rightarrow R_L C <<< \frac{1}{f_m}$

$|Area|_{max} = 8$

**Example:**




Calculate maximum value of $\mu$ in order to avoid diagonal clipping.

**Solution:**

$R_L C \le \frac{\sqrt{1-\mu^2}}{\omega_m \mu}$ $\quad \mu \le \frac{1}{\sqrt{1+(2\pi f_m R_L C)^2}}$

$\mu \le \frac{1}{\sqrt{1+\left(2\pi \times 10^4 \times 10^3 \times 10^4 \times 10^{-12}\right)^2}}$

$\mu \le 0.846$

$\mu_{max} = 0.847$

**Example:** AM signal is detected by using E.D. When $f_m = 2KHz, f_c = 1MHz$. Appropriate value of time constant of E.D.

**Solution:** $\frac{1}{f_c} <<< R_L C << \frac{1}{f_m}$

**Example:**

$m(t) = \cos(2000\pi t) + 4\cos(4000\pi t)$

($2000\pi t$: 1kHz; $4000\pi t$: 2KHz)

modulates the carrier $c(t) = \cos(2\pi \times 10^6 t)$. AM signal uses ED. The time constant of ED should satisfy.

(A) $0.5ms < RC < 1ms$

(B) $1\mu sec < RC < 0.5ms$

(C) $RC << 1\mu Sec$

(D) $RC >> 0.5ms$

**Solution:** $\frac{1}{f_c} <<< R_L C << \frac{1}{f_m}$

$f_c = 1000KHz$

$f_m = 2KHz$

**Example:** $A_c \cos\omega_c t + 2\cos\omega_c t \cos\omega_c t$ is to be demodulated by using ED calculate minimum value of $A_c$ for proper demodulation.

**Solution:**

$$s(t) = A_c\left(1+\frac{2}{A_c}\cos\omega_m t\right)\cos\omega_c t$$

$$s(t) = A_c\left(1+K_a m(t)\right)\cos\omega_c t$$

*OR*

$$A_c\left(1+\mu\cos\omega_m t\right)\cos\omega_c t$$

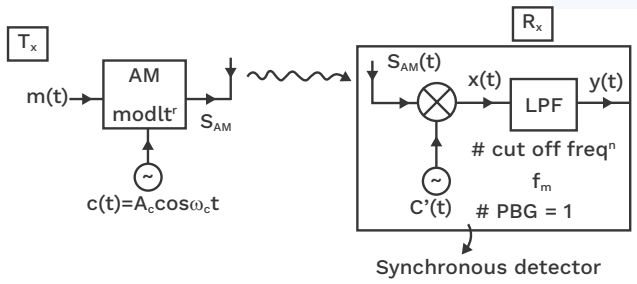
$$\mu = \frac{2}{A_c}$$

$$\mu = k_a\left|m(t)\right|_{max}$$

$$A_c \geq 2, \left(A_c\right)_{min} = 2v$$

## Synchronous Detector

'It is multiplier followed by L.P.F'.




- The synchronous detector requires complete synchronisation in both frequency and phase between local oscillator in detector and the transmitter's carrier.
- This type of detector is complex. Frequency synchronisation is easy to maintain but phase synchronisation requires additional circuitry which makes the detector complex.

**At Tx:** $c(t) = A_c\cos\omega_c t$ : Carrier before modulation

$S_{AM}(t) = \left(A_c + m(t)\right)\cos\omega_c t$ : Modulated signal or carrier after modulation

**At Rx:** $C'(t) = A_c'\cos\left(\left(\omega_c + \Delta\omega\right)t + \Delta\phi\right) \rightarrow$ Locally generated carrier at Rx.

**Mathematical analysis:**

1. Output of multiplier:

$$x(t) = S_{AM}(t)C'(t)$$

$$x(t) = \left[A_c + m(t)\right]\cos\omega_c t\, A_c'\cos$$

$$\left(\omega_c t + \Delta\omega t + \Delta\phi\right)$$

$$x(t) = \left[A_c\cos\omega_c t + m(t)\cos\omega_c t\right]$$

$$A_c'\cos\left(\omega_c t + \Delta\omega t + \Delta\phi\right)$$

$$x(t) = A_c A_c'\cos\omega_c t\cos\left(\omega_c t + \Delta\omega t + \Delta\phi\right) +$$

$$A_c' m(t)\cos\omega_c t\cos\left(\omega_c t + \Delta\omega t + \Delta\phi\right)$$

$$x(t) = \frac{A_c A_c'}{2}\cos\left(2\omega_c t + \Delta\omega t + \Delta\phi\right) +$$

$$\frac{A_c A_c'}{2}\cos\left(\Delta\omega t + \Delta\phi\right) +$$

$$\frac{A_c' m(t)}{2}\cos\left(2\omega_c t + \Delta\omega t + \Delta\phi\right) +$$

$$\frac{A_c' m(t)}{2}\cos\left(\Delta\omega t + \Delta\phi\right)$$

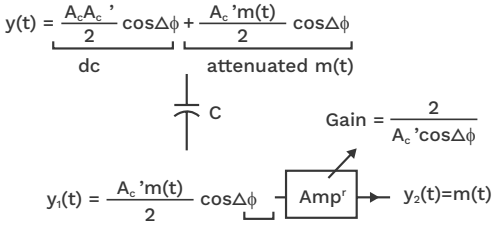
**LPF output:**

$$y(t) = \frac{A_c A_c'}{2}\cos\left(\Delta\omega t + \Delta\phi\right)$$

$$+\frac{A_c' m(t)}{2}\cos\left(\Delta\omega t + \Delta\phi\right)$$

**Case 1: Freqn synchronisation**

$\Delta\omega = 0 \qquad \Delta\phi \neq 0$



$$y_1(t) = \frac{A_c'}{2} m(t) \cos \Delta\phi$$

$$\Delta\phi = (2n+1)\frac{\pi}{2} \rightarrow y_1(t) = 0$$

**Quadrature null effect:**

**Case 2: Phase synchronisation**

$\Delta\phi = 0 \qquad \Delta\omega \neq 0$

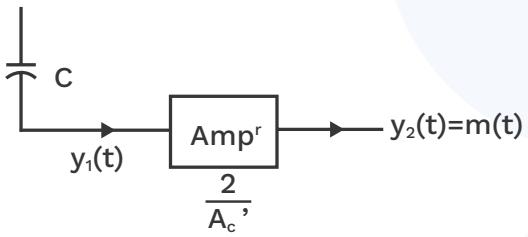
$$y(t) = \underbrace{\frac{A_c A_c'}{2}\cos(\Delta\omega t)}_{\text{Inerference}} + \underbrace{\frac{A_c'}{2} m(t)\cos(\Delta\omega t)}_{\text{distorted msg signal.}}$$

'm(t) cannot be recovered'.

**Case 3: Perfect synchronisation**

$\Delta\omega = 0 \qquad \Delta\phi = 0$

$$y(t) = \underbrace{\frac{A_c A_c'}{2}}_{dc} + \underbrace{\frac{A_c' m(t)}{2}}_{\text{atlenuated } m(t)}$$

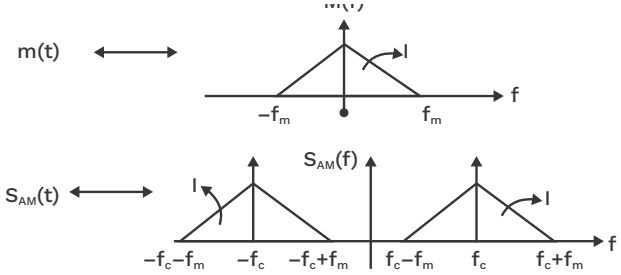



**Important**

| $\Delta\omega$ | $\Delta\phi$ | $\Delta\phi$ | Recovery |
|---|---|---|---|
| = 0 | $\neq 0$ | $= (2n+1)\frac{\pi}{2}$ | Yes |
| ¯ 4 | ʜ = | $m(t) = A_m$ c | Q.N.E. |
| $\neq 0$ | = 0 | = 0 | No |
| = 0 | = 0 | = 0 | Yes |

| **Demodulator** | | |
|---|---|---|
| Square law demodulator | Not used practically | |
| Envelope detector | Used only when $(\mu \leq 1)$ | Cost effective |
| Synchronous detector | Used for $\mu < 1$ $\mu = 1$ $\mu > 1$ | Costly |

**Important observations:**




1. In case of DSB-FC, m(t) is stored in the envelope of s(t) $(\mu \leq 1)$
2. In case of DSB-FC, M(f) is stored in the sidebands of S(f).

**m(t): Sinusoidal:** $P_{AM} = P_c\left(1 + \frac{\mu^2}{2}\right)$

$$P_c = \left(\frac{P_{AM}}{1+\frac{\mu^2}{2}}\right).$$

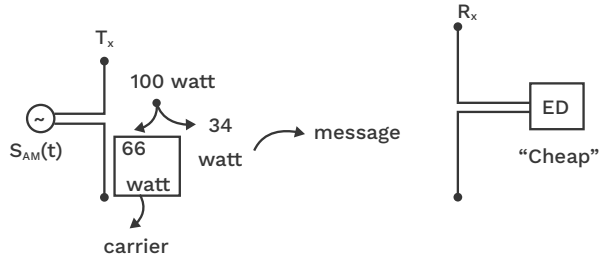
$\mu = 0 \quad P_c = P_{AM} \qquad P_{SB} = 0$

$\mu = 1 \quad P_c = \frac{2}{3}P_{AM} \qquad P_{SB} = \frac{1}{3}P_{AM}$

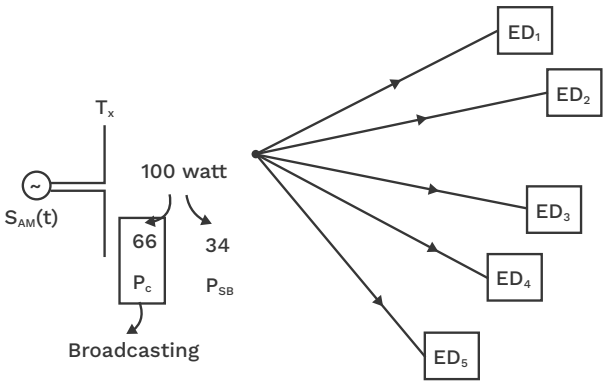
*no message* information

$\rightarrow$ *message* information.

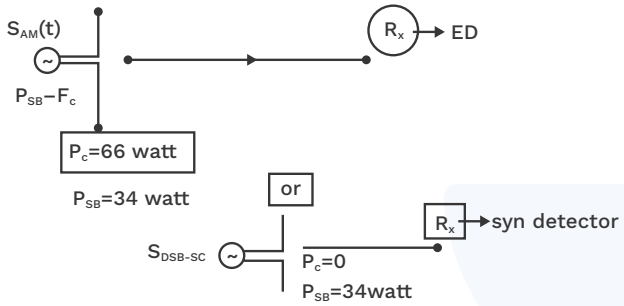
$$S_{AM}(t) = \left[A_c + m(t)\right]\cos 2\pi f_c t$$




**Case 1:** Point to multipoint commⁿ



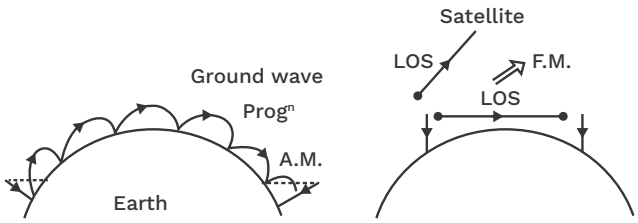

**Case 2: 'Point-to-Point communication'**
'Suppressed carrier systems.'




## Standard Data for AM(DSB-FC)

1. Carrier frequency: [535 kHz-1640 kHz]
2. Message signal BW: 5 kHz 'Cut-off freq. of LPF.'
3. Modulated signal BW: 10 kHz
4. Propagation mode: Ground wave propagation.

| Freq. | Propagation Mode | |
|---|---|---|
| 0–2 MHz | Ground wave | A.M. |
| 2 MHz–30 MHz | Ionospheric wave | |
| 30 MHz–300 MHz | Space wave or LOS | F.M. |
| μ wave freq. | Duct wave | |




**Point-to-point commn:** Suppressed carrier systems are used

## Double Side Band-Suppressed Carrier (DSB-SC)

$P_{AM} \rightarrow$ constant if

$$s(t) = A(t)\left[A_c \cos 2\pi f_c t\right]$$

A(t) = m(t)

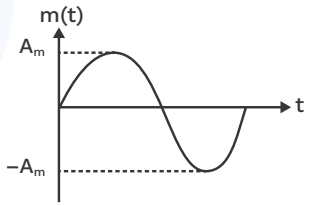
$$s(t) = m(t)\left(A_c \cos 2\pi f_c t\right) = m(t) \times c(t)$$

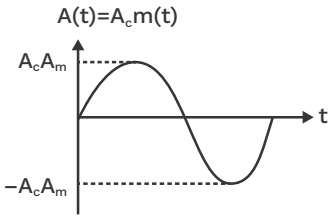
$$S_{DSB-SC}(t) = m(t) \times c(t)$$

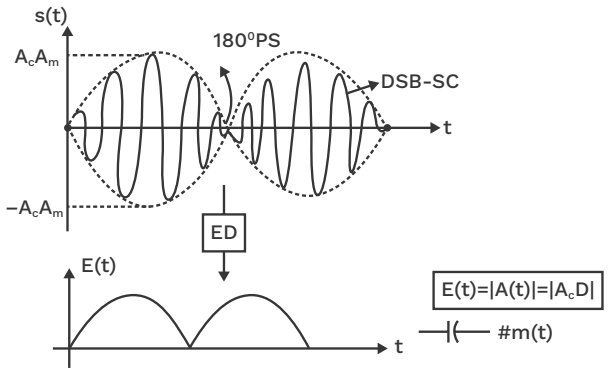
$$S_{DSB-SC}(t) = m(t)\left(A_c \cos 2\pi f_c t\right)$$

**Time domain analysis:**

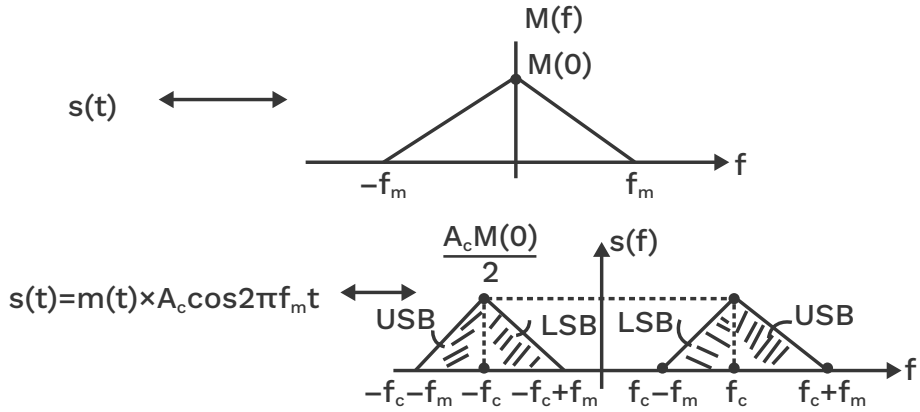
$$s(t) = \left(A_c m(t) \cos 2\pi f_c t\right)$$



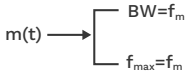



**Frequency domain analysis:**




**Frequency related calculation:**




$$c(t) \rightarrow f_c$$

$$s(t) \rightarrow BW = 2f_m = 2 \times \max freq.comp.\ of\ \ m(t)$$

**Power related calculation:**

$$m(t) \rightarrow \overline{m^2(t)} = P_m$$

$$c(t) \rightarrow \overline{c^2(t)} = P_c$$

$$s(t) = m(t) \times c(t)$$

$$P_{DSB} = \overline{m^2(t)} \times \overline{c^2(t)} = P_m P_c = P_{SB}$$

$$P_{USB} = P_{LSB} = \frac{P_{SB}}{2} = \frac{P_m P_c}{2}$$

**When μ is given:** $P_{AM} = P_c + \frac{P_c \mu^2}{2}$

$$P_{DSB} = \frac{P_c \mu^2}{2} = P_{SB}$$

**Case 1:** m(t) is single tone sinusoidal signal:

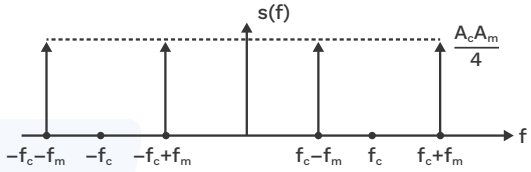
$$m(t) = A_m \cos 2\pi f_m t$$

1. **Time domain:**

$$s(t) = A_m \cos 2\pi_m t . A_c \cos 2\pi f_c t$$

$$s(t) = \frac{A_c A_m}{2} \cos\left(2\pi\left(f_c + f_m\right)t\right) + \frac{A_c A_m}{2} \cos\left(2\pi\left(f_c - f_m\right)t\right)$$

2. **Freqⁿ domain:**




**Freqⁿ related:**

$$m(t) = f_m$$

$$c(t) = f_c$$

$$s(t) = f_c \pm f_m \begin{cases} f_c + f_m \\ f_c - f_m \end{cases}$$

**Power related calculation:**

$$P_{DSB} = P_m P_c = \frac{A_c^2 A_m^2}{4}$$

**Case 2:** $m(t) = A_{m_1} \cos 2\pi f_{m_1} t + A_{m_2} \cos 2\pi f_{m_2} t$

$$P_{DSB} = P_c P_m = \frac{A_c^2}{2}\left\{\frac{A_{m1}^2}{2} + \frac{A_{m2}^2}{2}\right\}$$

$$m(t) \rightarrow f_{m_1}, f_{m_2}$$

$$c(t) : f_c$$

$$s(t) \rightarrow f_c \pm f_{m_1}$$

$$\mu_1 = 0.3$$

**Case 3:** m(t): Square wave

$$P_{DSB} = P_c P_m = \left(\frac{A_c^2}{2}\right)\left(A_m^2\right)$$

$$m(t) \to f_0, 3f_0, 5f_0, 7f_0 ---$$

$$c(t) \to f_c$$

$$s(t) \to f_c \pm f_0$$

$$f_c \pm 3f_0$$

$$f_c \pm 5f_0 \ldots$$

**Case 4:** m(t) is triangular wave:

$$P_{DSB} = P_c P_m = \left(\frac{Ac^2}{2}\right)\left(\frac{A_m^2}{3}\right)$$

$$m(t) \to f_0, 3f_0, 5f_0 \ldots$$

$$c(t) \to f_c$$

$$X(f) = \frac{S_{vsb}(f - fe) + S_{vsb}(f + fe)}{2}$$

**Case 5:** m(t) is sawtooth:

$$D_{DSB} = P_c P_m = \frac{A_c^2}{2}\frac{A_m^2}{3}$$

$$m(t) \to 0, f_0, 2f_0, 3f_0 \ldots$$

$$\cos 2\pi f_c t$$

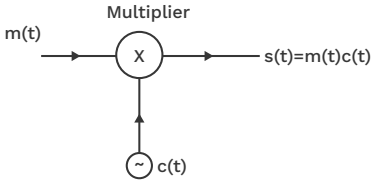
$$s(t) \to f_c \pm 0$$

$$f_c \pm f_0$$

$$f_c \pm 2f_0$$

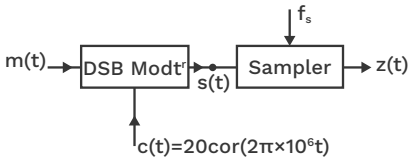
$$\vdots$$

**DSB-modulators:**



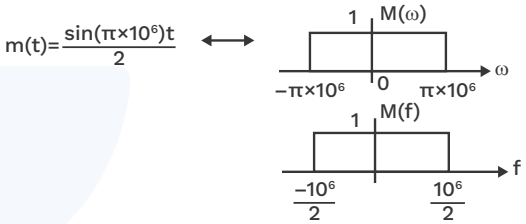

1. Multiplier Modtr
2. Product Modtr
3. DSB-SC Modtr

**Example:** $m(t) = \dfrac{\sin\left(\pi \times 10^6 t\right)}{\pi t}$



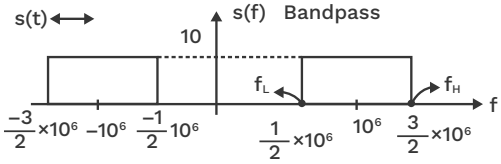

Calculate NR of z(t) =?

**Solution:**




$$s(t) = m(t)\left(20\cos\left(2\pi \times 10^6 t\right)\right)$$

$$s(f) = 10\left[M\left(f - 10^6\right) + M\left(f + 10^6\right)\right]$$




$s(t)$ : *Bandpass signal*

$$f_s \geq \frac{2f_H}{k}$$

$$k = \left[\frac{f_H}{f_H - f_c}\right] \quad [\bullet] \to \text{Imme previous integer}$$

$$N_R = 2f_{max}$$

[2.3] = 2

[2] = 2

$[-2.3] = -3$

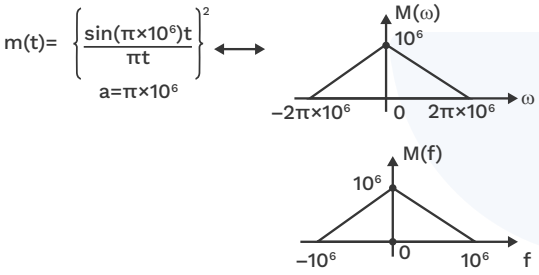
$$f_s \geq \frac{2f_H}{k} \qquad k = \left[\frac{f_H}{f_H - f_L}\right] = \left[\frac{\frac{3}{2}\times 10^6}{\frac{3}{2}\times 10^6 - \frac{1}{2}\times 10^6}\right]$$

$$f_s \geq 2\times\frac{3}{2}\times 10^6 \qquad k = [1.5] = 1$$

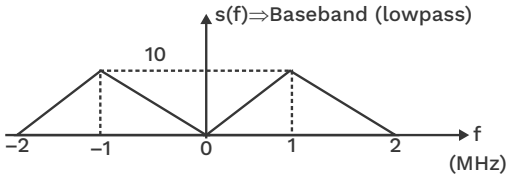
$$N_R = 2\times f_{max} = 2\times\frac{3}{2}\times 10^6 = 3MHz$$

**Example:** $m(t) = \left(\frac{\sin(\pi\times 10^6 t)}{\pi t}\right)^2$ 'use diagram of previous question'.

**Solution:**




$$s(t) \leftrightarrow 10\left[M(f+10^6) + M(f-10^6)\right] = S(f)$$




$s(t) = Lowpass$ :

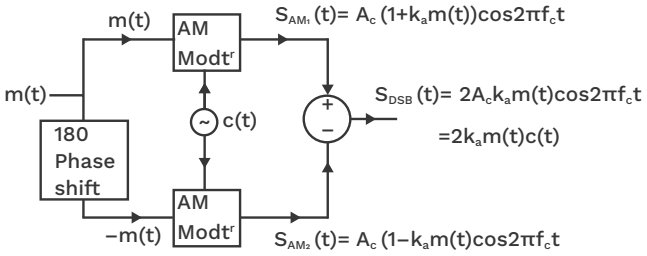
$f_s \geq 2f_{max}$ $\qquad f_s \geq 4\,MHz$

$N_R = 2f_{max}$ $\qquad f_s = 4\,MHz$

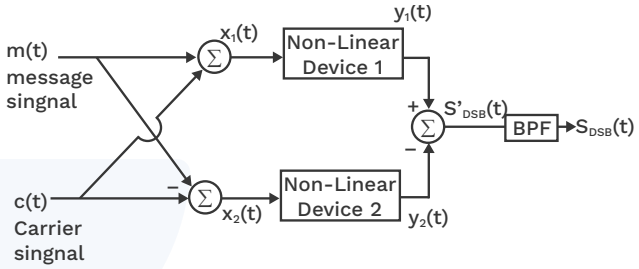
## Types of DSB-SC Modlt$^r$

1. Balanced modulator
2. Ring modulator

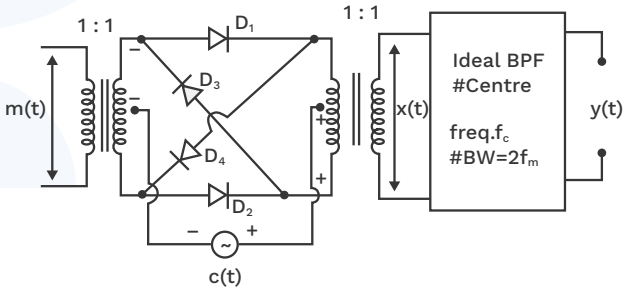
**Balanced modlt$^r$:**



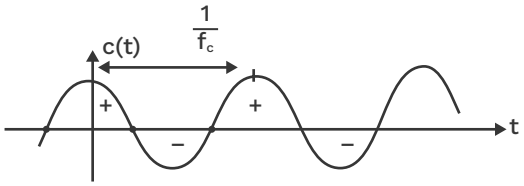

The balanced modulator can also be shown by using a circuit connecting two non-linear devices such as diodes to generate DSB-SC. The basic block diagram is




**Ring modulator:**

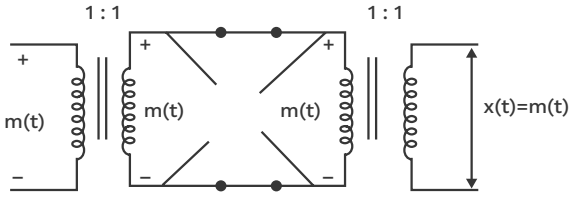



1. 4 diodes are used in ring form:
2. Transformer is 1:1 centre tap type.
3. c(t) is chosen such that it controls switching operation of diode.



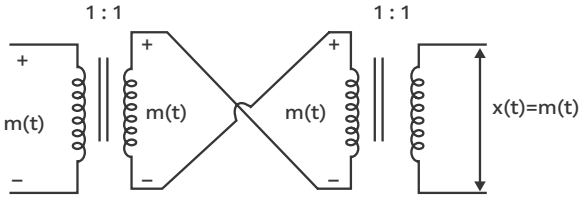

**Case 1:** $c(t) > 0$ : $\qquad D_1, D_2 \rightarrow F.B.$

$D_3, D_4 \rightarrow F.B.$




**Case 2:** $c(t) < 0$ : $D_1, D_2 \rightarrow R.B.$

$D_3, D_4 \rightarrow F.B.$




$$x(t) = \begin{cases} m(t) & c(t) > 0 \\ -m(t) & c(t) < 0 \end{cases}$$

$$x(t) = m(t) \begin{cases} 1 & c(t) > 0 \\ -1 & c(t) < 0 \end{cases}$$

$m^2(t)$,

$$g_p(t) = a_0 + \sum_{n=1}^{\infty} \left( a_n \cos n\omega_0 t + b_n \sin n\omega_0 t \right)$$

$$\omega_0 = \omega_c$$

$$g_p(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)} \cos(2n-1)\omega_c t$$

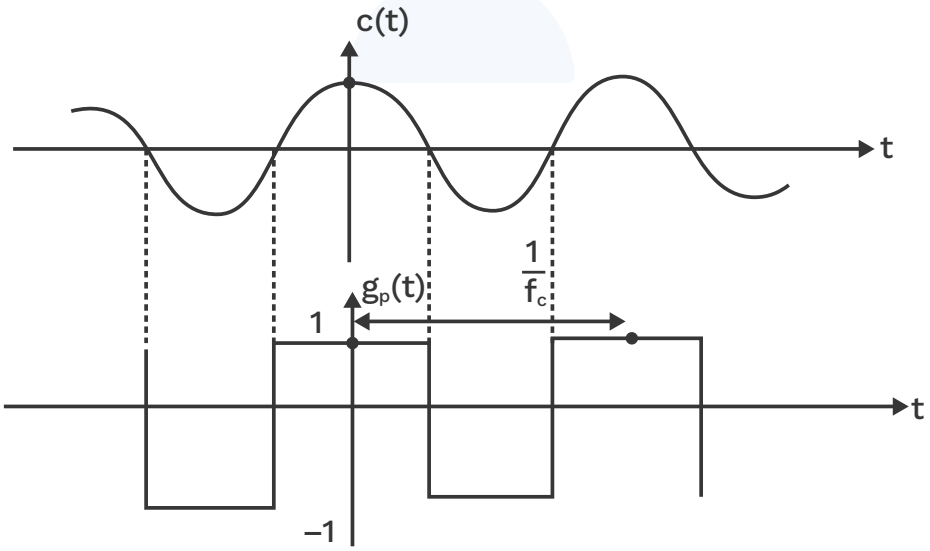
$$x(t) = m(t) g_p(t)$$

$$x(t) = m(t) \left[ \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)} \cos(2n-1)\omega_c t \right]$$

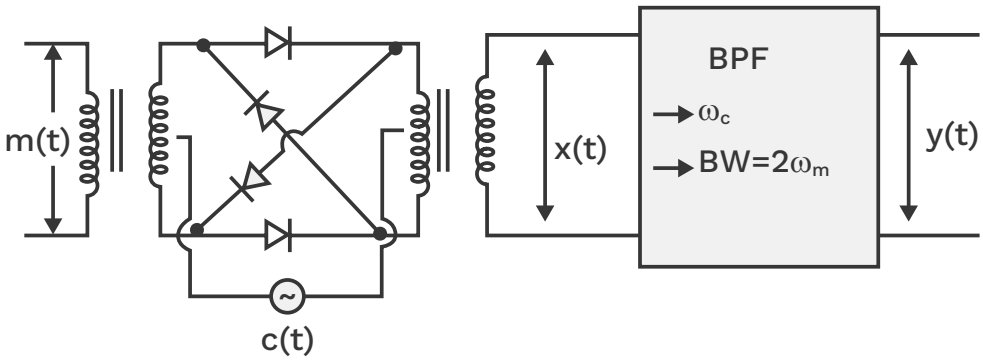
$$x(t) = m(t) \left[ \frac{4}{\pi} \cos \omega_c t - \frac{4}{3\pi} \cos 3\omega_c t + \frac{4}{5\pi} \cos 5\omega_c t \ldots \right]$$

Output of BPF:

$$y(t) = m(t) \frac{4}{\pi} \cos \omega_c t$$



## Conclusion




1. $x(t)\alpha\, m(t)\left[a_1\cos\omega_c t + a_2\cos 3\omega_c t + ....\right]$
2. $y(t)\ \alpha\ m(t)\cos\omega_c t$

   If m(t): sinusoidal:
3. $x(t)\rightarrow f_c \pm f_m$

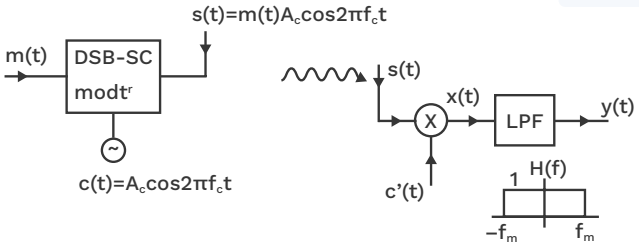
   $3f_c \pm f_m$

   $5f_c \pm f_m$
4. $y(t): f_c \pm f_m$

**DSB-SC demodulator:**

$$c'(t) = A_c'\cos\left(\omega_c t + \Delta\omega t + \Delta\phi\right)$$

**Synchronous detector:**




$$x(t) = s(t)c'(t)$$

$$x(t) = A_c m(t)\cos\omega_c t\left[A_c'\cos\left(\left(\omega_c + \Delta\omega\right)t + \Delta\phi\right)\right]$$

$$x(t) = A_c m(t)\cos\omega_c t\ A_c'\cos\left(\omega_c t + \Delta\omega t + \Delta\phi\right)$$

$$x(t) = \frac{A_c A_c'}{2} m(t)\cos\left(2\omega_c t + \Delta\omega t + \Delta\phi\right)$$

+

$$\frac{A_c A_c'}{2} m(t)\cos\left(\Delta\omega t + \Delta\phi\right).$$

**Output of LPF:**

$$y(t) = \frac{A_c A_c'}{2} m(t)\cos\left(\Delta\omega t + \Delta\phi\right)$$

**Case 1:** $\Delta\omega = 0 \qquad \Delta\phi \neq 0$

$$y(t) = \frac{A_c A_c'}{2} m(t)\cos\Delta\phi$$

Attenuated m(t)

If $\Delta\phi = (2n+1)\frac{\pi}{2}$

$$y(t) = 0 \rightarrow Q.N.E.$$

**Case 2:** $\Delta\omega \neq 0 \qquad \Delta\phi = 0$

$$y(t) = \frac{A_c A_c'}{2} m(t)\cos\left(\Delta\omega t\right)$$

distorted m(t)

**Case 3:** $\Delta\omega = 0 \rightarrow \Delta\phi = 0$

$$y(t) = \frac{A_c A_c'}{2} m(t) \rightarrow attenuated$$

**Advantages of AM**

1. It is used for long distance communication.
2. Demodulation methods are simple, i.e., by using diodes.
3. Transmitters are less complex.

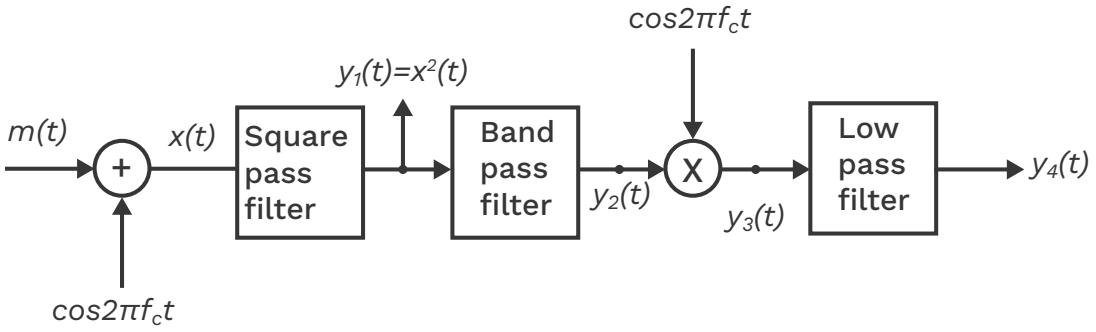
**Drawbacks**

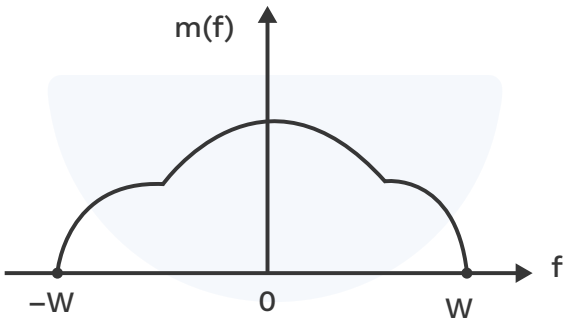
1. Transmitted power is wasted in the form of carrier power.
2. The channel bandwidth requirement is high.
3. The AM receivers are sensitive towards noise.

**Applications**

1. AM broadcasting is preferred and popular.
2. The communication for point to multipoint uses AM.

**Example:** The message signal m(t) whose spectrum is shown in figure (B) is passed through the system shown in figure (A)



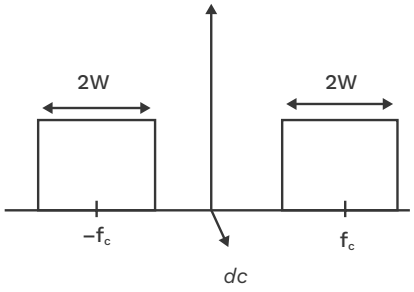

The bandpass filter has a B.W. of 2W centred at $f_c$ and the LPF has a bandwidth of W. The bandwidth of the output signal $y_4(t)$ is

(A) 2 W

(B) 3 W

(C) 4 W

(D) W




**Solution.** $x(t) = \left[m(t) + \cos 2\pi f_c t\right]$

$$y_1(t) = m^2(t) + \frac{1}{2} + \frac{1}{2}\cos 4\pi f_c t + 2m(t)\cos 2\pi f_c t$$

$$\boxed{\text{BPF o/p}} \rightarrow \boxed{y_2(t) = 2m(t)\cos 2\pi f_c t}$$

$$y_3(t) = \underbrace{m(t)}_{} + \underbrace{m(t)\cos 4\pi f_c t}_{\downarrow\ (LPF)_{output}}$$

$$\boxed{y_4(t) = m(t)}$$

**Example:** Let $c(t) = A_c \cos\left(2\pi f_c t\right)$ and $m(t) = \cos\left(2\pi f_m t\right)$. It is given that $f_c >> 5f_m$. The signal $c(t) + m(t)$ is applied to the input of a non – linear device, whose output $v_0(t)$ is related to the input $v_i(t)$ as $v_0(t) = av_i + bv_i^2(t)$, where $a$ and $b$ are positive constants. The output of the non-linear device is passed through an ideal band-pass filter with center frequency $f_c$ and bandwidth $3f_m$, to produce an amplitude modulated (AM) wave.

If it is desired to have the sideband power of the AM wave to be half of the carrier power, then $a / b$ is

(A) 0.25 (B) 0.5

(C) 1 (D) 2

**Solution:**

$$S_{AM}(t) = A_c'\left(1+K_a m(t)\right)\cos 2\pi f_c t$$

$$A_c' = a_0 A_c = aA_c$$

$$K_a = \frac{2a_1}{a_0} = \frac{2b}{a}$$

$$\mu = K_a \left|m(t)\right|_{max} = \frac{2b}{a}$$

$$P_{AM} = P_c'\left(1+\frac{\mu^2}{2}\right)$$

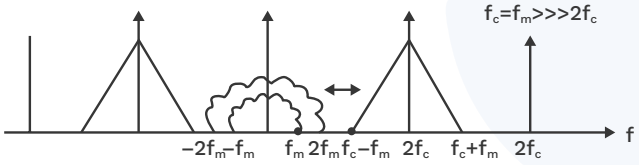
$$P_{SB} = \frac{P_c'}{2}$$

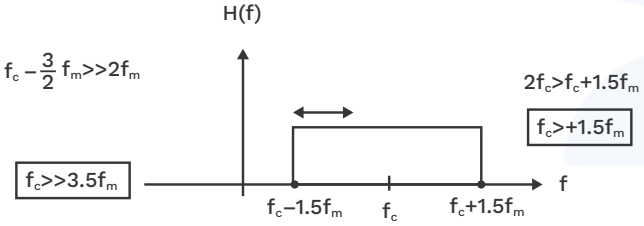
$$\frac{P_c'\mu^2}{2} = \frac{P_c'}{2}$$

$$\mu^2 = 1$$

$$\frac{4b^2}{a^2} = 1$$

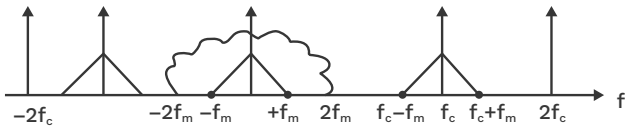
$\delta(t) =$

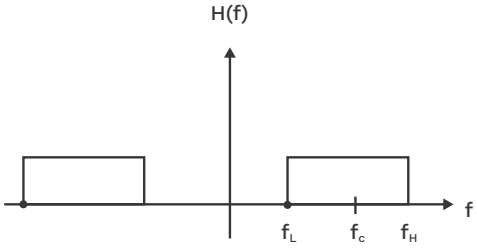



$$\left(f_c >>> 3.5f_m\right) \cap \left(f_c - f_m >>> 2\right) \Rightarrow f_c >>> 3.5f_m$$

**Note: Square LAW modulator**




1. $\left(f_c - f_m >> 2f_m\right)$
2. $2f_m < f_L < f_c - f_m$
3. $f_c + f_m < f_H < 2f_c$

**Example:** The unmodulated carrier power in an AM transmitter is 5 kW. This carrier is modulated by a sinusoidal modulating signal. The maximum percentage of modulation is 50%. If it is reduced to 40%, then the maximum unmodulated carrier power (in kW) that can be used without overloading the transmitter is _____.

**Solution:**

$P_c = 5K$ watt

$\mu_{max} = 0.5$

$S-1$ $\quad P_{AM} = P_c\left(\frac{1+\mu^2}{2}\right)$

$$(P)_{max} = P_c\left(1+\frac{\mu_{max^2}}{2}\right)$$

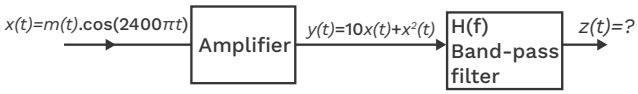
$$P_{max} = 5K\ \text{Watt}\left(1+\frac{(0.5)^2}{2}\right)$$

$$P_{max} = 5.625K\ \text{watt}$$

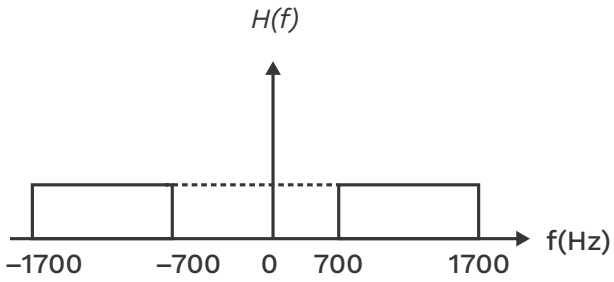
$S-2$ $\quad p_{max} = 5.625\times 10^3 = (P_c)_{max}\left(1+\frac{\mu_1^2}{2}\right)$

$$(P_c)_{max} = 5.208K\ \text{watt}$$

**Example:** In the system shown figure (A), m(t) is a low-pass signal with bandwidth *W* Hz. The frequency response of the band-pass filter *H*(*f*) is shown in figure (B). If it is desired that the output signal $z(t) = 10x(t)$, the maximum value of *W* (in Hz) should be strictly less than _____.

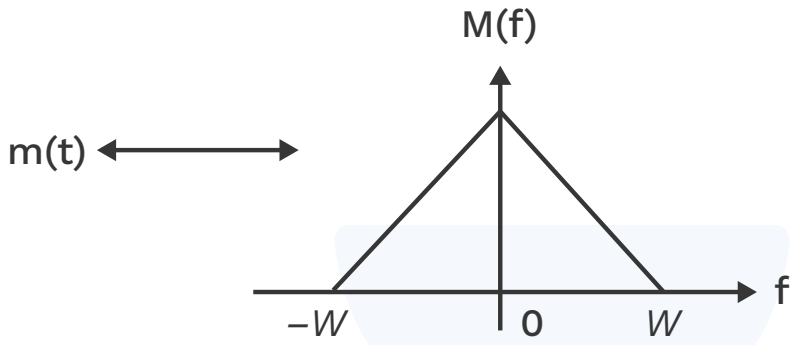


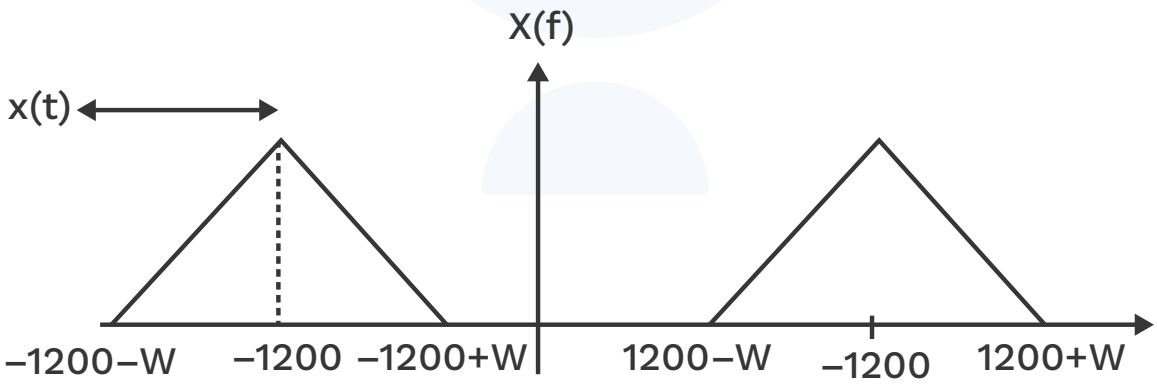

$$x(t) = m(t).\cos\left(2400\pi t\right)$$




**Solution:**

**Given:**



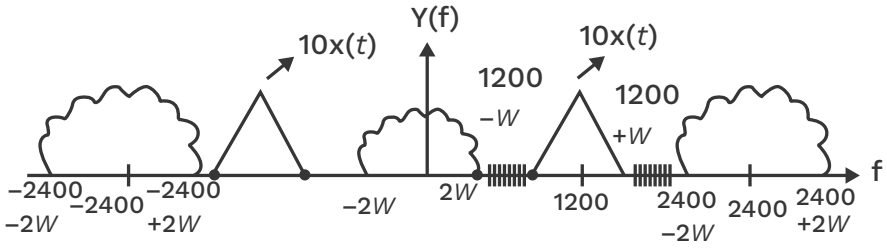

$$y(t) = 10x(t) + x^2(t)$$

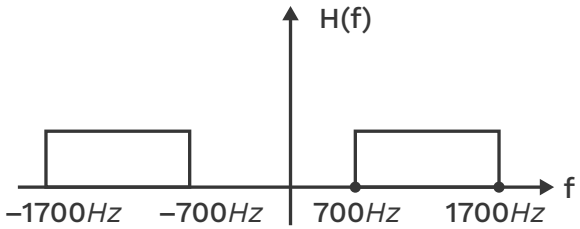
$$y(t) = 10m(t)\cos\left(2400\pi t\right) + m^2(t)\cos^2\left(2400\pi t\right)$$

$$y(t) = 10m(t)\cos 2400\pi t + \frac{1}{2}m^2(t)\cos\left(4800\pi t\right) + \frac{m^2(t)}{2}$$

**Note:**

'Spectrum of $x^2(t)$ cannot be directly obtained from spectrum of x(t) in this Case'.




1. $1200 - W > 2W$
2. $1200 + W < 2400 - 2W$
3. $2W < 700 < 1200 - W$ ↗ $2W < 700 \rightarrow W < 350$; ↘ $W < 500$
4. $1200 + W < 1700 < 2400 - 2W$ ↗ $W < 500$; ↘ $2W < 700$

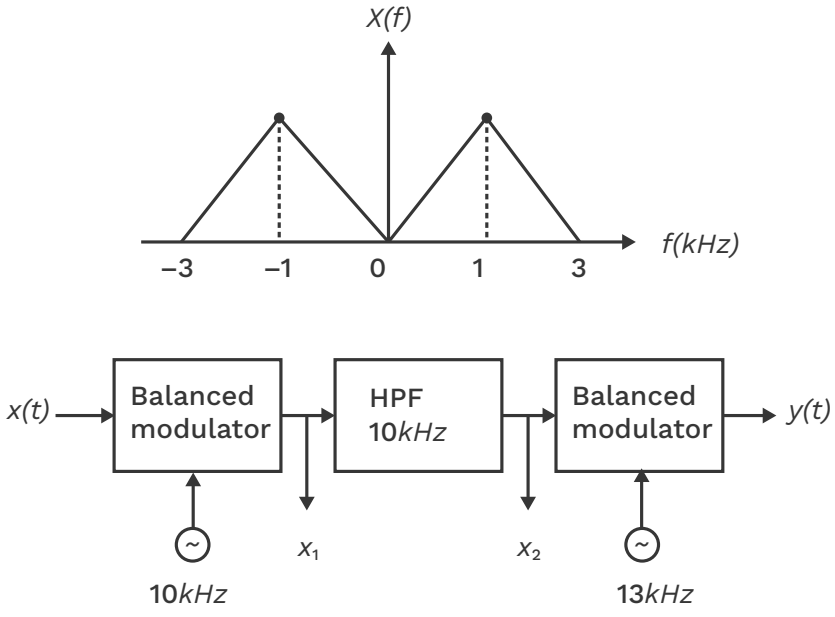
$\boxed{(1) \cap (2) \cap (3) \cap (4)}$

$(W < 400) \cap (W < 400) \cap \left[(W < 350) \cap (W < 500)\right]$

$\cap \left[(W < 350) \cap (W < 500)\right]$

$\boxed{W < 350Hz}$

**Example:** Consider a system shown in figure. Let $x(f)$ and $Y(f)$ denote the Fourier transforms of $x(t)$ and $y(t)$ respectively. The ideal HPF has the cut-off frequency 10kHz.



**Example:** The positive frequencies where $Y(f)$ has spectral peaks are

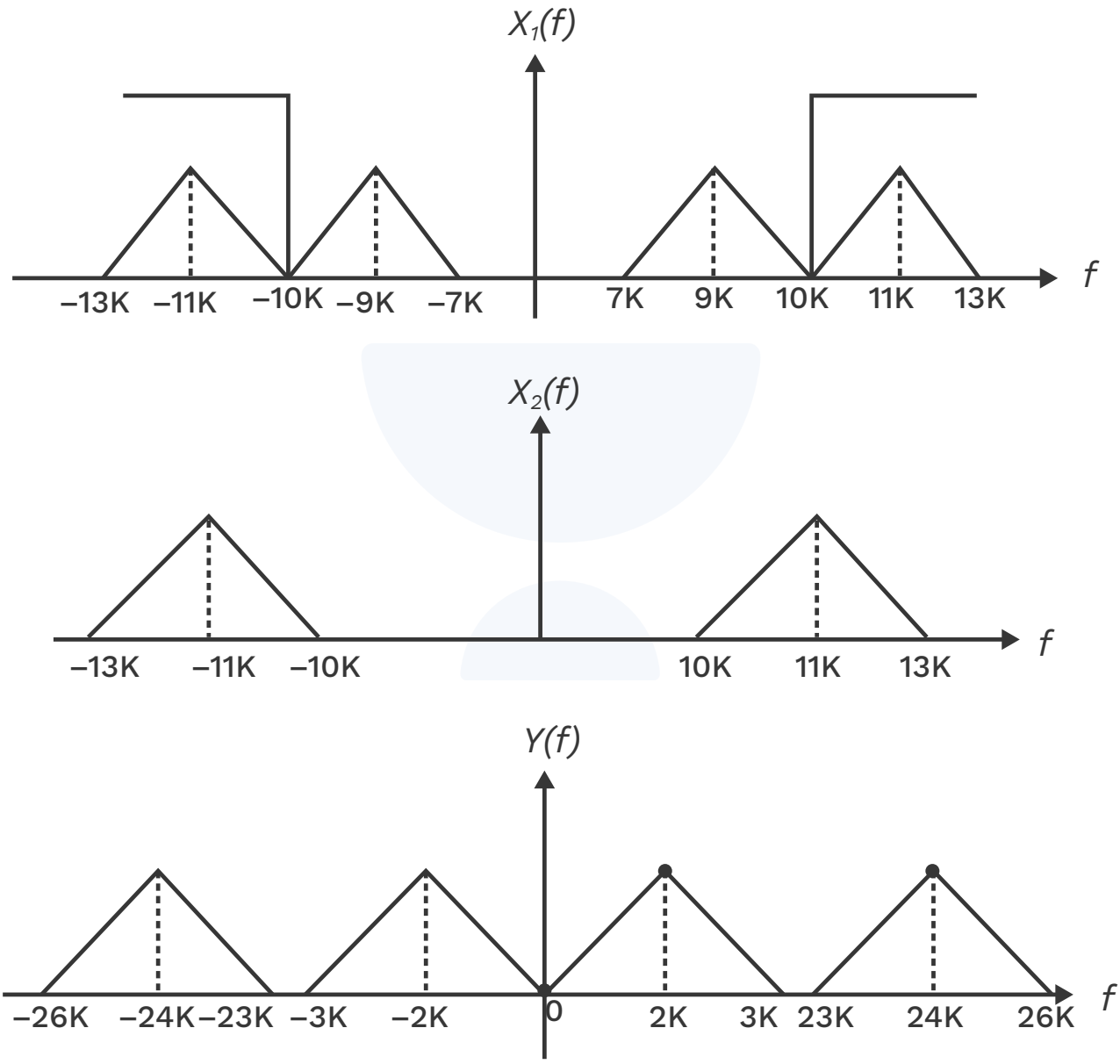
(A) 1 kHz and 24 kHz

(B) 2 kHz and 24 kHz

(C) 1 kHz and 14 kHz

(D) 2 kHz and 14 kHz

**Solution:**




**Example:** A DSB-SC signal is to be generated with a carrier frequency $f_c = 1\text{MHz}$ using a nonlinear device with the input–output characteristics $v_0 = a_0 v_i + a_1 v_i^3$ where $a_0$ and $a_1$ are constants. The output of the nonlinear device can be filtered by an appropriate band–pass filter. Let $v_i = A_c' \cos\left(2\pi f_c' t\right) + m(t)$ where $m(t)$ is the message signal. Then the value of $f_c'$ (in MHz) is

(A) 1.0

(B) 0.333

(C) 0.5

(D) 3.0

**Solution:**

$$\rightarrow \quad V_i = A_c' \cos 2\pi f_c' t + m(t)$$

$$V_0 = a_0 A_c' \cos 2\pi f_c' t + a_0 m(t) + a_1 \left[ \begin{array}{l} A_c' \cos^3 2\pi f_c' t + m^3(t) + 3A_c' m(t) \cos 2\pi f_c' t \\ \left[ m(t) + A_c' \cos 2\pi f_c' t \right] \end{array} \right]$$

$$V_0 = a_0 A_c' \cos 2\pi f_c' t + a_0 m(t) + a_1 A_c'^3 \cos^3 2\pi f_c' t + m^3(t) a_1 + a_1 3A_c' m^2(t) \cos 2\pi f_c' t + 3A_c'^2 m(t) \cos^2 2\pi f_c' t \; a_1$$

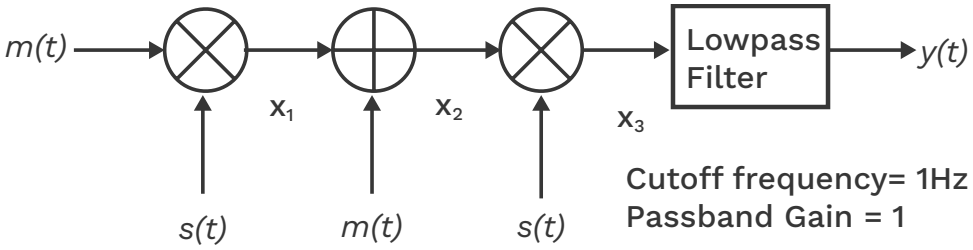
$$+ \frac{3A_c'^2 m(t)}{2} + \frac{3A_c'^2 m(t)}{2} \cos 4\pi f_c' t$$

Output BPF: $y(t) = \frac{3A_c'^2}{2} m(t) \cos\left(4\pi f_c' t\right)$

$2f_c' = 1MHz$

$f_c' = 0.5MHz$

**Example:** In figure $m(t) = \frac{2\sin 2\pi t}{t}$, $s(t) = \cos 200\pi t$ and $n(t) = \frac{\sin 199\pi t}{t}$, the output y(t) will be




(A) $\frac{\sin 2\pi t}{t} + \frac{\sin \pi t}{t} \cos 0.75\pi t$

(B) $\frac{\sin 2\pi t}{t} + \frac{\sin \pi t}{t} \cos 3\pi t$

(C) $\frac{\sin 2\pi t}{t} + \frac{\sin 0.5\pi t}{t} \cos 1.5\pi t$

(D) $\frac{\sin 2\pi t}{2t} + \frac{\sin \pi t}{t} \cos 0.75\pi t$

**Solution:**

$$x_1(t) = m(t)s(t) = \frac{2\sin 2\pi t}{t} \cos 200\pi t$$

$$= \frac{\sin(202\pi t)}{t} - \frac{\sin(198\pi t)}{t}$$

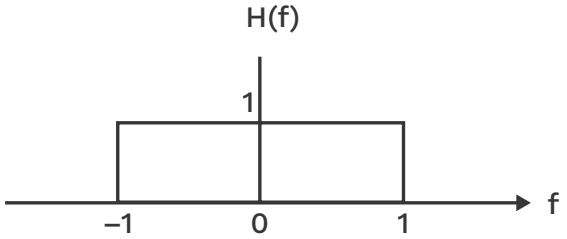
$$x_2(t) = \frac{\sin(202\pi t)}{t} - \frac{\sin(198\pi t)}{t} + \frac{\sin(199\pi t)}{t}$$

$$\rightarrow x_3(t) = x_2(t) \cos 200\pi t$$

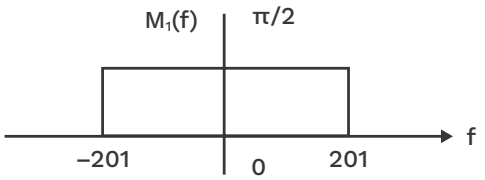
$$x_3(t) = \frac{\sin(202\pi t)}{t} \cos 200\pi t = \frac{1}{2t}\overset{(1)}{\sin}(402\pi t) + \frac{1}{2t}\overset{(2)}{\sin}(2\pi t) -$$

$$\frac{\sin 198\pi t}{t} \cos 200\pi t = -\frac{1}{2t}\overset{(3)}{\sin}(398\pi t) + \frac{1}{2t}\overset{(4)}{\sin}(2\pi t)$$

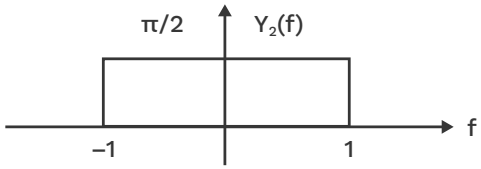
$$\frac{\sin(199\pi t)\cos 200\pi t}{t} = \frac{1}{2t}\overset{(5)}{\sin}(399\pi t) - \frac{1}{2t}\overset{(6)}{\sin}(\pi t)$$

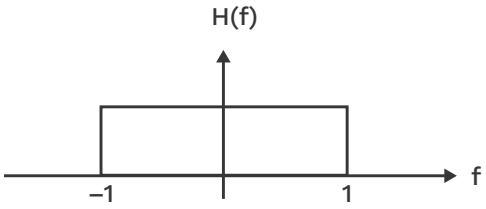


$$m_1(t) = \frac{\pi}{2}\left\{\frac{\sin(402\pi t)}{\pi t}\right\} \longleftrightarrow$$

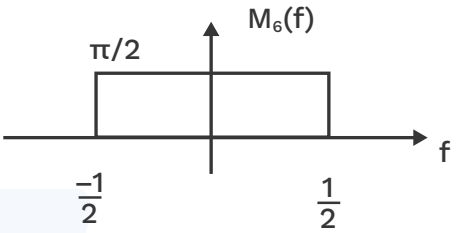



$$y_2(t) = \frac{1}{2}\frac{\sin 2\pi t}{2t} \longleftrightarrow$$

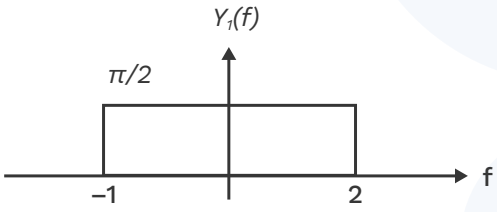


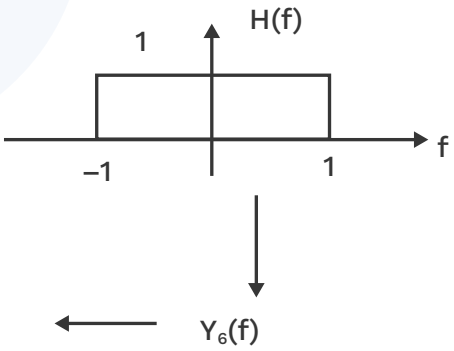




$$m_6(t) = \frac{\sin \pi t}{2t} \longleftrightarrow$$

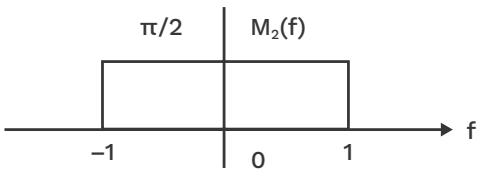



$$y_1(t) = \frac{\pi}{2}\left\{\frac{\sin(\overset{\nearrow a}{2}\pi t)}{\pi t}\right\} \longleftrightarrow$$

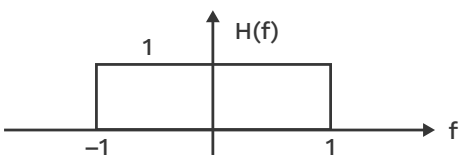



$$y_1(t) = \frac{1}{2}\frac{\sin(2\pi t)}{t}$$




$$m_2(t) = \frac{\sin 2\pi t}{2t} = \frac{\pi}{2}\left\{\frac{\sin 2\pi t}{\pi t}\right\} \longleftrightarrow$$




$$y_6(t) = \frac{\sin \pi t}{2t}$$

$$y(t) = \frac{1}{2}\frac{\sin(2\pi t)}{t} - \frac{1}{2}\frac{\sin(2\pi t)}{t} + \frac{1}{2}\sin\frac{(2\pi t)}{t}$$

$$y_1(t) \quad - \quad y_3(t) \quad + \quad y_5(t)$$

$$+\frac{1}{2}\frac{\sin(2\pi t)}{t} + \frac{1}{2}\frac{\sin(2\pi t)}{t} - \frac{\sin \pi t}{2t}$$

$$y_2(t) \qquad y_4(t) \quad y_c(t)$$

$$y(t) = \frac{3}{2}\frac{\sin(2\pi)t}{t} - \frac{\sin \pi t}{2t}$$

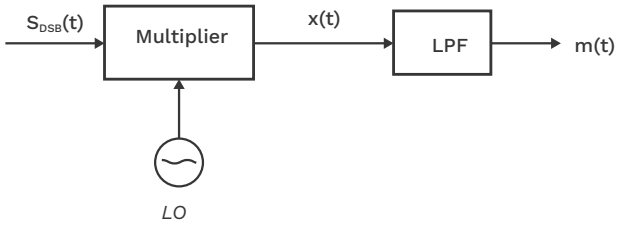
$$y(t) = \frac{\sin(2\pi t)}{t} + \frac{1}{2}\frac{\sin 2\pi t}{t} - \frac{\sin \pi t}{2t}$$

$$y(t) = \frac{\sin 2\pi t}{t} + \frac{1}{2t}[2\cos(15\pi t)\sin(0.5\pi t)]$$

$$\boxed{y(t) = \frac{\sin 2\pi t}{t} + \frac{\cos(15\pi t)\sin(0.5\pi t)}{t}}$$

## Demodulation of DSB-SC

**Synchronous detector:**




Now, $s_{DSB}(t) = A_c m(t)\cos 2\pi f_c t$

**Case 1:**

Assuming that the local oscillator is in perfect synchronisation with the carrier signal, so it's output is $A_c \cos 2\pi f_c t$ .

Now the multiplier output is:

$$x(t) = s_{DSB}(t) \times (\text{output of L.O.})$$

$$= (A_c m(t)\cos 2\pi f_c t) \times (A_c \cos 2\pi f_c t)$$

$$= A_c^2 \times m(t)\cos^2 2\pi f_c t$$

$$= \frac{A_c^2 m(t)}{2}\left\{1 + \cos 4\pi f_c t\right\}$$

When it is passed through a LPF the output is $= \frac{A_c^2 m(t)}{2}$ and finally it is passed through an amplifier to given message signal $m(t)$.

**Case 2:**

Let the local oscillator output not be in phase synchronisation. In this case the output is not $c(t)$; it can be written as: $A_c \cos(2\pi f_c t + \phi)$

Now, the multiplier output is

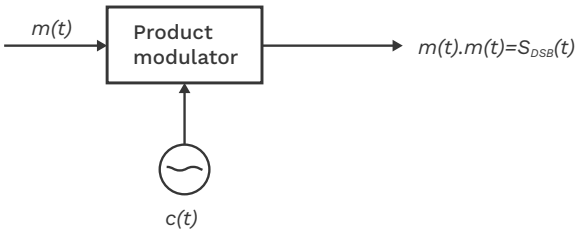
$$x(t) = A_c m(t)\cos 2\pi f_c t \times A_c \cos(2\pi f_c t + \phi)$$

$$= A_c^2 m(t)\cos 2\pi f_c t \cos(2\pi f_c t + \phi)$$

$$= \frac{A_c^2 m(t)}{2}\cos(4\pi f_c t + \phi) + \frac{A_c^2 m(t)}{2}\cos\phi$$

The output of the LPF is $\frac{A_c^2 m(t)}{2}\cos\phi$

- To reconstruct the message signal m(t), $\phi$ has to be maintained as a constant, to do this an additional circuit is required which makes this detector complex.
- When $\phi = 90^0$, the output is zero (QNE)
- DSB-SC is used in Quadrature Carrier Multiplexing (QCM)
- The product modulator is the name given to any modulator capable of generating DSB-SC signal.



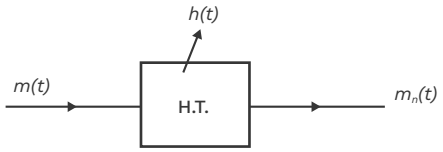

**Advantages of DSB–SC over AM:**

1. Compared to AM, it provides 100% modulation efficiency that is $\eta = 1$.
2. Due to suppression of carrier, the power is saved in DSB-SC.

**Hilbert Transform**

1. It converts time domain signal into time domain signal.
2. Hilbert transformer is an LTI system

$$\boxed{h(t) = \frac{1}{\pi t}}$$



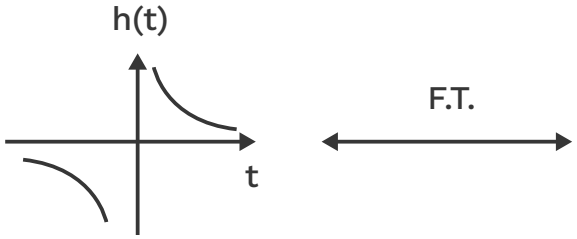

$$m_h(t) = m(t) * h(t)$$

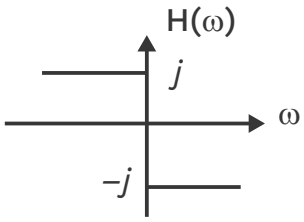
$$\boxed{m_h(t) = m(t) * \frac{1}{\pi t}}$$

$$\boxed{m_h(t) = \int_{-\infty}^{\infty} \frac{m(\tau)}{\pi(t-\tau)} d\varepsilon}$$

**Unit impulse response of HT:**

$$h(t)=\frac{1}{\pi t}\xleftrightarrow{F.T.}H(\omega)=-jsgn(\omega)=\begin{cases}-j & \omega>0\\ 0 & \omega=0\\ j & \omega<0\end{cases}$$




$$h(t)\neq 0 \ \ For \ \ t<0$$

Noncausal system:

$$H(\omega)=-j\,\mathrm{sgn}(\omega)=\begin{cases}-j & : \ \omega>0\\ 0 & : \ \omega=0\\ j & : \ \omega<0\end{cases}=\begin{cases}1e^{-j\pi/2} & \omega>0\\ 0 & \omega=0\\ 1e^{j\pi/2} & \omega<0\end{cases}$$

$$H(\omega)=\begin{cases}1 & : \ \omega>0\\ 0 & : \ \omega=0\\ +1 & : \ \omega<0\end{cases}\quad :\quad \angle\{H(\omega)=\begin{cases}-\pi/2 & \omega>0\\ 0 & \omega=0\\ \pi/2 & \omega<0\end{cases}$$

$$h(t)=\frac{1}{\pi t}\longleftrightarrow H(f)=-j\,\mathrm{sgn}[f]$$

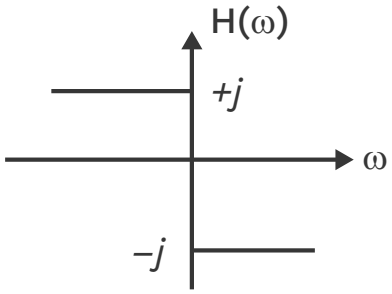
$$\mathrm{sgn}(at)=\mathrm{sgn}(t)\qquad u(at)=u(t)$$

$$\mathrm{sgn}[\omega]=u(\omega)-u(-\omega)=2u(\omega)-1$$

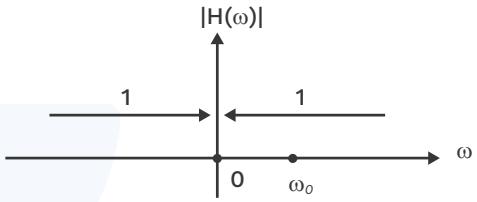
$$\mathrm{sgn}(2\pi f)=u(2\pi f)-u(-2\pi f)=2u(2\pi f)-1$$

$$\mathrm{sgn}(2\pi f)=u(f)-u(-f)=2u(f)-1=\mathrm{sgn}(f)$$

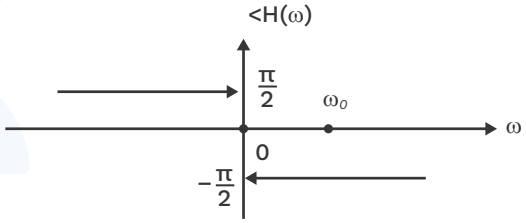
**1. Amplitude spectrum:**

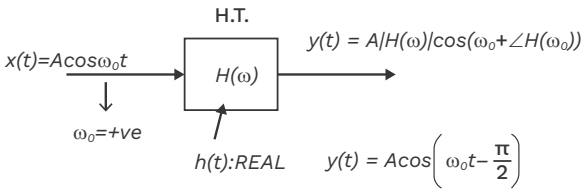



**2. Magnitude spectrum:**



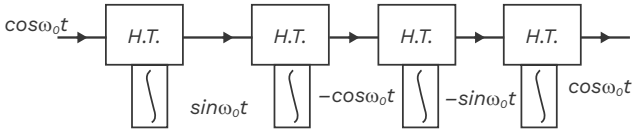

**3. Phase spectrum:**




**Note:** 'HT is $-90^0$ phase shifter'

**HT of sinusoids:**

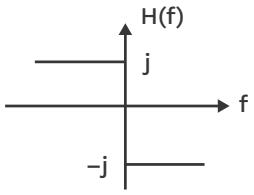



**HT of non-sinusoidal signal:**

$m(t)\rightarrow$ non-sinusoidal

# $m_h(t) = m(t) * \frac{1}{\pi t} = \int_{-\infty}^{\infty} \frac{m(\tau)}{\pi(t-\tau)} d\tau$

$h(t) = \frac{1}{\pi t} \longleftrightarrow$




# $M_h(\omega) = M(\omega)\left[-j\,\text{sgn}(\omega)\right]$

$\updownarrow$ *I.F.T.*

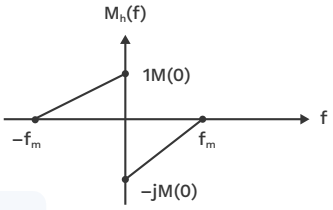
$m_n(t)$

**Example:** $m(t) = \delta(t-2) \overset{HT}{\longleftrightarrow} ?$

$$m_h(t) = m(t) * \frac{1}{\pi t} = \delta(t-2) * \frac{1}{\pi t} = \frac{1}{\pi(t-2)}$$

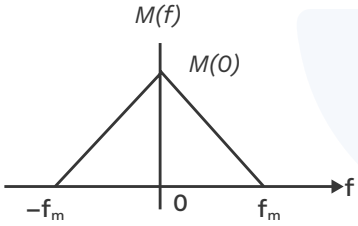
$$\delta(t-a) * x(t-b) = x(t-a-b)$$

$m_h(t) = m(t) * \frac{1}{\pi t} \longleftrightarrow$




→ **HT of baseband signals:**

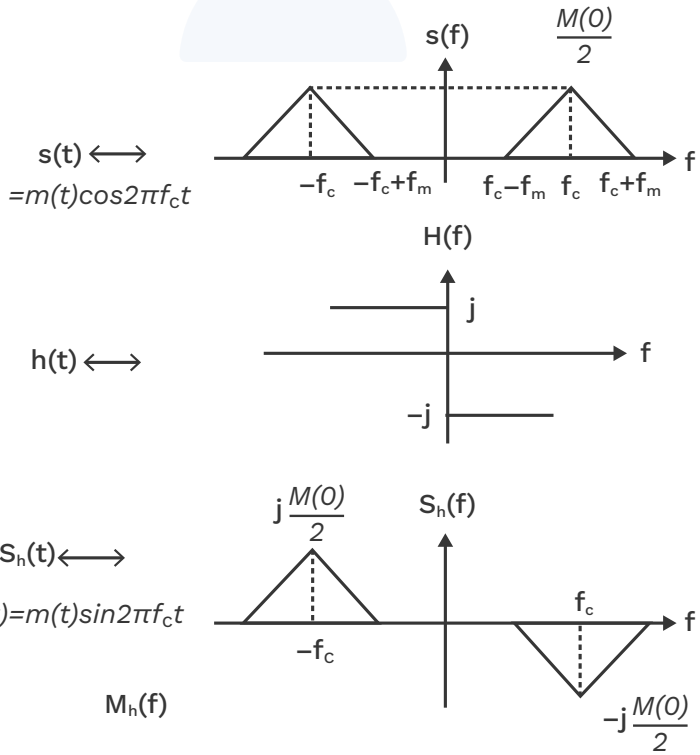
$m(t) \overset{FT}{\longleftrightarrow}$




**HT of band pass signals:**

$s(t) = m(t)\cos 2\pi f_c t \longleftrightarrow S_h(t) = ?$

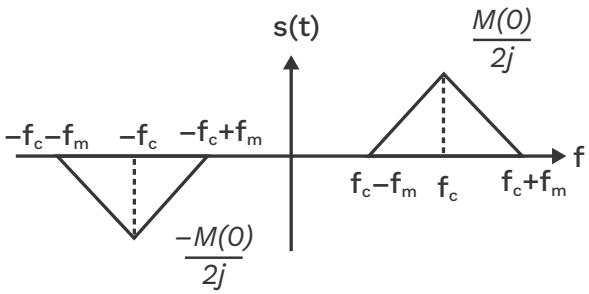
$s_h(t) = s(t) * \frac{1}{\pi t} \longleftrightarrow S_h(f) = \underline{S(f)}\,[-j\,\text{sgn}[f]]$



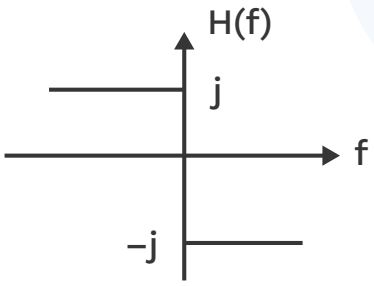
$$s(t) = m(t)\sin 2\pi f_c t \longleftrightarrow s_h(t) = ?$$

$$s(t) \longleftrightarrow S(f) = \frac{M(f - f_c)}{2j} - \frac{M(f + f_c)}{2i}$$

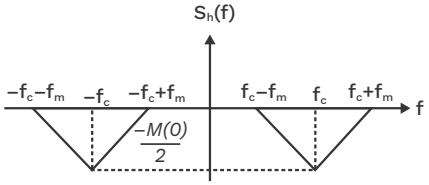
$$s_h(t) = s(t) * \frac{1}{\pi t} \longleftrightarrow S_h(f) = S(f)[-j\,\text{sgn}[f]]$$




$$s(t) \longleftrightarrow$$
$$= m(t)\sin 2\pi f_c t$$



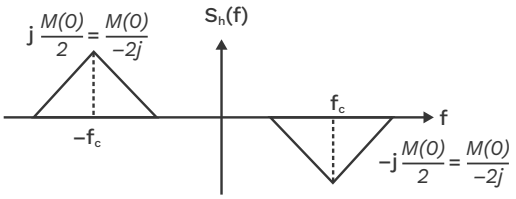

$$h(t) \longleftrightarrow$$



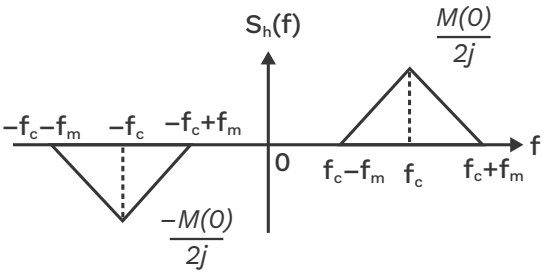

$$s_h(t) \longleftrightarrow$$
$$= s(t) * h(t)$$
$$= -m(t)\cos 2\pi f_c t$$

**Important Observation**

$$S(t) = m(t)\cos 2\pi f_c t \longleftrightarrow S_h(t) = S(t) * \frac{1}{\pi t}$$




$$s_h(t) = m(t)\sin 2\pi f_c t \quad \xleftrightarrow{I.F.T.}$$




## Conclusions

1. $m(t)$ : BASE BAND

$$\boxed{m_h(t) = m(t) * \frac{1}{\pi t}} \text{ or } \boxed{M_h(f) = M(f)\left[-j\,\text{sgn}(f)\right]}$$

2. $m(t)$ : BASE BAND

a) $S(t) = m(t)\cos 2\pi f_c t \rightarrow$ Band pass

$$\boxed{\begin{aligned} s_h(t) &= HT\left[s(t)\right] = HT\left[m(t)\cos 2\pi f_c t\right] \\ &= m(t)HT\left[\cos 2\pi f_c t\right] \end{aligned}}$$

b) $s(t) = m(t)\sin 2\pi f_c t \rightarrow$ BANDPASS

$$\boxed{\begin{aligned} s_h(t) &= HT\left[s(t)\right] = HT\left[m(t)\sin 2\pi f_c t\right] \\ &= m(t)HT\left[\sin 2\pi f_c t\right] \end{aligned}}$$

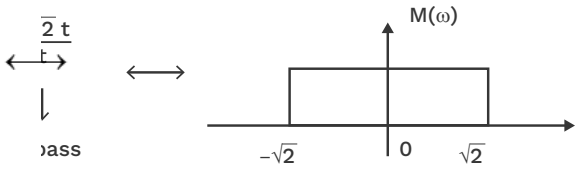
**Example:** $x(t) = \left(\frac{\sin\sqrt{2}t}{\pi t}\right)\sin\omega_c t$ . Calculate HT of $x(t) = ?$

**Solution:**

$$x(t) = \left(\frac{\sin\sqrt{2}t}{\pi t}\right)\underbrace{\sin\omega_c t}_{\substack{\text{High frequency} \\ \text{carrier}}} = m(t)\sin\omega_c t$$

$\downarrow$

Lowpass




$$x_h(t) = HT\left[x(t)\right] = HT\left[m(t)\right]\sin 2\pi f_c t$$

$$x_h(t) = m(t)HT\left[\sin 2\pi f_c t\right] = m(t)\left[-\cos 2\pi f_c t\right]$$

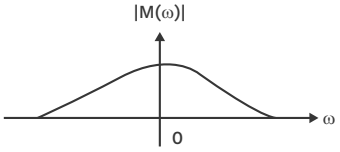
$$\boxed{x_h(t) = \frac{-\sin\sqrt{2}t}{\pi t}\cos 2\pi f_c t}$$

**Example:** $s(t) = \underbrace{e^{-at}u(t)}_{\text{Lowpass}}\underbrace{\cos\left(\omega_c t + \Delta\omega + +\phi\right)}_{\text{High Frequency carrier}}$

'Calculate HT of s(t)'

$$m(t) = \underset{\text{Low pass signal}}{e^{-at}}\,u(t) \longleftrightarrow M(f) = \frac{1}{a + j\omega}$$

$$\left|M(\omega)\right| = \frac{1}{\sqrt{a^2 + \omega^2}}$$




$$\boxed{s_h(t) = e^{-at}u(t)\sin\left(\omega_c t + \Delta\omega t + \phi\right)}$$

**Properties of HT**

1. It is $-90^0$ phase shifter.
2. It is noncausal LTI system.
3. Magnitude response of H.T. resembles the magnitude response of A.P.F. except it does not allow 0Hz.
4. $\boxed{\begin{aligned} x(t) &\xleftrightarrow{HT} x_h(t) \\ x_h(t) &\xleftrightarrow{HT} -x(t) \end{aligned}}$
5. Magnitude spectrum of x(t) and $x_h(t)$ will be same

   $$X_h(\omega) = X(\omega)\left[-j\,\text{sgn}(\omega)\right]$$

   $\boxed{\left|X_h(\omega)\right| = \left|X(\omega)\right|}$ $\boxed{1 - \left|\text{sgn}(\omega)\right| = 1}$
6. If x(t): Band limited then $x_h(t)$ is also band limited.
7. If x(t): Periodic $\longrightarrow x_h(t)$ is also periodic
8. If x(t): Nonperiodic $\longrightarrow x_h(t)$ is also nonperiodic
9. x(t) and $x_h(t)$ are orthogonal signals
10. x(t): Energy signal $\longrightarrow x_h(t)$: Energy signal

    $x(t)$ and $x_h(t)$ will have same

    a) Energy

    b) ESD

    c) Autocorrelation function
11. x(t): Power signal $\longrightarrow x_h(t) \rightarrow$ Power signal

    $x(t)$ and $x_h(t)$ will have same

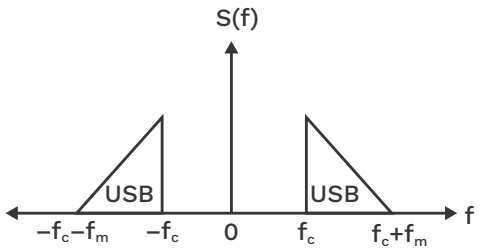
    a) Power

    b) PSD

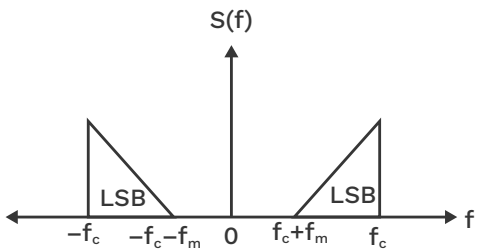
    c) Average ACI
12. $\underset{\text{Lowpass}}{m}(t)\underset{H.F.C.}{\cos\omega_c t} \xrightarrow{HT} m(t)HT\left(\cos\omega_c t\right)$

**Single-sideband Supressed Carrier Modulation (SSB-SC)**

- In single-sideband modulation, only upper or lower side band is transmitted. As only one sideband is transmitted, it reduces the bandwidth requirement.




OR




Spectrum for SSB-SC

- 50% of channel bandwidth will be saved in SSB-SC as compared to DSB-SC & AM.
- 50% of power will be saved in SSB-BC as compared to DSB-SC.

**SSB-SC (Single sideband suppressed carrier):**

1. Point-point communication
2. Two methods of Generation
   a) Phase discrimination method
   b) Frequency discrimination method

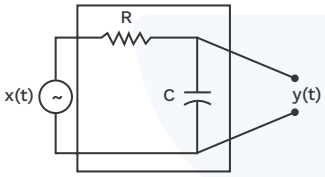
**Phase Discrimination Method**

1. It is valid only for single tone sinusoidal message signal.
2. Because it is practically not possible to design -90° wideband phase shifter (-90° wideband HT)

$$H(s)=\frac{1}{1+SRC}$$

$$H(\omega)=\frac{1}{1+j\omega RC}$$

$$\angle H(\omega)=-\tan^{-1}\omega RC$$




**Case 1:** Constant -90° phase shift

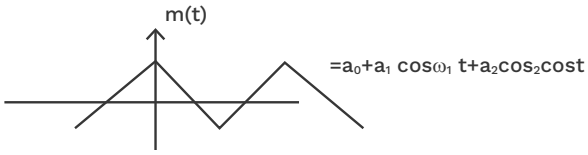
$$x(t)=A\sin\omega_0 t \Rightarrow y(t)=A|H(\omega_0)|\sin(\omega_0 t+\angle H(\omega_0))$$

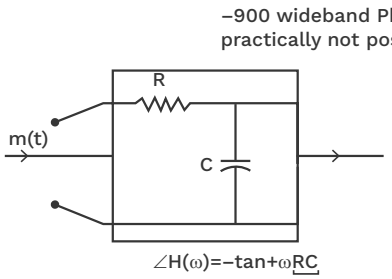
Single frequency $\angle H(\omega_0)=-\tan^{-1}\omega_0 RC$

$RC\uparrow\uparrow\rightarrow -90^0$

R→choose C→choose

**Case 2:**




**Note:** 'A constant phase shifter can be practically designed for single tone sinusoidal inputs'.

**General Expression of SSB-SC**

DSB-FC: $S(t)=[A_c+m(t)]\cos\omega_c t$

DSB-SC: $S(t)=A_c m(t)\cos\omega_c t$

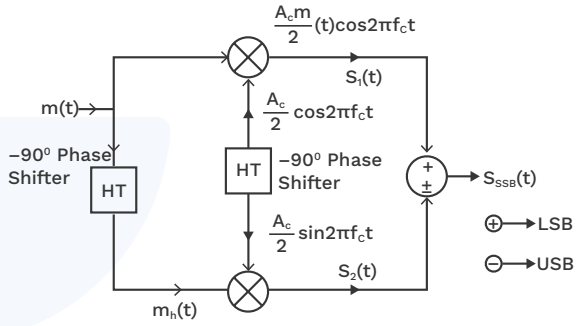
SSB-SC:

$$S(t)=\frac{A_c m(t)}{2}\cos 2\pi f_c t \pm \frac{A_c m_h(t)}{2}\sin 2\pi f_c t$$

$m_h(t)=HT(m(t))$ +→LSB

–→USB

**Phase discrimination setup:**



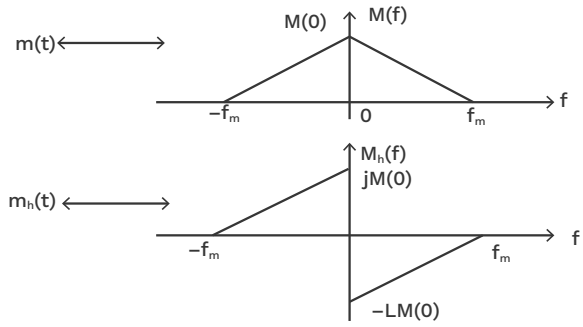

$$S(t)=\frac{m(t)A_c}{2}\cos 2\pi f_c t \pm \frac{m_h(t)A_c}{2}\sin 2\pi f_c t$$

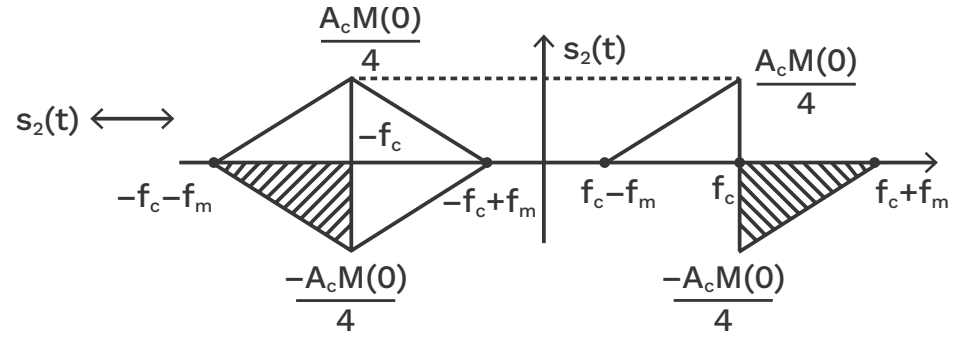
$$S_1(t)=\frac{A_c}{2}m(t)\cos 2\pi f_c t \longleftrightarrow S_1(f)$$

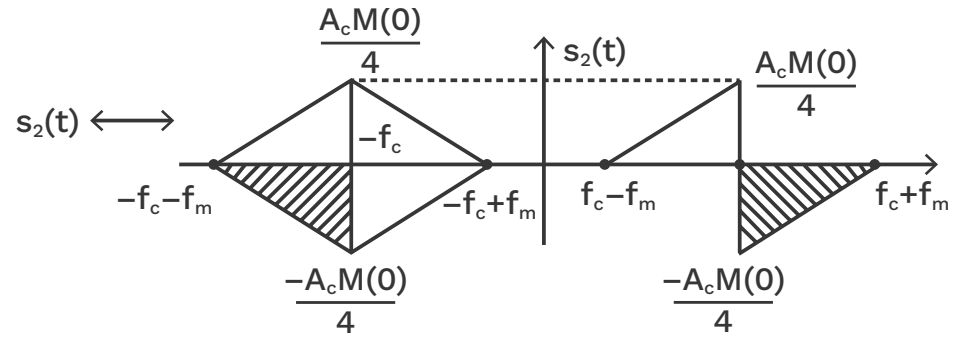
$$=\frac{A_c}{4}[M(f-f_c)+M(f+f_c)]$$

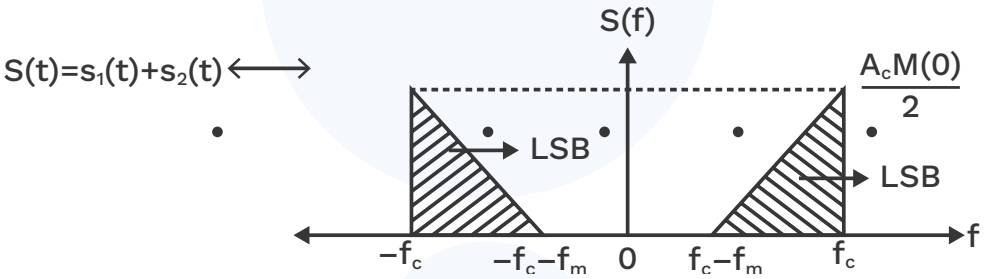
$$S_2(t)=\frac{A_c}{2}m_h(t)\sin 2\pi f_c t \longleftrightarrow$$

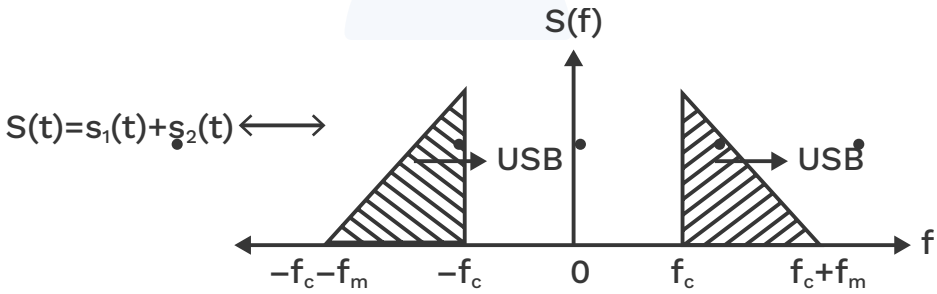
$$S_2(f)=\frac{A_c}{4j}[M_h(f-f_c)-M_h(f+f_c)]$$





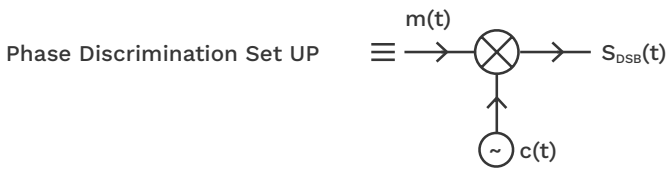



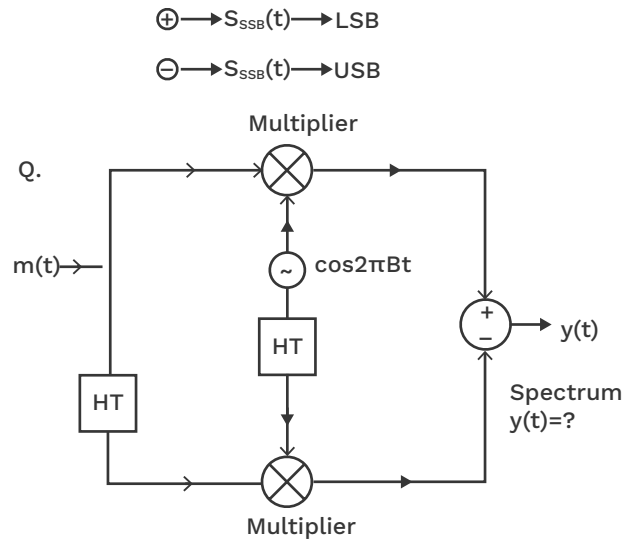

**Problems Solving**

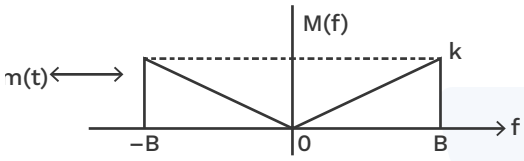
S.1 Identify the phase discrimination setup

S.2



S.3 Phase discrimination setup:

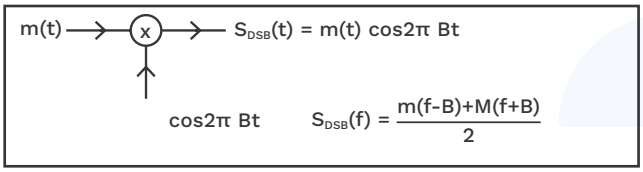


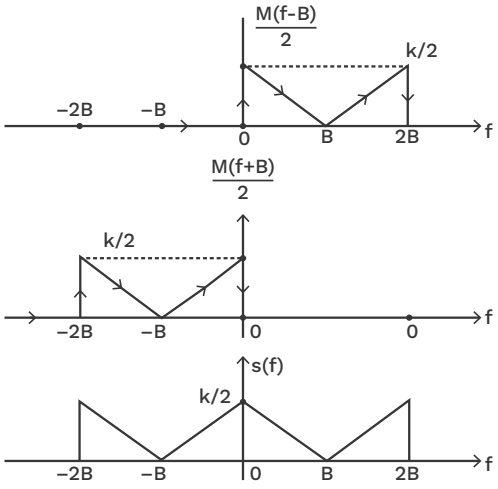




**Solution:**

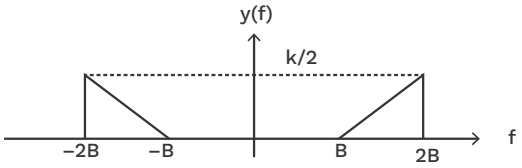
S.1 Phase discrimination method

S.2

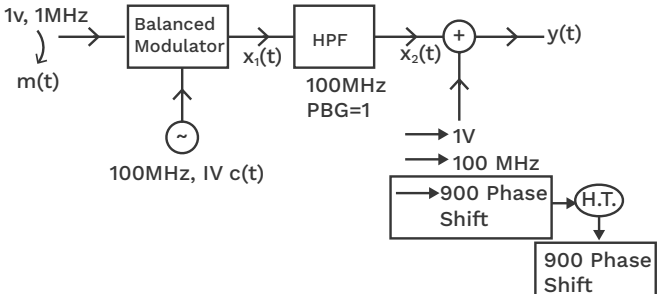



S.3 –Sign→USB




**Solution:**




Envelope of y(t) =?

**Solution:**

$m(t) = A_m \cos 2\pi f_m t \qquad A_m=1V, \qquad f_m=1MHz$

$C(t) = A_c \cos 2\pi f_c t \qquad A_c=1V, \qquad f_c=100\ MHz$

$$x_1(t) = \frac{A_c A_m}{2}\cos\left(2\pi \overbrace{(f_c+f_m)}^{101MHz} t\right) + \frac{A_c A_m}{2}\cos\left(2\pi \overbrace{(f_c-f_m)}^{99Mhz} t\right)$$

↓ *HPF*

$$x_2(t) = \frac{A_c A_m}{2}\cos\left(2\pi(f_c+f_m)t\right)$$

$$C_1(t) = A_c \cos\left(2\pi f_c t - 90^o\right) = A_c \sin 2\pi f_c t$$

$$y(t) = x_2(t) + c_1(t)$$

$$= \frac{A_c A_m}{2}\cos\left(2\pi(f_c+f_m)t\right) + A_c \sin 2\pi f_c t$$

$$y(t) = \frac{A_c A_m}{2}\cos 2\pi f_c t \cos 2\pi f_m t - \frac{A_c A_m}{2}\sin 2\pi f_c t \sin 2\pi f_m t + A_c \sin 2\pi f_c t$$

$$y(t) = \underbrace{\cos 2\pi f_c t}_{\cos\theta}\overbrace{\left[\frac{A_c A_m}{2}\cos 2\pi f_m t\right]}^{A} + \underbrace{\sin 2\pi f_c t}_{\sin\theta}\overbrace{\left(A_c - \frac{A_c A_m}{2}\sin 2\pi f_m t\right)}^{B}$$

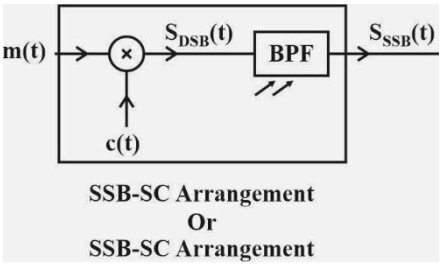
**Envelope**

$$E=\sqrt{\frac{A_c^2A_m^2}{4}\cos^2 2\pi f_m t + A_c^2 + \frac{A_c^2A_m^2}{4}\sin^2 2\pi f_m t - A_c^2A_m \sin 2\pi f_m t}$$

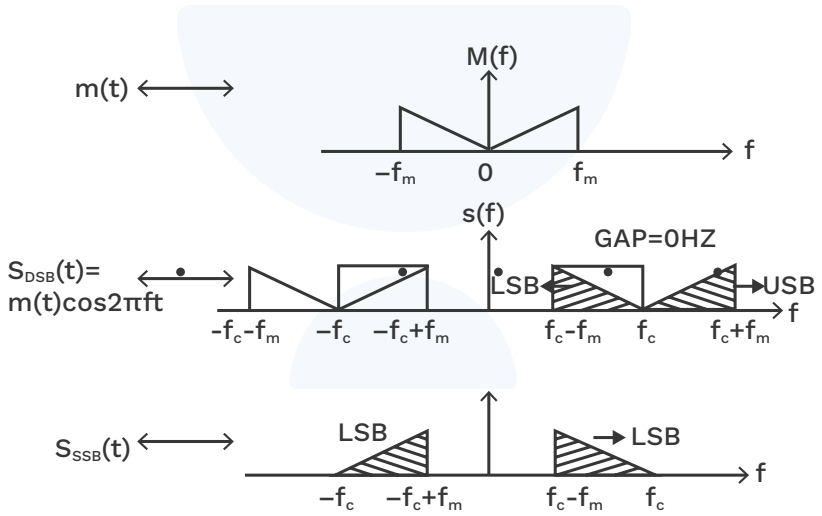
$$E=\sqrt{\frac{A_c^2A_m^2}{4} + A_c^2 - A_c^2A_m \sin 2\pi f_m t} \quad A_c = 1 \;\; A_m = 1$$

$$E = \sqrt{\frac{5}{4} - \sin\left(2\pi x 10^6 t\right)}$$

**Frequency Discrimination Method**




SSB-SC Arrangement
Or
SSB-SC Arrangement



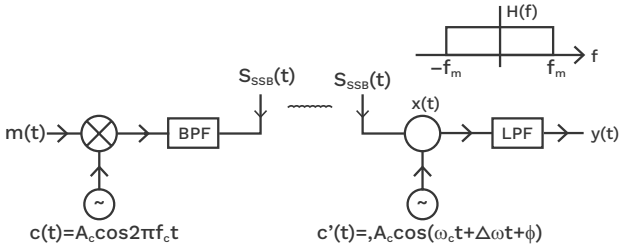

SSB-SC: Modulator → SSB-SC arrangement

| Spectral GAP in DSB | BPF | Signal |
|---|---|---|
| 0 Hz | Ideal | SSB-SC |
| 0 Hz | Practical | VSB-SC |
| ≠ 0 Hz | Ideal | SSB-SC |
| ≠ 0 Hz | Practical | *Depends on practical BPF* ↗ *VSB – SC*, ↘ *SSB – SC* |

Let SSB-SC has been generated:

Then synchronous detector can be used as demodulator.

**Synchronous detector:**




$$S_{SSB}(t)=\frac{A_c}{2}m(t)\cos\omega_c t \pm \frac{A_c}{2}m_h(t)\sin\omega_c t$$

$$x(t)=S_{SSB}(t)\cos(\omega_c t+\Delta\omega t+\phi)A_c'$$

$$x(t)=\frac{A_c' A_c}{4}m(t)\cos(2\omega t+\Delta\omega t+\phi)+$$
$$\frac{A_c A_c'}{4}\cos(\Delta\omega t+\phi)m(t)$$
$$\pm$$
$$\frac{A_c A_c'}{4}m_h(t)\sin(2\omega_c t+\Delta\omega t+\phi)$$
$$-\frac{A_c A_c'}{4}\sin(\Delta\omega t+\phi)m_h(t)$$

$$y(t)=\frac{A_c A_c'}{4}m(t)\cos(\Delta\omega t+\phi)$$
$$\mp\frac{A_c A_c'}{4}m_h(t)\sin(\Delta\omega t+\phi)$$

1. Freq Sync $\boxed{\begin{matrix}\Delta\omega=0\\ \Delta\phi\neq 0\end{matrix}}$

$$y(t)=\underbrace{\frac{A_c A_c'}{4}m(t)\cos\phi}_{\downarrow msg}\mp\underbrace{\frac{A_c A_c'}{4}m_h(t)\sin\phi}_{\downarrow \text{Interference}}$$

$m(t)$: *Recovery not possible*

2.

$\phi=0$ $\quad\Delta\omega\neq 0$ *Phase* sync:

m(t): Recovery not possible

3. Perfect Sync: $\Delta\omega=0$ $\quad\Delta\phi=0$

$$y(t)=\frac{A_c A_c'}{4}m(t)\rightarrow\text{Recovery possible}$$

4. $\Delta\omega=0$ $\quad\Delta\phi=\pi/2$

$$y(t)=\mp\frac{A_c A_c'}{4}m_h(t)\rightarrow N_0 \quad Q.N.E.$$

| Scheme | $\Delta\omega$ | $\Delta\phi$ | $m(t)$ Recovery |
|---|---|---|---|
| *A.M.* $\rightarrow$ | 0 | $\Delta\phi\neq 0$<br>$\Delta\phi\neq(2n+1)\pi/2$ | *Yes* |
| *DSB – SC* $\rightarrow$ | 0 | $\Delta\phi\neq 0$<br>$\Delta\phi\neq(2n+1)\pi/2$ | *Yes* |
| *SSB – SC* $\rightarrow$ | 0 | $\Delta\phi=0$ | *Yes* |

**Advantages of SSB-SC modulation over AM:**

- It uses less power, i.e., power is saved.
- It uses less bandwidth, i.e., bandwidth is saved.
- No quadrature null effect

**Disadvantages of SSB-SC modulation:**

- It is limited for voice transmission
- Demodulation of SSB is complex

**Efficiency $(\eta)$**

$$Efficiency\ \eta=\frac{\text{modulated power}}{\text{total power}}$$

**Efficiency of AM $(\eta)$**

$$S_{AM}=A_c\left(1+k_a m(t)\right)\cos 2\pi f_c t$$

$$Total\ Power\ (P_t)=\frac{A_c^2}{2}\left[1+K^2{}_a P_m\right]$$

Where $P_m$ = Message signal power

$$P_t=\underbrace{\frac{A^2{}_c}{2}}_{P_c}+\underbrace{\frac{A^2{}_c}{2}K_a{}^2 P_m}_{P_{SB}}$$

$$\eta=\frac{P_{SB}}{P_t}=\frac{\frac{A^2{}_c}{2}K^2{}_a P_m}{\frac{A^2{}_c}{2}\left(1+K^2{}_a P_m\right)}=\frac{K^2{}_a P_m}{1+K^2{}_a P_m}$$

Efficiency of AM

$$\eta_{AM}=\frac{K^2{}_a P_m}{1+K^2{}_a P_m}$$

**For sinusoidal message signal:**

RMS value $= \frac{A_m}{\sqrt{2}}$

Power value, $P_m = \frac{A_m^2}{2}$

$$\eta_{AM} = \frac{K_a^2 P_m}{1 + K_a^2 P_m} \quad (\because K_a A_m = \mu)$$

$$= \frac{\frac{\mu^2}{2}}{1 + \frac{\mu^2}{2}} \Rightarrow \eta = \frac{\mu^2}{2 + \mu^2}$$

For maximum efficiency, $\mu = 1$

$$\eta = \frac{1}{2+1} = 0.33$$

$\eta_{percentage} = 33\%$ for sinusoidal message signal

**For rectangular message signal:**

$$R_{MS}\ value = A_m$$

$$P_m = A^2{}_m$$

$$\eta = \frac{K_a^2 A_m^2}{1 + K_a^2 A_m^2} = \frac{\mu^2}{1 + \mu^2}$$

For maximum efficiency $\mu = 1$

$$\eta = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

$\eta_{percentage} = 50\%$ for rectangular message signal

**Efficiency of DSB-SC:**

$$\eta = \frac{Power\ used}{Total\ power}$$

For DSB-SC

$$P_t = \frac{A_c^2 \mu^2}{4}$$

$$P_{used} = \frac{A_c^2 \mu^2}{4}$$

$$\eta = 100\%$$

**Efficiency of SSB-SC:**

$$P_t = \frac{A_c^2 \mu^2}{8}$$

$$P_{used} = \frac{A_c^2 \mu^2}{8}$$

$$\eta = \frac{\frac{A_c^2 \mu^2}{8}}{\frac{A_c^2 \mu^2}{8}} = 1$$

$$\eta = 100\%$$

**Percentage Power Saving**

$\Rightarrow \%$ **Power saving in DSB-SC compared to AM**

$$= \frac{Power\ \ saved}{Total\ \ power}$$

In DSB-SC carrier power is saved compared to AM. Hence,

$$\% \text{ Power saving} = \frac{P_c}{P_t} = \frac{P_c}{P_c\left(1 + \frac{\mu^2}{2}\right)}$$

$$\% \text{ Power saving} = \frac{2}{2 + \mu^2} \times 100 \left( \begin{matrix} DSB - SC \\ compared \\ to\ AM \end{matrix} \right)$$

$\Rightarrow \%$ **Power saving in SSB-SC compared to AM**

In SSB-SC carrier power and half of side band power is saved as compared to A.M signal

$$\%\text{Power saving} = \frac{Power\ Saved}{Total\ Power}$$

$$= \frac{P_c + \frac{P_{SB}}{2}}{P_t} = \frac{P_c + \frac{P_c \mu^2}{4}}{P_c\left(1 + \frac{\mu^2}{2}\right)}$$

$$\% \text{ Power saving} = \frac{4 + \mu^2}{4 + 2\mu^2} \times 100$$

## Test Your Understanding

**5.** Power of amplitude modulated signal is given by 100 watts. Find the amount of power saved if carrier and one of the sideband is suppressed.
$[given\ \mu = 0.8]$

**Solution: 87.88**

**6.** Given a message signal $m(t) = 4\ \sin 100\pi t$ is under amplitude modulation with efficiency of 25%. Calculate the value of amplitude sensitivity of modulator $x(t)$ =

**Solution: 0.204**

**Vestigial Sideband Modulation (VSB)**

As in case of SSB modulation when sideband is passed through the band pass filter, the band pass filter may not work perfectly in practice, as a result of this SSB modulated signal may not be recovered properly. Hence for solving this problem VSB (Vestigial side band) technique is used.

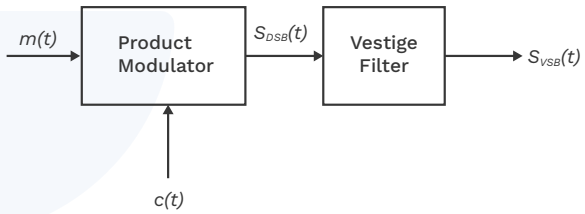
VSB (Vestigial side band) is the process where a part of the signal called as vestige is modulated along with one sideband.

This modulation technique is famous for video signal transmission.

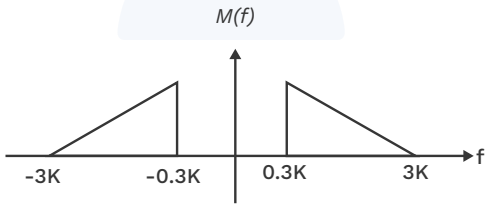
The bandwidth and power requirement of VSB signal is almost same as SSB-SC.

$$S_{VSB}(f) = S_{DSB}(f).H_V(F)$$

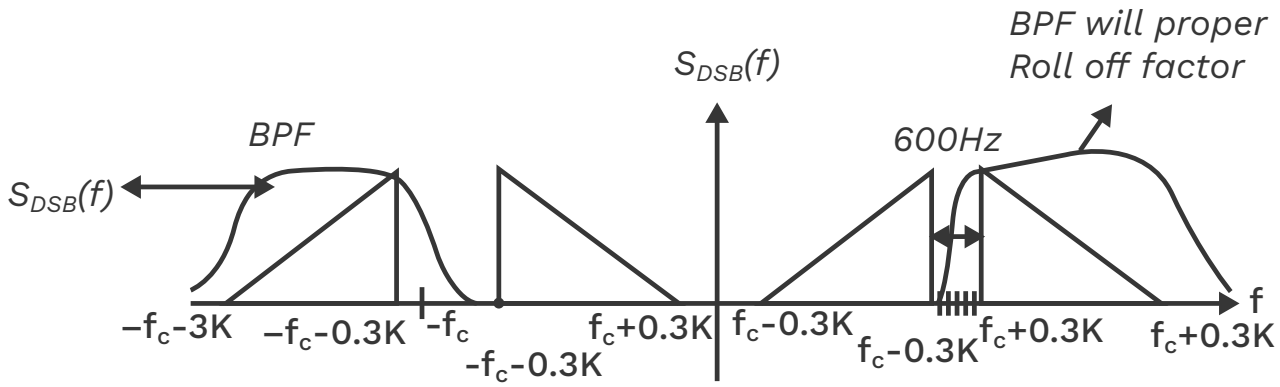
**Generation of VSB:**

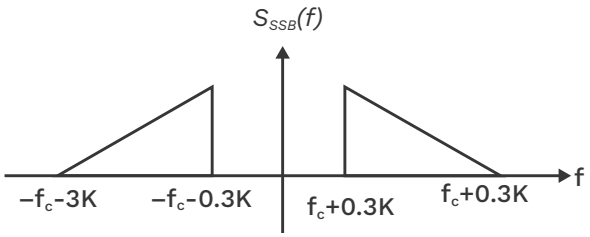



**Transmission of Voice Signal Through SSB-SC Modlt$^r$**

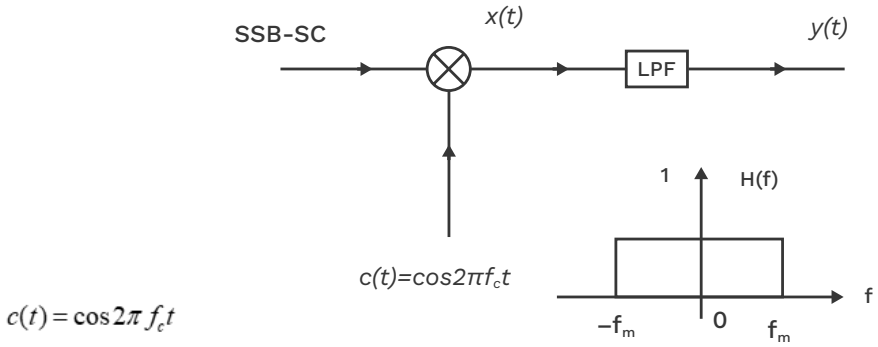


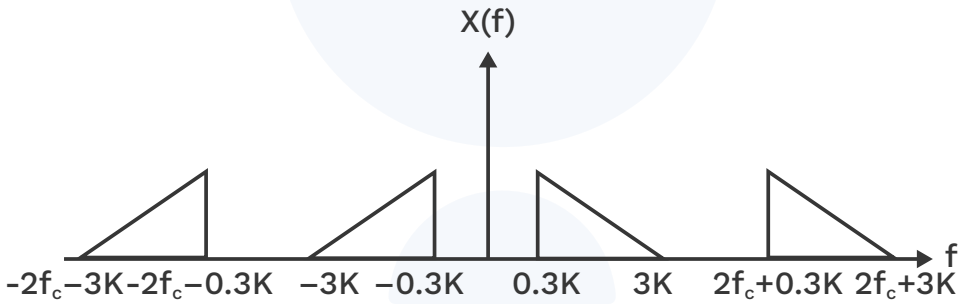

$m(t) \xleftrightarrow{F.T}$

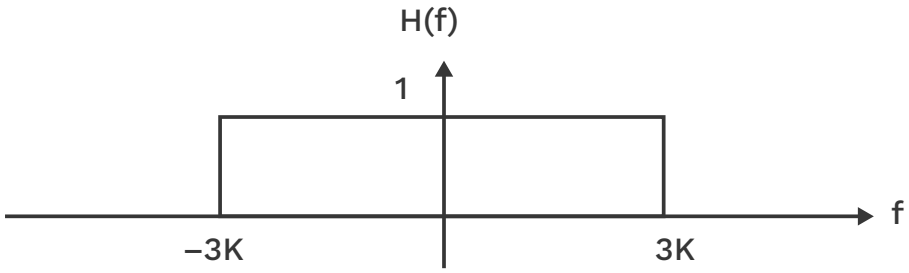


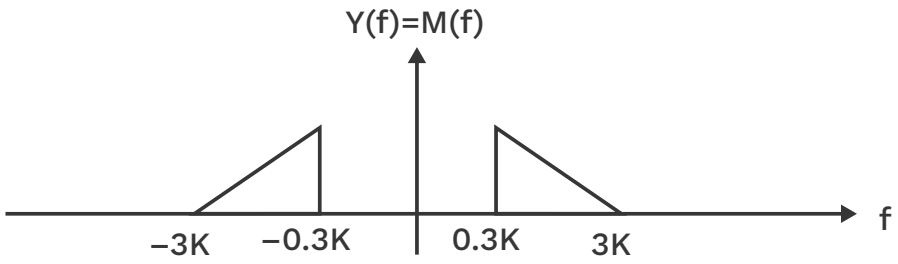



**Reception of voice signal through synchronous detector:**



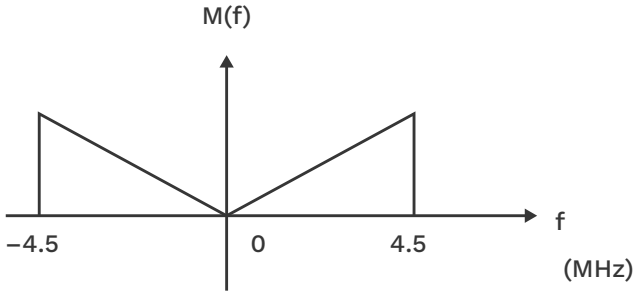



1. When voice signal is Txed through SSB-SC modulator → SSB-SC is obtained when BPF with proper Roll off factor is chosen (Spectral gap = 600Hz allows the above SSB-SC)
2. Synchronous detector recovers m(t) (voice) from above SSB-SC signal.

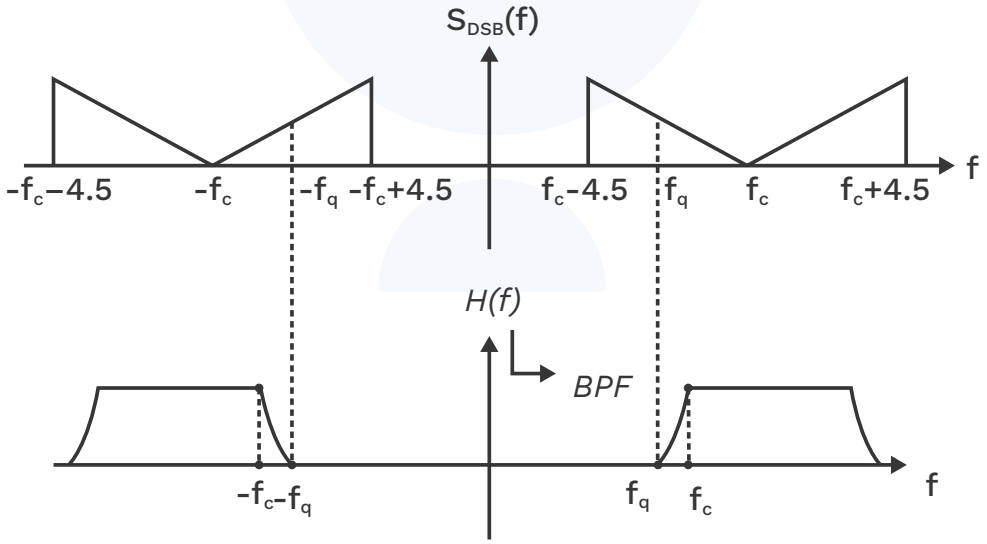
**Transmission of Video Signal through SSB-SC Modltr**

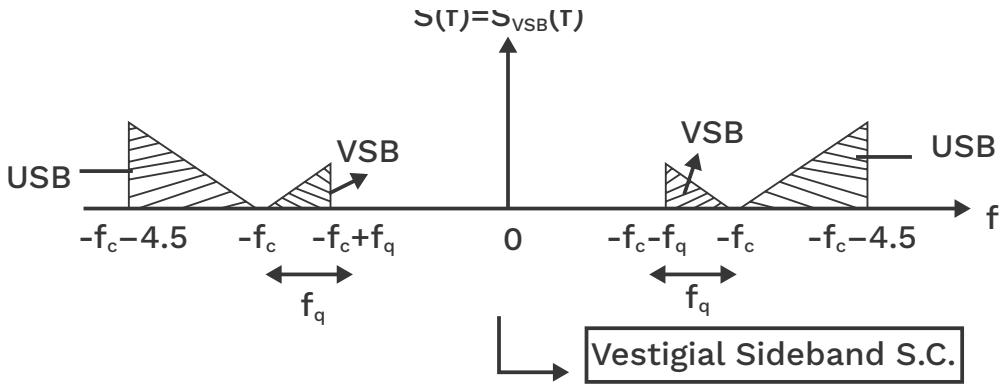
Tx:

$m(t) \longleftrightarrow$




$S_{DSB}(t) \longleftrightarrow$

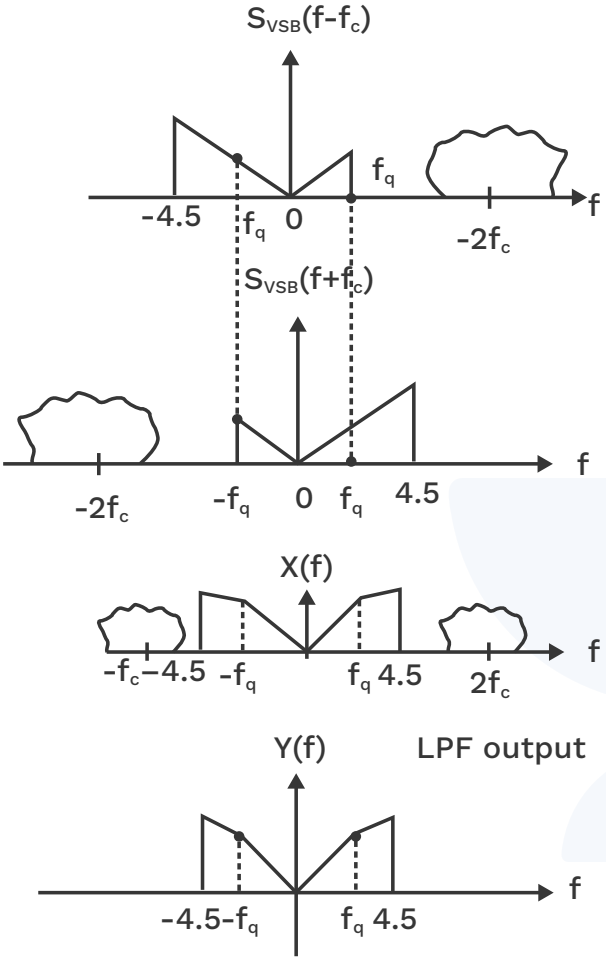


**Reception of video signal using synchronous detector:**

$$x(t) = S_{VSB}(t)\cos 2\pi fet$$

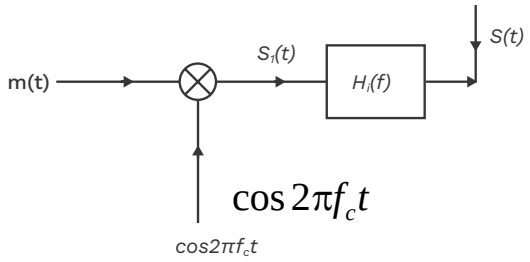
$$X(f) = \frac{S_{VSB}(f - fe) + S_{VSB}(f + fe)}{2} \neq M(f)$$




**Note:**

1. When video signal is Txed through SSB-SC modlt$^r$ → VSB-SC is generated.
2. Synchronous detector cannot recover m(t) (video signal) from above generated VSB-SC.

**Alternate Arrangement**

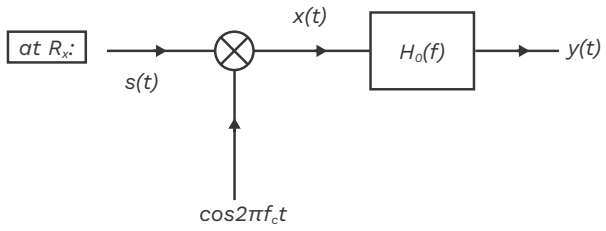



$$S_1(t) = m(t)\cos 2\pi f_c t \longleftrightarrow$$

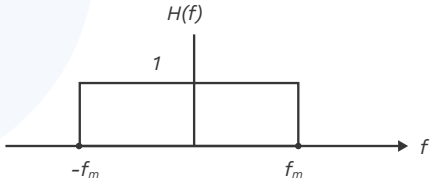
$$S_1(f) = \frac{M(f - f_c) + M(f + f_c)}{2}$$

$$S(t) = \underbrace{[m(t)\cos 2\pi f_c t]}_{S_1(t)} * h_i(t) \longleftrightarrow s(f) = S_1(f)H_i(f)$$

$$S(f) = \left\{\frac{M(f - f_c) + M(f + f_c)}{2}\right\} H_i(f)$$




**Case 1:** Low pass filter is ideal:




$$x(t) = s(t)\cos 2\pi f_c t$$

$$s(f) = \left\{\frac{M(f - f_c) + M(f + f_c)}{2}\right\} H(f)$$

$$X(f) = \frac{S(f - f_c) + S(f + f_c)}{2}$$

$$X(f) = \frac{[M(f - 2f_c) + M(f)]H_i(f - f_c)}{4} + \frac{[M(f) + M(f + f_c)]H_i(f + f_c)}{4}$$

$$X(f) = \frac{[M(f - 2f_c) + M(f)]}{4} H_i(f - f_c) + \frac{[M(f) + M(f + 2f_c)]H_i(f + f_c)}{4}$$

LPF output: $Y(f) = X(f)H_0(f)$

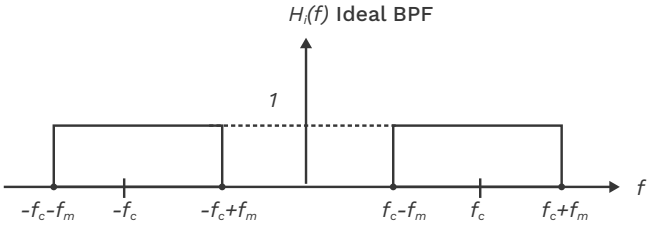
$$Y(f) = \left\{\frac{M(f)H_i(f - f_c) + M(f)H_i(f + f_c)}{4}\right\} : |f| \leq f_m$$

$\mu_1 = 0.8$ $\mu_2 = 0.6$

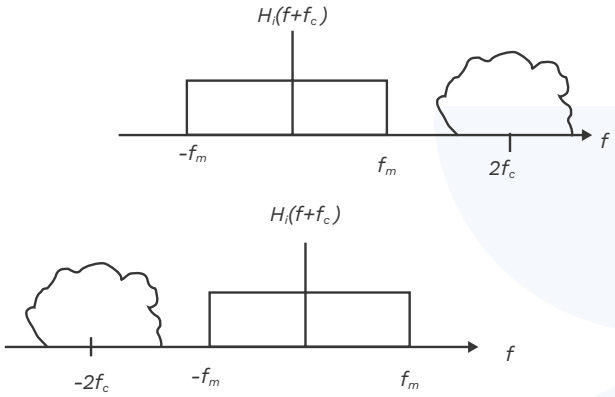
If $Y(f) \alpha M(f)$

Then $\left|H_i(f - f_c) + H_i(f + f_c)\right| = \text{constant} : |f| \le f_m$

$f_m \le f \le f_m$

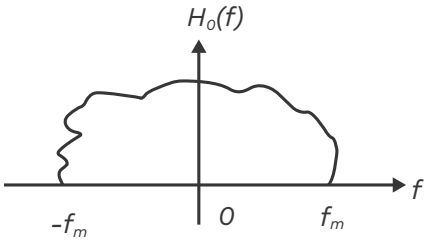



If $\boxed{Y(f) \ \alpha \ M(\phi)}$




$H_i(f + f_c) + H_i(f - f_c) = \text{contant} : -f_m \le f \le f_m$

**Case 2:** Non-ideal LPF as synch detector:




$$X(f) = \frac{[M(f - 2f_C) + M(f)]H(f - f_C)}{4} + \frac{[M(f + 2f_C) + M(f)H_i(f + f_C)]}{4}$$

$$Y(f) = X(f)H_0(f)$$

$$= \frac{M(f)}{4}\underbrace{[H_i(f - f_C) + H_i(f + f_C)]H_0(f)} : |f| \le f_m$$

$$D_{DSB} = P_c P_m = \frac{A_c}{2}$$

$$\boxed{|H_0(f)| \alpha \frac{1}{|H_i(f - f_c) + H_i(f + f_c)|}} \text{ in } |f| \le f_m$$

## Important Points

1. If spectral spacing between sidebands of DSB-SC is very less or 0Hz then BPF of SSB-SC arrangement is replaced by another filter with $H_i(f)$ for proper $P_c$ and $R_x$.
2. $\boxed{\left|H_i(f - f_c) + H_i(f + f_c)\right| = \text{constant} : |f \le f_m| \quad \text{when } H_0(f) : \text{Ideal LPF}}$
3. $|H_0(f)|\left|H_i(f + f_c) + H_i(f - f_c)\right| = \text{constant} : |f| \le f_m$

   When $H_0(f)$ is non-ideal

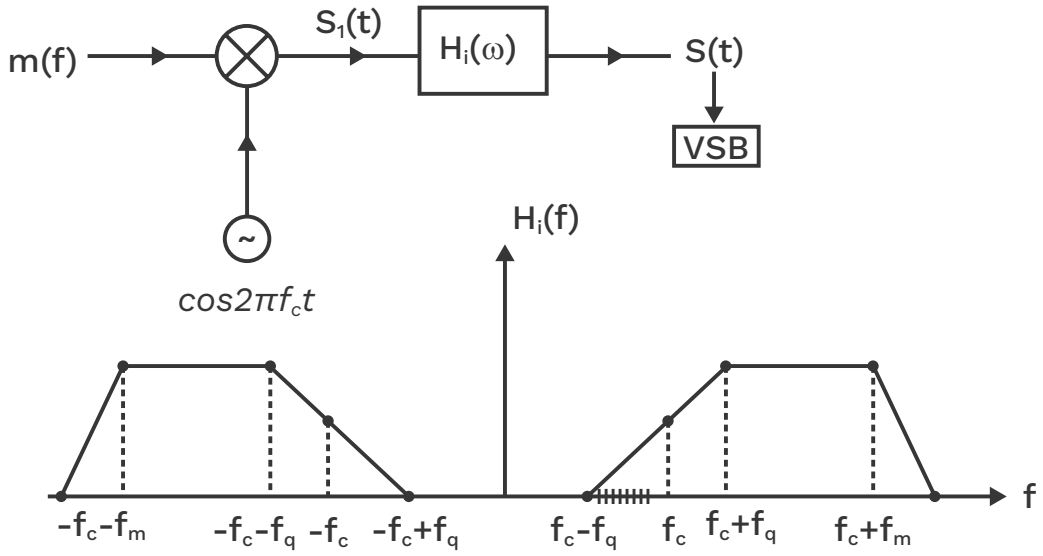
## Points To Remember

| Modulation | BW | Power | Application |
|---|---|---|---|
| DSB-FC | $2f_{max}$ | $P_c + P_{SB} \longrightarrow$ | Broadcasting |
| DSB-SC | $2f_{max}$ | $P_{SB} \longrightarrow$ | x |
| SSB-SC | $f_{max}$ | $P_{SB} / 2 \longrightarrow$ | Point-to-point voice communication |
| VSB-SC | $f_{max} < f < 2f_{max}$ | $\frac{P_{SB}}{2} < P_{VSB} < P_{SB} \longrightarrow$ | Point-to-point video communication |

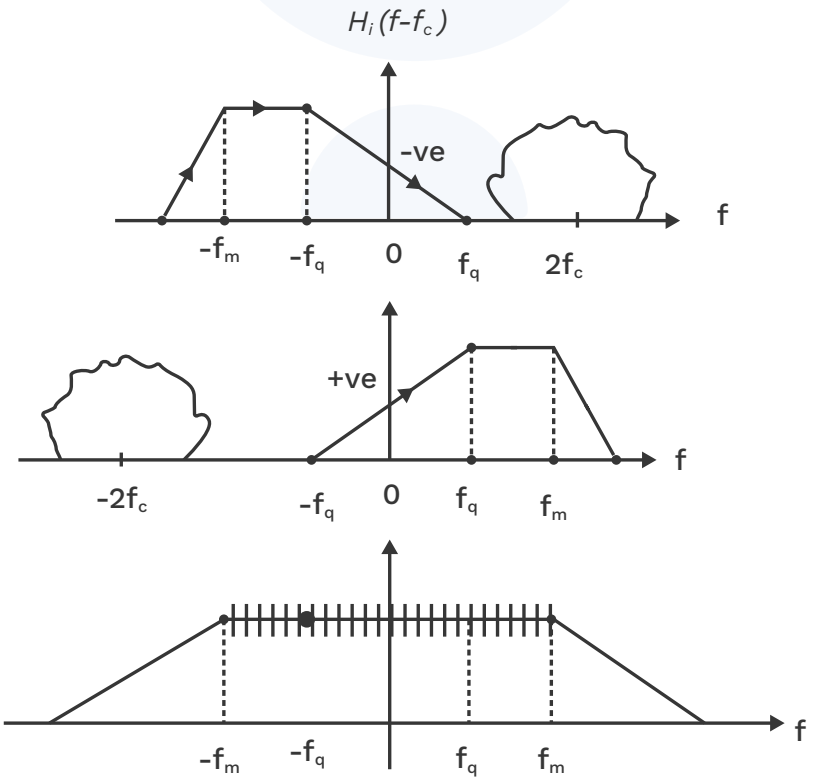
**Example:**

Spectrum of $S_1(t)$ is having 0Hz spectral GAP.

Will $m(t)$ be recovered by using system detector?




**Solution:** $Hi\left(f-f_c\right)+Hi\left(f+f_c\right)=\text{constant} \quad -f_m \le f \le f_m$



**Example:** $m(t) = \dfrac{1}{1+t^2}$

1. If $K_a = \dfrac{1}{20}$, calculate $\mu = ?$
2. General expression of SSB – SC

$$\rightarrow \begin{matrix} \nearrow \text{USB- SC} \\ \searrow \text{LSB- SC} \end{matrix}$$

**Solution:**

1. $E\left[X(t_1)\right] = \mu_x \qquad E\left[X(t_2)\right] = \mu_x$

$$\boxed{\mu = \frac{1}{20}} \qquad \boxed{\frac{dm(t)}{dt} = 0} \longrightarrow$$

$$\boxed{m(0) = 1}$$

$$\frac{\left(1+t^2\right)0 - 2t}{\left(1+t^2\right)} = 0$$

$$\left|m(0)\right|_{\max} = 1 \qquad \boxed{t = 0}$$

2. $S(t) = \dfrac{A_c}{1} m(t)\cos 2\pi f_c t + \dfrac{A_c}{2}\hat{m}(t)\sin 2\pi f_c t$

$$\boxed{\hat{m} = m_h}$$

$$\boxed{m(t) = \frac{1}{1+t^2} \overset{H.T.}{\longleftrightarrow} \hat{m}(t)}$$

$$\boxed{\frac{A_c}{2}\left\{\frac{1}{\left(1+t^2\right)}\cos 2\pi f_c t + \frac{t}{\left(1+t^2\right)}\sin 2\pi f_c t\right\}}$$

$$\hat{M}(\omega) = -j\,\text{sgn}(\omega)M(\omega) = \begin{cases} -jM(\omega) & \omega > 0 \\ jM(\omega) & \omega < 0 \end{cases}$$

$$\hat{M}(\omega) \overset{IFT}{\longleftrightarrow} \hat{m}(t)$$

$$m(t) = \frac{1}{1+t^2} \qquad e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+\omega^2}$$

$$\frac{e^{-|t|}}{2} \longleftrightarrow \frac{1}{1+\omega^2}$$

$$m(t) = \frac{1}{1+t^2} \longleftrightarrow \frac{2\pi e^{-|-\omega|}}{2} = M(\omega) \longrightarrow \pi e^{-|\omega|} = \begin{cases} \pi e^{-\omega} & \omega > 0 \\ \pi e^{\omega} & \omega < 0 \end{cases}$$

$$\hat{M}(\omega) = \begin{cases} -j\pi e^{-\omega} & \omega > 0 \\ j\pi e^{\omega} & \omega < 0 \end{cases}$$

$$\boxed{M(\omega) = \begin{cases} \pi e^{-\omega} & \omega > 0 \\ \pi e^{\omega} & \omega < 0 \end{cases}}$$

$$\boxed{\hat{M}(\omega) = +j\frac{d}{d\omega}M(\omega)}$$

$$\downarrow$$

$$\boxed{\hat{m}(t) = tm(t)}$$

$$\boxed{\begin{matrix} m(t) \longleftrightarrow M(\omega) \\ tm(t) \longleftrightarrow +j\dfrac{dm(\omega)}{d\omega} \end{matrix}}$$

$$\boxed{\begin{matrix} \dfrac{1}{1+t^2} \overset{H.T.}{\longleftrightarrow} \dfrac{t}{1+t^2} \\ \dfrac{t}{1+t^2} \overset{H.T.}{\longleftrightarrow} \dfrac{-1}{1+t^2} \end{matrix}}$$

## Pre-envelope and Complex Envelope

**1. Pre-envelope**

a) Pre-envelope is calculated for both baseband (lowpass) and bandpass signal.

b) Which means it is calculated for narrow hand and wideband signal.

Let x(t) is a real valued signal

$x_t(t) =$ pre-envelope of $x(t)$

$$\boxed{x_t(t) = x(t) + j\hat{x}(t)} \qquad \boxed{\hat{x}(t) = HT(x+1)}$$

**Note:** Pre-envelope of real valued signal results in complex signal.

**Spectrum of pre-envelope:**

$$x_t(t) = x(t) + j\hat{x}(t)$$

$$\boxed{\hat{x}(f) = -j\,\text{sgn}\,n(f)x(f)}$$

$$X+(f) = X(f) + j\hat{X}(f)$$

$$X+(f) = X(f)\left[1+\text{sgn}(f)\right]$$

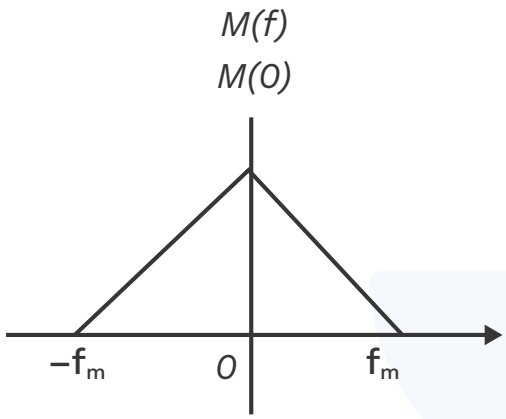
$$X+(f)=\begin{cases}0 & f<0\\ X(f)=X(0) & f=0\\ 2X(f) & f>0\end{cases}=X(f)\left(1+\operatorname{sgn}(f)\right)$$

**Note:** Spectrum of pre-envelope of real valued signal $x(t)\rightarrow$ exist only for +ve frequencies.

**Case 1:** m(t) is loco pass signal:

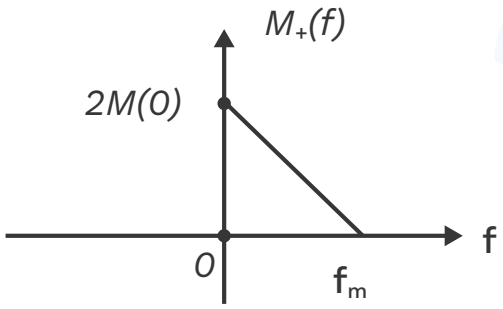
$m(t)\longleftrightarrow$

locopass




$m+(t)\longleftrightarrow$

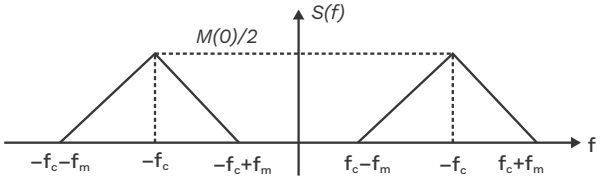
locopass




**Case 2:** $s(t)=m(t)\cos 2\pi f_c t$

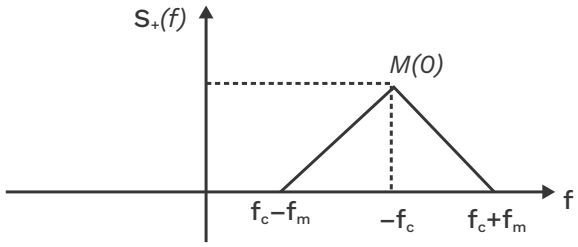
$s(t)\xleftrightarrow{F.T}$

Bandpass




$S_t(t)\xleftrightarrow{F.T}$

Bandpass




**Complex envelope:** It is calculated only for bandpass (Narrowband) signal:

$\rightarrow \underbrace{x(+)}_{\text{Bandpass}}=m(t)\cos w_c t$ High frequency carrier

lowpass resutsin lowpass sig.

Let x(t) is B.P.S

S.1 Calculate: $x_+(t)=x(t)+j\hat{x}(t)$

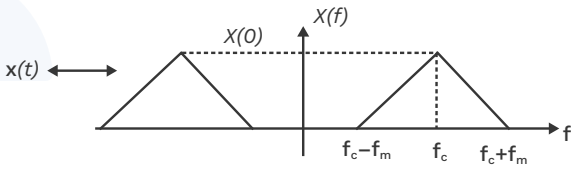
S.2 $x_c(t)=x_+(t)e^{-jw_c t}$

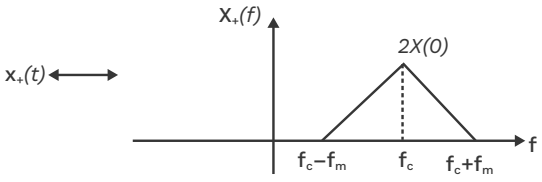
Spectrum of complex envelope:

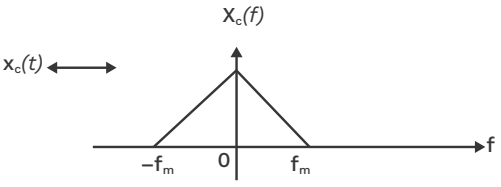
$x_c(t)=x_+(t)e^{-1\omega_c t}$

$X_c(f)=X_+(f+f_c)$

$\rightarrow$Left shift preenvelop by $f_c$




**Example:**

$x(t)=m(+)\cos(\omega_c t+\theta)\rightarrow$ Complex envelope:

Bandpass

**Solution:**

S.1 $x_+(t) = x(t) + j\hat{x}(t)$

$X_+(t) = m(t)\cos(\omega_c t + \theta) + j\, m(t)HT(\cos(\omega_c t + \theta))$

$X_+(t) = m(t)\cos(\omega_c t + \theta) + j\, m(t)\sin(\omega_c t + \theta)$

$X_+(t) = m(t)e^{j\omega_c t}e^{j\theta} \quad \rightarrow X_c(t) = m(t)e^{j\theta}$

**Example:**

$x(t) = m(t)\sin\omega_c t + \theta) \rightarrow$ Complex Envelop = ?

**Solution:**

$X_+(t) = m(+)\sin(\omega_c t + \theta) + jm(t)(-\cos(\omega_c t + \theta))$

$X_+(t) = -j\, m(+)[\cos(\omega_c t + \theta) + j\sin(\omega_c t + \theta)]$

$X_+(t) = -j\, m(+)e^{j\omega_c t}e^{j\theta} \rightarrow x_c(t) = -j\, m(t)e^{j\theta}$

$m(t)\cos(\omega_c t + \theta) \xrightarrow{C.E.} m(t)e^{j\theta}$

$m(t)\cos(\omega_c t + \theta) \xrightarrow{C.E.} m(t)e^{j\theta}$

$e^{-\frac{j\pi}{2}} m(+)e^{j\theta}$

$m(t)e^{j(\theta-\pi_2)}$

**Example:** $x(t) = -\dfrac{\sin 2\pi t}{\pi t}\cos(w_c t - \pi/2)$

Complex envelope = ?

**Solution:**

$x_c(t) = \left\{\dfrac{-\sin 2\pi t}{\pi t}\right\} e^{j(-\pi/2)} \qquad -1 = e^{-j\pi}$

$x_c(t) = \dfrac{\sin 2\pi t}{\pi t} e^{-j\left(\frac{\pi}{2}+\pi\right)}$

**Example:** $x(t) = -\dfrac{\sin 2\pi t}{\pi t}\sin(\omega_c t - \dfrac{\pi}{4})$

**Solution:**

$x(t) = m(t)\sin(\omega_c t + \theta)$

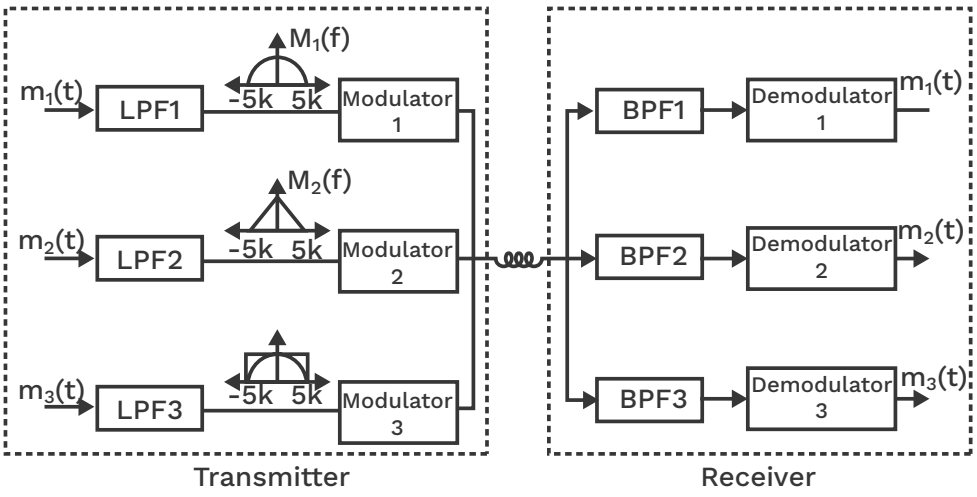
$x_c(t) = -jm(t)e^{j\theta} = -j\left(\dfrac{-\sin 2\pi t}{\pi t}\right)e^{-j\pi/4}$

$= \left(\dfrac{\sin 2\pi t}{\pi t}\right)e^{\frac{j\pi}{2}}e^{-\frac{j\pi}{4}}$

$x_c(t) = \left(\dfrac{\sin 2\pi t}{\pi t}\right)e^{\frac{j\pi}{4}}$

## Frequency Division Multiplexing (FDM)

- FDM is a multiplexing technique in which multiple separate information signals can be transmitted over a single communication channel by occupying different frequency slots within common channel bandwidth.
- This technique is used to multiplex continuous signals.
- In frequency division multiplexing, simultaneous transmission of the signals take place over a common channel in which the channel bandwidth is divided into various sub-channels comprised of different frequency slots for carrying individual signal during transmission.

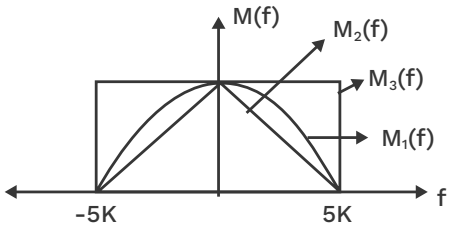


**Why FDM is used?**

For the transmission of signals $m_1(t), m_2(t)$ *and* $m_3(t)$ shown above, this method is used.

**Case 1:**

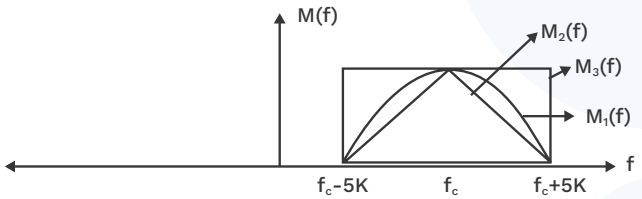
If no modulator is used, then interference between signals occurs as shown below:




Due to interference between signals, we are not able to recover message signals.

**Case 2:**

If modulator is used, but we are using the same frequency $f_c$.



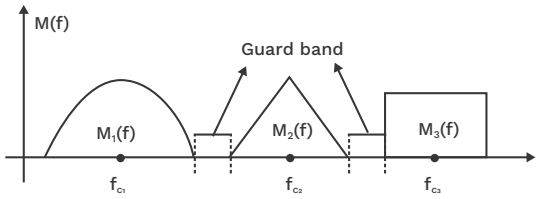

Transmitted output

With modulator:

Again, same problem arises and we are not able to recover the message signals.

**Case 3:**

Now to overcome this problem, FDM technique is used, as it multiplexed signal over different frequency slots as shown in figure below.




Transmitted output with FDM

Since, the band pass filters are not ideal hence for successful multiplexing guard band should be maintained.

**Example:** Five message signals which includes 3 AM, 1 DSB-SC and 1 SSB-SC signals each band limited to 10 KHz are multiplexed using FDM technique with a guard band of 1 KHz. Find out the multiplexed bandwidth.

**Solution: 94 KHz**

For multiplexing 5 signals, 4 guard bands are used multiplex bandwidth = (total msg bandwidth + guard band)

3 AM signals, bandwidth

$= 3 \times (2f_m) = 3 \times 2 \times 10 = 60\,KHz$

1 DSB-SC signal, bandwidth

$= 1 \times 2f_m = 10 \times 2 = 20\,KHz$

1 SSB-SC signal, bandwidth

$= 1 \times f_m = 10\,KHz$

Now, multiplexed bandwidth

$= (60 + 20 + 10 + 4 \times 1)\,KHz = 94\,KHz$

Hence, multiplexed bandwidth for this is 94 KHz.

## Exercise

**1.** The antenna current of an AM transmitter is 8 A when only the carrier signal is transmitted. What is the antenna current when the carrier signal is modulated by sinusoidal signal $V(t) = 1.4 \sin(2\pi \times 500t)$ with modulation index 0.8? **[ESE-2021]**

(A) 3.2 A

(B) 7.2 A

(C) 9.2 A

(D) 11.2 A

**Solution: (C)**

**2.** A 360 W carrier is simultaneously modulated by two audio waves with modulation percentages of 55 and 65 respectively. The effective modulation index and the total power radiated are **[ESE-2019]**

(A) 0.85 and 490.5 W
(B) 0.65 and 490.5 W
(C) 0.85 and 450.5 W
(D) 0.65 and 450.5 W

**Solution: (A)**

**3.** The Vestigial Sideband (VSB) modulation is preferred in TV systems because
I. It reduces the bandwidth requirement to half
II. It avoids phase distortion at low frequencies
Which of the above statement(s) is/are correct? **[ESE-2016]**
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

**Solution: (A)**

**4.** A single tone 4 kHz message signal is sampled with 10 kHz and 6 kHz. Aliasing effect will be seen in the reconstructed signal when the signal is sampled with **[ESE-2016]**
(A) 10 kHz
(B) 6 kHz
(C) Both 10 kHz and 6 kHz
(D) Neither 10 kHz nor 6 kHz

**Solution: (B)**

**5.** A sinusoidal audio signal is given by $e_s = 15\sin 2\pi(2000t)$. It modulates a sinusoidal carrier wave $e_c = 60\sin 2\pi(10^5 t)$. The value of amplitude modulation index must be **[ESE-2015]**
(A) 0.5
(B) 4
(C) 0.6
(D) 0.25

**Solution: (D)**

## Solutions

**1. (C)**

We have $I_t = I_C\sqrt{1+\frac{\mu^2}{2}}$

Given, $I_C = 8A,\ \mu = 0.8$

Substituting, we get,

$$I_t = 8\sqrt{1+\frac{(0.8)^2}{2}}$$

$$I_t = 9.2\,\text{A}$$

**2. Solution: (A)**

$$P_C = 360\text{W}$$

$$\mu_1 = 0.55$$

$$\mu_2 = 0.65$$

$$\mu_t = \sqrt{\mu_1^2 + \mu_2^2} = 0.85$$

$$P_t = P_C\left[1+\frac{\mu_t^2}{2}\right] = 360\left[1+\frac{(0.85)^2}{2}\right]$$

$$= 490.5\ \text{W}$$

**3. Solution: (A)**

**4. Solution: (B)**

To avoid aliasing

$$f_s \geq 2f_m$$

$$f_s \geq 2\times 4$$

$$f_s \geq 8Hz$$

$\Rightarrow$ Aliasing effect is seen if $f_s$ is 6 kHz.

**5. Solution: (D)**

Modulating signal $e_s = 15\sin(2\pi\times 2000t)$

Carrier signal $e_c = 60\sin(2\pi\times 10^5 t)$

$$A_c = 60 \quad A_m = 15$$

$$\mu = \frac{A_m}{A_c} = \frac{15}{60} = 0.25$$